AK High-Dimensional Projection Structural Theory

Appendix UB

Universal Bridge Component Library and Validated Return Portfolio

Reusable Dual-Core Adapters, Coefficient/Tail/Descent/Coherence/Rigidity Components,

and Twelve Source-Bound Validation Cases

Canonical v21 Global-Reaudited Maximal Derived-Theorem Edition

Theory version: v21.0.0

Canonical authority rule. AK-HPDST v21.0.0 Main Chapter 9 owns the universal Bridge datum, generic component calculus, target-relative strength calculus, defect/result sorts, and Unified HT–UB Bridge Closure theorem. Main Chapters 5 and 7 own the stable generic tail and finite-diagram/globalization theorems. Appendix DG owns their expanded finite-diagram and descent proofs. Appendix HT owns the complete HC–PE–SS–RL–IW/Return research spine. Appendix UB owns reusable executable component implementations, component-local proof expansions not duplicated by Main, regression witnesses, source-specific adapters, and the twelve-case validation portfolio. A UB locator is not a competing canonical theorem UID.

Conservative migration rule. The complete v20 UB source is retained as immutable predecessor mathematics at its exact source-supported strength. Generic statements promoted into the v21 Main or Core are read through their canonical Main identities; retained UB codes become implementation, proof-locator, regression, or provenance identifiers. Named arithmetic, PDE, representation-theoretic, and p-adic Returns remain source-specific and are not promoted into the stable Core by co-location or successful validation.

Release boundary. This artifact is a construction-stage v21 synchronization and extraction audit. Final Claim Register assignment, release-manifest closure, source-binding acceptance, and release designation remain downstream governance acts.

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Abstract

Appendix UB is the v21 synchronized reusable-component and validation artifact for the AK-HPDST Unified Bridge architecture. It consumes the Chapter 8 BridgeDAGInput handoff, the dual-Core outputs of Core_P and Core_{DG}, the Appendix DG descent contract, and the Appendix HT transfer spine. It does not duplicate the canonical Main Chapter 9 component grammar or closure theorem. Instead it records executable component schemas, exact input/output types, complete hypotheses, assumption transport, downward-closed target-relative strength sets, single-owner failure leaves, regression witnesses, and immutable replay obligations.

The re-audit separates four kinds of content that were conflated in the v20 predecessor: generic mathematics now canonically owned by Main Chapters 5, 7, and 9; UB-local reusable implementations and proof expansions; source-specific Bridge/Return theorems; and downstream validation cases. The generic finite-state, finite-diagram, descent, Kunneth, duality, determinant, coherence, rigidity, and closure statements are therefore linked to their canonical v21 owners rather than reissued with competing theorem authority. The twelve validated cases remain independently replayable leaves: VC1, VC2, and VC6 test finite reconstruction and descent; VC3 and VC4 treat periodic Tachikawa lanes; VC5 and VC9 treat Galois–Leopoldt and certified regulator lanes; VC7, VC7L, VC10, and VC11 treat inverse-limit and Iwasawa lanes; and VC8 treats a fixed Serrin compactness/continuation lane.

The v21 hardening adds the exact Chapter 8 handoff boundary, target-indexed Bridge results, separation of atomic status from readiness and Return result, the canonical top-level Bridge defect stack, one selected tail mechanism for each infinite quantifier, the canonical nine-field DescentInput with one active lane, Core-P/Core-DG/HT/UB interface firewalls, benchmark-quarantine and Chapter 10 no-backflow rules, componentwise migration/ownership records, and a lossless execution schema. No validation success strengthens a source theorem, no determinant or observation Return becomes object Return without rigidity, and no successful build or replay substitutes for proof.

The present derived-theorem extension additionally proves an Appendix-local implementation paracategory and substitution theorem, a finite Pareto strength frontier, a unique minimal causal failure antichain, an independent product-category Return, strict descent of finite-state derived inverse limits, faithful field-extension transport for such limits, and an exact mixed regular–singular one-variable Iwasawa finite-layer theorem. The second global re-audit proves quotient-composition congruence, makes descent stabilization stalkwise, removes any hidden product-existence assumption, and separates missing-evidence undefined from counterexample-certified method obstruction. These additions are derived only from already proved components and exact source classes; they do not promote the validation portfolio to universal completeness.

Contents

Re-audit integration and optimization map

Cluster	Validated cases	Unified reusable kernel	Non-compressible boundary
Finite reconstruction and descent	VC1, VC2, VC6	scalar zeta inversion; enriched generator equivalence; strict object/morphism/exact/derived descent	scalar probes do not recover enriched structure; higher coherence is not inferred from pairwise data
Periodic homological Return	VC3, VC4	finite syzygy orbit; stable identity; periodic or twisted-periodic one-degree self-Ext detection	general Tachikawa remains open outside the declared periodic/orbit classes
Arithmetic descent and nonvanishing	VC5, VC9	subgroup-fixed reconstruction; certified DVR valuation; regulator-to-Leopoldt Return	primitive regulator nonvanishing and new infinite families require external arithmetic input
Inverse-limit and Iwasawa Return	VC7, VC7L, VC10, VC11	stable-image limit; measure/Amice equivalence; certified Weierstrass data; regular and singular finite-layer profiles	multivariable/ noncommutative pseudo-null and higher-Fitting errors are not controlled by one determinant
Analytic compactness and continuation	VC8	finite time atlas; Aubin–Lions compactness; nonlinear product Return; Serrin continuation	the global Serrin witness is not generated from energy or persistence data

Optimization decision. The universal component library remains stated once, and each validated case remains an independently replayable theorem package. The former pre-validation “Application Theorems and Expansion Paths” part is removed because its Tachikawa, finite-poset, and local-to-global statements are strictly superseded by VC3, VC2, and VC6. The Iwasawa cases are logically integrated through the dependency chain

$$\mathrm{VC7} \longrightarrow \mathrm{VC7L} \longrightarrow \mathrm{VC9} \longrightarrow \mathrm{VC10} \longrightarrow \mathrm{VC11},$$

but are not physically collapsed into one proof block: VC10 is the cyclotomic-free exact lane, whereas VC11 is the positive-rank singular lane and depends on different certificate data.

Re-audit invariants. All positive statements retain their original hypotheses and Return ceilings. Source-theorem obligations remain local to the validated case that consumes them. Stable textual locators are normalized below so that the canonical registry refers to theorem environments that actually occur in the source.

Part I

v21 Constitutional Synchronization, Component Extraction, and Ownership Audit

1 Scope, authority, and conservative migration

Classification 1.1 (Appendix UB v21 role)**.** **Theory version** *AK-HPDST v21.0.0.*

Primary layer *Bridge/Return reusable component library plus Validation portfolio.*

Canonical generic authority *Main Chapter 9; Main Chapters 5 and 7 for stable tail and Core-DG mathematics.*

Expanded-proof authorities *Foundation Appendices A–G, Appendix DG, and Appendix HT at their declared scopes.*

UB authority *Executable component implementations, nonduplicative local lemmas, source-specific adapters and Returns, validation cases, counterexamples, tests, migration records, and theorem locators.*

Workflow status *v21 reaudited maximal construction candidate; final Claim Register and release acceptance pending.*

Axiom 1.2 (Immutable predecessor boundary)**.** *Every theorem, proof, counterexample, source binding, non-claim, and artifact inherited from the frozen v20 UB source remains immutable at its predecessor strength. A v21 item that repairs, strengthens, retypes, relocates, or operationalizes a predecessor statement has a distinct v21 role record. Migration is not retroactive mutation.*

Protocol 1.3 (Canonical precedence)**.** *For a v21 run the normative order is:*

1. *AK-HPDST v21 Main Chapters 1–10 at their canonical identities;*
2. *Foundation Appendices A–G and Appendix DG as proof/record owners for their Main chapters;*
3. *the Chapter 8* BridgeDAGInput *handoff;*
4. *Main Chapter 9 for the generic component, composition, defect, result, and closure statement;*
5. *Appendix HT for detailed HC–PE–SS–RL–IW/Return research-spine material;*
6. *Appendix UB for a separately registered executable component or source-specific validation theorem;*
7. *Problem Interface, Validation, Search/Platform, Companion, and governance artifacts at their typed roles.*

If sources disagree, the weakest source-supported type-correct reading governs.

Warning 1.4 (No third theorem authority). A generic theorem may have a canonical Main statement, an Appendix HT research expansion, and a UB executable implementation. It still has one canonical theorem identity. UB component codes are implementation/proof-locator identifiers unless a row below explicitly declares an Appendix-local theorem not duplicated by Main.

2 Incoming Chapter 8 handoff and v21 component schema

Definition 2.1 (UB consumer of the Chapter 8 handoff)**.** A UB run begins with a BridgeDAGInput record containing source and pre-Bridge boundary nodes, exact Bridge goal and target record, requested context and strength, propagated conditions, route-supported strengths, node/edge registries, source bindings, quarantine cut, artifact closure, atomic verdicts, failure routing, and manifest identity. The token `bridge_input_ready` may be emitted only after the pre-Bridge boundary passes. It is not a Bridge theorem and contains no final Return node.

Definition 2.2 (v21 UB implementation record)**.** A reusable implementation is

$$\mathrm{UBImpl} = (\mathrm{id}, \mathrm{owner}, \mathcal{C}_{\mathrm{src}}, \mathcal{C}_{\mathrm{tgt}}, F, \mathrm{Hyp}, \mathrm{Pres}, \mathrm{Out}, \mathrm{Cond}, \mathfrak{S}, \mathrm{Fail}, \mathrm{Test}, \mathcal{A}),$$

where owner names the canonical mathematical theorem or the Appendix-local theorem identity; F is the construction; Hyp is the complete typed hypothesis bundle; Pres is the proved preservation/reflection theorem; Out is the exact output type; Cond is the propagated assumption context; $\mathfrak{S}$ is a downward-closed set of target-relative strengths; Fail contains local single-owner failure leaves; Test is a positive and negative regression suite; and $\mathcal{A}$ is the immutable source/proof/execution artifact bundle.

Verifier contract 2.3 (UB extraction acceptance)**.** A statement is accepted as a reusable UB component only when:

1. it is independent of a named external problem, or every source-specific hypothesis is exposed as an input;
2. source and target types, variance, coefficients, amplitude/completion/topology, and equivalence relation are explicit;
3. construction provenance, theorem provenance, and execution artifacts are separate;
4. the output schema is stable and executable;
5. the supported strength set and prohibited converses are explicit;
6. every causal failure leaf has one primary owner;
7. at least one positive regression and one boundary/counterexample witness are attached;
8. migration and canonical-owner records are complete.

Copying prose from HT or Main does not satisfy this contract.

Theorem 2.4 (UB implementation compatibility)**.** *Let* $\mathrm{UBImpl}_1, \mathrm{UBImpl}_2$ *be certified implementations. They compose only when output/input types and variance match, coefficient and target records agree or are connected by a certified component, all assumptions propagate or are discharged, required coherence cells exist, and the requested strength lies in* $\mathfrak{S}_1 \cap \mathfrak{S}_2$*. The composite assumption context is the propagated union after named discharges and its supported strength set is the intersection. No stronger output and no smaller context is licensed.*

3 Atomic status, readiness, and target-indexed Bridge results

Definition 3.1 (Status-axis separation)**.** Each component obligation has exactly one Chapter 8 atomic status

$$\texttt{pass}, \quad \texttt{reject}, \quad \texttt{undefined}, \quad \texttt{not\_invoked}.$$

Readiness tokens such as `bridge_input_ready` and `descent_input_ready` are derived handoff tokens, not additional atomic statuses. For each fixed Return target record the Bridge primary result is exactly one of

$$\texttt{reject}, \texttt{undefined}, \texttt{obstructed}, \texttt{partial\_return}, \texttt{regularized\_return}, \texttt{return}.$$

The three sorts are never serialized in one field.

Definition 3.2 (Target-relative strength)**.** Strength is compared only relative to a fixed target record or along a registered target functor. On one reconstruction spine one may have

$$\Sigma_0 \preceq \Sigma_{\mathrm{pred}} \preceq \Sigma_{\mathrm{inv}} \preceq \Sigma_{\mathrm{obj}} \preceq \Sigma_{\mathrm{mor}} \preceq \Sigma_{\mathrm{ex}} \preceq \Sigma_{\mathrm{der}} \preceq \Sigma_{\mathrm{str}}.$$

Determinants, ideals, graded pieces, stable classes, pseudo-isomorphism classes, and regularized targets are distinct target records and may be incomparable. No universal scalar ceiling or unique greatest strength is assumed.

Proposition 3.3 (No assumption erasure)**.** *Every condition in an implementation record remains in the route context until a named theorem discharges it. Passing a later component, validating one example, or identifying a weaker target does not erase an earlier assumption.*

4 Canonical owner partition for the predecessor UB kernel

Predecessor family	Canonical v21 owner	UB v21 role	Prohibited reading
Universal datum, strength, compatibility	Main Definitions 9.4–9.10 and Theorem 9.11	executable schema, migration and regression	no competing universal datum or total strength order

Predecessor family	Canonical v21 owner	UB v21 role	Prohibited reading
Source realization, coefficients, completion, UCT, and Bockstein	Main 9.12–9.18 or named theorem owner	typed implementations and source-specific coefficient adapters	field, reduction, or completion is not lossless by notation
Core-P and P-to-DG	Main 9.19–9.21; Chapters 2–4	consumer adapters and finite replay	barcode/rank data do not manufacture diagram morphisms
Finite diagrams, generators, cones, Morita	Main Chapter 7 and Appendix DG	executable generator/probe implementations and tests	scalar probes do not replace enriched diagrams
Noetherian, finite-state, and tail components	Main Chapter 5; Main 9.28–9.29	implementation records and source-specific tower cases	finite prefix or state summary is not an infinite theorem
Descent and globalization	Main Chapter 7; Main 9.30–9.31; Appendix DG	lane-specific implementation and validation	readiness is not effectivity; pairwise maps are not a cocycle
Kunneth, perfect duality, determinant	Main 9.32–9.34	executable components, ring and PID boundaries, tests	determinant or invariant Return is not object Return
Coherence and rigidity	Main 9.35–9.38	generator-level implementation and promotion tests	metric proximity is not a coherence cell; no promotion without rigidity
Unified DAG, defects, results, closure	Main 9.39–9.50, especially Theorem 9.47	implementation, orchestration test, failure localization	UB-CLOSE is not a second canonical closure theorem
Source-specific Returns and twelve cases	UB / Problem Interface / Validation per case	theorem, adapter, execution and regression owner	successful case does not promote a generic Core theorem

Proposition 4.1 (Owner partition is non-overlapping)**.** *The preceding mapping preserves the complete predecessor mathematics while preventing duplicate v21 theorem identity. A retained UB theorem that coincides with a Main theorem is consumed as a proof/implementation locator; an Appendix-local theorem remains local only at its printed source-supported strength.*

5 Dual-Core and HT/UB interface firewalls

Definition 5.1 (Core-P export consumed by UB)**.** A base Core_P export contains only the passing Chapter 2–4 output actually proved: a repaired persistence object or predicate, finite detector witness, endpoint-complete finite-grid object, barcode/rank/Mobius data, admissible representative, or eligible higher-edge conclusion. Profile stage, comparison mode, endpoint policy, completeness, reflection, and information-loss fields remain independent.

Definition 5.2 (Core-DG export consumed by UB)**.** A Core_{DG} export contains the exact finite category presentation, relations and variance; actual objects and morphisms; generator/cone/Morita data; selected descent lane and effectivity result where invoked; conservativity theorem; output equivalence strength; supported lower set; and artifacts. An objectwise isomorphism class is not an actual diagram, and a metric record is not a natural transformation.

Verifier contract 5.3 (P-to-DG handoff)**.** A UB component crossing $\mathrm{Core}_P \to \mathrm{Core}_{DG}$ records the endpoint-complete finite grid/category, object and morphism action modes, exactness, reflection and information profiles, coefficient/variance/terminal conventions, category relations, supported strengths, artifacts, and prohibited readings. A barcode-only branch may Return only its registered objectwise barcode strength and cannot supply cones, natural transformations, cocycles, or descent maps.

Verifier contract 5.4 (HT–UB division)**.** Appendix HT owns research narratives, integrated module proofs, source-specific adapters, and successor investigations. UB owns small reusable implementations with stable schemas and tests. An HT output may enter UB only through the extraction acceptance contract; a UB output may enter HT only through an exact local input type and an explicit interface certificate. Co-location creates no arrow.

6 Tail, descent, coherence, and rigidity contracts

Definition 6.1 (One selected tail mechanism)**.** Every infinite quantifier selects exactly one certified mechanism for that quantifier: finite-state reduction, Noetherian bound, eventual constancy, Mittag–Leffler/Roos derived-limit control, completion theorem, or another named reduction. Different mechanisms may coexist for different quantifiers, but hypotheses or conclusions are not borrowed across mechanisms.

Definition 6.2 (Canonical descent input and active lane)**.** The only canonical descent input is

$$\mathrm{DescentInput} = (\{E_\alpha\}, \{g_{\alpha\beta}\}, \{c_{\alpha\beta\gamma}\}, \{h_\sigma\}, \mathcal{W}, \mathcal{C}, \mathfrak{M}_{\mathrm{coh}}, \mathcal{T}_{\mathrm{eff}}, \mathcal{A}).$$

Here $\mathcal{C}$ is object-level cover/restriction compatibility, not certificate-atlas completeness. One run selects exactly one lane:

$$\texttt{strict\_alexandrov\_sheaf},$$
$$\texttt{strict\_bounded\_complex},$$
$$\texttt{model\_certified\_derived}.$$

No unselected lane supplies assumptions, effectivity, comparison strength, or uniqueness.

Proposition 6.3 (Separation of readiness, effectivity, and conservativity)**.** *`descent_input_ready` certifies only the typed handoff. It does not imply effectivity, a descended object, uniqueness, joint conservativity, global zero/isomorphism, or external Return. Each is a separate theorem edge.*

Definition 6.4 (Coherence and rigidity implementation boundary)**.** A coherence implementation stores the actual strict equality, specified isomorphism, or higher cell and all boundary equations in a named ambient model. A rigidity implementation stores the observation/reconstruction functor, essential source class, external equivalence, finite presentation/generator theorem, normalization, and the promotion theorem. Without the latter, the Return ceiling is the observation target.

7 Canonical defect ownership and noncompensation

Definition 7.1 (Canonical top-level Bridge defect stack)**.** The v21 top-level stack is

$$\mathfrak{D}_{\mathrm{Bridge}} = (D_{\mathrm{src}}, D_{\mathrm{real}}, D_{\mathrm{coeff}}, D_P, D_{DG}, D_{HT},\\ D_{\mathrm{tail}}, D_{\mathrm{desc}}, D_{\mathrm{coh}}, D_{\mathrm{rig}}, D_{\mathrm{repl}}, D_{\mathrm{return}}, D_{\mathrm{cert}}).$$

The predecessor UB stack is retained only as an implementation-local refinement mapped into these owners. It does not add competing top-level defect sorts.

Protocol 7.2 (Single primary owner)**.** *Each concrete causal failure is stored once under the earliest typed owner that can state it. Missing descent effectivity belongs to D_{desc}; a proved descended object whose restrictions are not jointly conservative for a requested global-zero inference belongs to D_{DG}; absent external interpretation belongs to D_{return}. Downstream nodes reference the same leaf rather than duplicate it.*

Proposition 7.3 (Noncompensation)**.** *No metric reserve, numerical agreement, AI confidence, finite sample count, successful compilation, replay, manifest completeness, or validation pass repairs a false, absent, ill-typed, circular, or unsupported necessary component.*

8 Twelve-case v21 validation migration

Case	Principal v21 layer	Preserved UB contribution	Nonpromotion boundary
VC1	Validation / Appendix-local finite class	exact box-rank inversion, sharp scalar-probe lower bound, replay	not wild multiparameter classification; scalar ranks do not reconstruct arbitrary enriched diagrams
VC2	Core-DG regression plus Validation	explicit enriched projective-probe equivalence; exact, Yoneda, and derived execution	generic generator and Morita authority remains Main Chapter 7/DG
VC3	source-specific Validation	finite-window Nakayama Tachikawa closure	no general Tachikawa theorem
VC4	source-specific Validation / Bridge	periodic and twisted-periodic one-degree Return	only named periodic/orbit families
VC5	arithmetic Problem Interface / Validation	finite Galois fixed-point reconstruction and defect propagation	primitive regulator nonvanishing remains external
VC6	Core-DG regression plus Validation	explicit finite Alexandrov, descent, and hypercohomology execution	generic descent authority remains Chapter 7/DG; one active lane only
VC7	Chapter 5/9 tail regression plus Validation	finite-state inverse-limit, stable-image and derived-apex execution	no arbitrary infinite tower conclusion
VC7L	arithmetic Problem Interface / Validation	measure, Amice, Fourier, and finite-precision and p-adic L interfaces	reciprocity, characteristic ideal and Leopoldt leaves remain named inputs
VC8	PDE Problem Interface / Validation	compactness, nonlinear-defect, finite Serrin atlas, continuation	no generated Serrin witness and no unconditional 3D regularity
VC9	arithmetic Validation	complete-DVR nonvanishing, logarithm and regulator replay	new uniform nonvanishing families require external proof

Case	Principal v21 layer	Preserved UB contribution	Nonpromotion boundary
VC10	arithmetic and Iwasawa Validation	exact regular finite-layer growth beyond certified threshold	no multivariable or noncommutative or pseudo-null control
VC11	arithmetic and Iwasawa Validation	singular rank, Bockstein and reduced-resultant Return	determinant-only data do not recover full module structure

Definition 8.1 (Validation case record)**.** Each case records its immutable ID, unique principal validation class, exact input and source binding, selected proof-DAG route, expected and observed outputs, target-relative strength, propagated assumptions, artifacts, atomic statuses, target-indexed result, failure witness, and prohibited stronger reading.

Axiom 8.2 (Chapter 10 leaf and benchmark quarantine)**.** *Validation results, benchmark truth values, target constants, conclusion-derived witnesses, and known final answers are downstream leaves. They may validate a completed derivation but may not construct the realization, detector, finite certificate, reduction, rigidity, or Return theorem of the route being validated.*

9 Lossless serialization and replay

Verifier contract 9.1 (v21 UB run schema)**.** A lossless run serializes at least:

```
ub_bridge_run:
  theory: AK-HPDST-v21.0.0
  appendix: UB
  revision_id: immutable
  predecessor_artifact: content_addressed
  bridge_dag_input:
    identity: immutable
    atomic_status: pass | reject | undefined | not_invoked
    readiness_token: bridge_input_ready | null
  component:
    implementation_id: immutable
    canonical_owner: Main_or_UB_local
    source_type: typed
    target_type: typed
    hypotheses: complete
    propagated_context: []
    supported_strengths: downward_closed
    failure_leaves: single_owner
    regression_artifacts: []
  selections:
    tail_mechanism: named | null
    descent_lane: strict_alexandrov_sheaf |
      strict_bounded_complex | model_certified_derived | null
  target_results:
    exact_target: {id: immutable, result: typed}
    related_targets: []
  quarantine_cut: immutable
  artifact_bundle: []
```

```
claim_register_status: pending_final_CR
```

Omission of target identity, canonical owner, assumptions, supported strengths, atomic/readiness separation, actual required morphisms, tail selection, nine-field descent input, active lane, effectivity, rigidity, replacement identity, quarantine, artifacts, or failure routing destroys verifier equivalence.

Proposition 9.2 (Replay hierarchy)**.** *Operational replay, semantic replay, and mathematical reconstruction are distinct. A successful XeLaTeX build or deterministic re-execution proves only its declared operational property unless interpretation and theorem obligations are separately verified.*

10 Migration/ownership ledger and regression suite

Predecessor family	Semantic status	Disposition	v21 canonical owner	UB role
Frozen v20 UB source	inherited	deprecated, source-only for exact historical runs	frozen source artifact	immutable predecessor corpus
v21 universal datum, status, and closure overlay	strengthened	merged, relocated	Main Chapter 9	implementation and audit schema
Finite generator, Morita, and cone kernel	strengthened or inherited per item	relocated where generic	Main Chapter 7 / DG	executable component and regression
Noetherian and finite-state tail kernel	strengthened or inherited per item	relocated where generic	Main Chapter 5; Main 9.28–9.29	implementation and source-specific cases
Finite descent kernel	strengthened or inherited per item	relocated where generic	Main Chapter 7 / DG; Main 9.30–9.31	lane implementation and validation
Kunneth, duality, determinant, coherence, and rigidity	strengthened	relocated, merged	Main 9.32–9.38	executable components and proof tests
Unified UB–HT closure	strengthened	relocated, merged	Main Theorem 9.47	orchestration implementation and regression
Twelve source-bound cases	inherited or strengthened per printed theorem	source-specific	UB / Problem Interface / Validation	exact theorem, adapter and replay owner
Single-artifact locator registry	inherited	merged	Appendix UB	navigation and provenance only

Protocol 10.1 (Claim Register deferral)**.** *Stable local locators, canonical Main references, predecessor locators, and migration rows support construction-stage cross-audit. They are not final Claim UIDs. Final extraction occurs only after UB, CM, TB, Technical Extension, Problem Interface, Validation, Search/Platform, Companion, and release-manifest closure are frozen.*

Classification 10.2 (Mandatory UB regression witnesses)**.** *The regression suite includes at least:*

1. *scalar ranks agreeing while enriched diagrams differ;*
2. *a barcode-only P-to-DG branch unable to produce a natural transformation or cone;*
3. *field Kunneth used over a general ring without Tor control;*
4. *a dualizable but noninvertible observation that fails reflection;*
5. *a finite tower prefix with a later divergent tail;*
6. *equality of finite state summaries without typed transition compatibility;*
7. *pairwise local isomorphisms without a cocycle;*
8. `descent_input_ready` *without effectivity;*
9. *determinant or characteristic-ideal agreement without rigidity for exact objects;*
10. *a regularized target incorrectly identified with the original exact target;*
11. *a benchmark conclusion or witness crossing the quarantine cut;*
12. *a validation pass copied into an upstream component premise.*

11 v21 UB closure and contribution ceiling

Theorem 11.1 (Appendix UB v21 synchronization closure)**.** *The present overlay yields a conservative v21 UB architecture in which:*

1. *every generic theorem has one canonical Main/Core owner;*
2. *UB component entries have stable executable schemas, exact hypotheses, strength sets, failure leaves, artifacts, and regressions;*
3. *the dual-Core and HT interfaces are typed without automatic reconstruction;*
4. *tail, descent, coherence, rigidity, replacement, and Return remain independent necessary edges;*
5. *all twelve source-bound validation cases remain available at their printed strengths without Core promotion;*
6. *predecessor mathematics and locators remain preserved;*
7. *final Claim Register and release acceptance remain downstream.*

The theorem asserts architecture and ownership closure, not proof of every source-specific external hypothesis and not release acceptance.

Non-claim 11.2. Appendix UB proves no general Tachikawa conjecture, Iwasawa main conjecture, unconditional Leopoldt theorem, unconditional Navier–Stokes regularity, universal regulator nonvanishing theorem, wild multiparameter classification, noncommutative Iwasawa control theorem, or universal finite model for arbitrary mathematics. Its reusable contribution is exact-strength componentization, implementation, testing, and source-bound finite Return validation.

Part II

Appendix-Local Derived Theorem Pack and Cross-Component Strengthening

12 Admission rule, theorem identity, and derivation boundary

Classification 12.1 (Derived-theorem status)**.** *The statements in this Part are new Appendix-UB-local v21 theorems derived from the already proved component and validation corpus. They do not replace Main Chapter 9 generic theorem identities. Each theorem below satisfies one of the following admissible forms:*

1. *a formal consequence of the certified implementation contract that is specific to executable UB records;*
2. *a nontrivial composition of two or more independently proved UB cases or Main-owned components;*
3. *an exact theorem on a strictly declared source class with an explicit failure boundary.*

No theorem is admitted solely because several validation cases succeeded, because a computation agreed with a benchmark, or because a stronger statement would be architecturally convenient.

Protocol 12.2 (No duplicate promotion)**.** *If a derived statement is extensionally identical to a canonical Main theorem, the Main theorem remains the owner and the present statement is only an implementation corollary. A genuinely Appendix-local theorem must expose either new executable structure, a new cross-component composition, or a strictly smaller source class carrying a stronger explicit conclusion than the generic Main statement.*

13 Implementation algebra and replacement invariance

Definition 13.1 (Interface-equivalent UB implementations)**.** Let UBImpl and UBImpl$'$ be certified implementations. They are *interface-equivalent*, written

$$\mathrm{UBImpl} \simeq_{\mathrm{int}} \mathrm{UBImpl}',$$

when:

1. they name the same canonical mathematical owner or the same Appendix-local theorem;
2. their source and target schemas, variance, coefficient data, target identity, and requested relation agree;
3. their constructions are connected by a typed natural isomorphism on the common hypothesis domain;

4. their hypothesis bundles are interderivable through registered discharge theorems and normalize to the same propagated context;
5. their exact output types and downward-closed supported strength sets agree under the natural isomorphism;
6. their primary failure leaves correspond bijectively with cause and scope preserved;
7. the replacement artifact bundle contains semantic replay evidence for all preceding identifications.

Operational file identity is not required to agree, but mathematical and verifier-facing semantics must agree.

Proposition 13.2 (Interface equivalence is a compositional congruence)**.** *On certified implementations, $\simeq_{\text{int}}$ is an equivalence relation. Moreover, whenever the displayed composites are defined,*

$$\mathrm{UBImpl}_1 \simeq_{\text{int}} \mathrm{UBImpl}_1', \qquad \mathrm{UBImpl}_2 \simeq_{\text{int}} \mathrm{UBImpl}_2'$$

imply

$$\mathrm{UBImpl}_2 \circ \mathrm{UBImpl}_1 \simeq_{\text{int}} \mathrm{UBImpl}_2' \circ \mathrm{UBImpl}_1'.$$

Thus compatible composition descends to interface-equivalence classes.

Proof. Reflexivity uses the identity natural isomorphism and identity replay artifact. Symmetry uses inverse natural isomorphisms, the symmetric interderivability of the normalized hypothesis bundles, and the inverse bijection of failure leaves. Transitivity is obtained by composing the typed natural isomorphisms, discharge records, failure-leaf bijections, and semantic replay artifacts. For the congruence statement, whisker the two natural isomorphisms through the adjacent constructions. Deterministic normalization gives the same propagated context, supported strength set, and single-owner failure semantics on both composites. The concatenated replay artifacts certify these identifications. □

Theorem 13.3 (UB-v21-DER-01: certified implementation paracategory)**.** *Fix a target-relative strength preorder and a deterministic normalization rule for propagated contexts, discharged assumptions, and failure-leaf names. Typed boundary schemas form the objects of a paracategory whose arrows are interface-equivalence classes of certified UB implementations and whose partial composition is the compatibility composition of the v21 UB implementation contract. Identity implementations exist, composition is well defined on equivalence classes, and every defined triple composite is associative up to interface equivalence.*

Proof. For a boundary schema X, the identity construction on X, with empty additional assumptions, the boundary-supported strength set, no new failure leaves, and the identity replay artifact, is certified and acts as a two-sided identity. If UBImpl_1 and UBImpl_2 are compatible, their composite construction is ordinary functorial composition; its context is the normalized propagated union, its supported strength set is the intersection, and its failure leaves are the normalized single-owner union. Proposition 13.2 makes this composition independent of the chosen representatives. For three compatible implementations, construction composition is associative, set intersection is associative, and normalized union is independent of parenthesization by the fixed normalization rule. Coherence isomorphisms and artifact concatenation provide the required interface equivalence. Composition is partial exactly because type, variance, coefficient, target, coherence, or strength compatibility may fail. □

Corollary 13.4 (UB-v21-DER-02: implementation substitution invariance)**.** *Let a certified proof-DAG contain an implementation* UBImpl*, and let* $\mathrm{UBImpl}' \simeq_{\mathrm{int}} \mathrm{UBImpl}$*. Replacing* UBImpl *by* UBImpl' *preserves every mathematical target result, propagated assumption context, supported Return strength, and primary failure semantics. Only implementation identity, execution provenance, and artifact hashes may change.*

Proof. Every incoming and outgoing interface sees the same source and target schemas, context normal form, output, strength set, and failure semantics. The natural isomorphism transports the internal construction, and semantic replay verifies the substitution. Apply associativity in Theorem 13.3 to the upstream and downstream composites. □

14 Finite strength frontiers and minimal causal failure cuts

Definition 14.1 (Finite route strength quotient)**.** For a finite proof-DAG route, quotient its target-relative strength preorder by mutual reachability:

$$\sigma \sim \tau \quad \Longleftrightarrow \quad \sigma \preceq \tau \text{ and } \tau \preceq \sigma.$$

Assume the resulting poset $\overline{\mathfrak{S}}$ is finite on the requested target record. For every route edge e, let $S_e \subseteq \overline{\mathfrak{S}}$ be its downward-closed supported set.

Theorem 14.2 (UB-v21-DER-03: finite Pareto strength-frontier theorem)**.** *For a finite compatible route with edge set E, the composite supported set*

$$S_{\text{route}} = \bigcap_{e \in E} S_e$$

is downward closed and is determined uniquely by the finite antichain

$$\text{Front(route)} := \text{Max}(S_{\text{route}}).$$

More precisely,

$$S_{\text{route}} = \downarrow \text{Front(route)}.$$

Adding a required component can only shrink the frontier-generated lower set. Replacing one component by a certified implementation with a larger supported set can only enlarge it. No total-order or unique-ceiling assumption is needed.

Proof. Intersections of downward-closed subsets are downward closed. If $S_{\text{route}} = \emptyset$, then its frontier is empty and $\downarrow \emptyset = \emptyset$. Otherwise, in a finite poset every element of S_{route} lies below a maximal element, so the set equals the lower closure of its maximal elements. Maximal elements are pairwise incomparable, hence form an antichain, and uniqueness follows because the maximal elements of a fixed set are intrinsic. Adding a required component adds an intersection factor. Replacing one component by a certified compatible implementation with a larger supported set enlarges that factor and therefore cannot shrink the composite supported set. □

Definition 14.3 (Causal blocker set)**.** Let G be a finite acyclic proof-DAG. Write $u \leq_G v$ when a directed path runs from u to v. Let F be the set of required nodes whose atomic status is `reject` or `undefined`. Define

$$B(G, F) = \{v \in F : \text{ no } u \in F \text{ satisfies } u <_G v\}.$$

Thus $B(G, F)$ contains the earliest unresolved causal failures, not every downstream node prevented from completing.

Theorem 14.4 (UB-v21-DER-04: unique minimal failure-antichain theorem)**.** *The causal blocker set $B(G, F)$ is the unique inclusion-minimal antichain contained in F such that every node of F lies on or downstream from an element of $B(G, F)$. In fact, every failed/undefined ancestry cover contained in F contains $B(G, F)$. Consequently:*

1. *if $B(G, F) \neq \emptyset$, the route cannot emit* `return`*;*
2. *changing only descendants of unchanged blocker nodes cannot repair the route;*
3. *a sound repair protocol first repairs or scope-excludes every blocker, then reevaluates its descendants; favorable sibling nodes never compensate for a blocker.*

Proof. Because G and F are finite, every $v \in F$ has a $\leq_G$-minimal failed or undefined ancestor, which belongs to $B(G, F)$. Two elements of $B(G, F)$ cannot be comparable, so it is an antichain. Any failed/undefined ancestry cover contained in F, whether or not initially presented as an antichain, must contain each minimal failed node. Removing comparable redundant elements then leaves the same blocker antichain, proving uniqueness and inclusion-minimality. A required blocker prevents every route containing it from passing. Altering a strict descendant does not change the blocker status, and noncompensation excludes repair by siblings. □

Theorem 14.5 (UB-v21-DER-05: independent target-product Return)**.** *Let two target routes R_1, R_2 share an identical certified upstream sub-DAG and have disjoint terminal target nodes. Suppose their union is acyclic, every shared node has one identity, one status, and one failure owner, and the terminal Returns have the same certified upstream source object. Declare the product-record target in the product category*

$$(\mathcal{T}_1, \mathcal{T}_2) \in \mathcal{C}_1 \times \mathcal{C}_2$$

with coordinatewise target relation and registered projection functors. If both routes pass and there is no additional cross-target compatibility obligation, then their union yields a Return to this product-record target at product strength (σ_1, σ_2). Its supported set is the product of the two route-supported lower sets, and shared failures are deduplicated. Conversely, a product-record Return projects to the two component Returns. A failure in either component route is not compensated by success in the other.

Proof. The product-record construction pairs the two terminal constructions on the same upstream source object. Its assumptions are the normalized union, and the declared coordinatewise relation makes its strength coordinates independent. Hence its supported lower set is the Cartesian product of the component lower sets. The projection functors recover the original target records and Return maps. Since shared upstream nodes are identified rather than copied, failure ownership remains single-valued. Noncompensation is coordinatewise. If a cross-target equation, coherence cell, diagonal condition, or common-target product object is requested, it is an additional component and the present independence theorem does not discharge it. □

15 Finite-state descent and derived inverse-limit Return

Definition 15.1 (Strict finite-state descent tower)**.** Let P be a finite poset, let $\mathcal{U} = \{U_i\}_{i=1}^m$ be a finite upper-Alexandrov open cover, and fix a field k. A strict finite-state descent tower consists of:

1. bounded local complexes
$$C^\bullet_{i,n} \in \mathrm{Ch}^b(\mathrm{Diag}_{U_i}(k)) \qquad (n \geq 0);$$
2. overlap chain isomorphisms satisfying the strict triple-cocycle equations;
3. bonding chain maps $d_{i,n} : C^\bullet_{i,n+1} \to C^\bullet_{i,n}$ commuting with every overlap isomorphism;
4. one deterministic finite-state presentation of the complete local, overlap, and bonding record, eventually periodic from N with period r.

Levelwise strict descent produces global complexes $C^\bullet_n \in \mathrm{Ch}^b(\mathrm{Diag}_P(k))$ and global bonding maps.

Theorem 15.2 (UB-v21-DER-06: strict descent of finite-state derived inverse limits)**.** *Let $(C^\bullet_{i,n})$ be a strict finite-state descent tower, and let $C^\bullet_n$ be its levelwise descended global tower. For each degree q, write $T_{i,q}$ for the period monodromy on the local cohomology diagram $H^q(C^\bullet_{i,N})$. Choose a stalkwise stabilizing exponent*
$$E_q \geq \max_{1\leq i\leq m} \max_{p\in U_i} \dim_k H^q(C^\bullet_{i,N})(p)$$
and set
$$W_{i,q} := \mathrm{im}(T_{i,q}^{E_q}) \quad \textit{in } \mathrm{Diag}_{U_i}(k).$$
Then:

1. *the $W_{i,q}$ and their restricted overlap maps form a strict descent datum;*
2. *the descended global tower is finite-state with certified eventual period r (not necessarily minimal);*
3. $\varprojlim^1 H^q(C^\bullet_n) = 0$ *in* $\mathrm{Diag}_P(k)$*;*
4. *the local stable images descend to the cohomology of the global derived inverse limit:*
$$\boxed{H^q\left(\mathbf{R}\varprojlim_n C^\bullet_n\right) \cong \mathrm{Desc}_{\mathcal{U}}(\{W_{i,q}\}).}$$
In particular, $\mathbf{R}\varprojlim_n C^\bullet_n$ is acyclic if and only if every local stable-image diagram $W_{i,q}$ is zero.

Proof. Overlap isomorphisms commute with the bonding maps and hence with period monodromy. They therefore carry $\mathrm{im}(T_{i,q}^{E_q})$ isomorphically to the corresponding stable image on each overlap, and the inherited triple cocycle gives strict descent. Levelwise descent is functorial in the complete local record, so equality of states after one period descends to equality of global states and bonding maps.

The finite diagram category is abelian, restriction to an upper open is exact, and kernels, images, inverse-system limits, derived-limit defects, and cohomology objects are computed pointwise. The displayed stalkwise bound is therefore a simultaneous Fitting-stabilization exponent. Apply the VC7 finite-state derived-limit theorem at every point of the global diagram. Restriction to U_i identifies the resulting global stable-image diagram with $W_{i,q}$, because exact restriction commutes with the image of the monodromy power. Strict descent uniqueness identifies that global stable image with $\mathrm{Desc}_{\mathcal{U}}(\{W_{i,q}\})$. The objectwise Milnor/Roos calculation gives $\varprojlim^1 = 0$ and identifies derived-limit cohomology with this stable image. Finally, the finite cover restrictions are jointly conservative, so a descended diagram is zero exactly when all cover restrictions are zero. □

Corollary 15.3 (UB-v21-DER-07: local-to-global finite-state quasi-isomorphism criterion)**.** *Let $f_{i,n} : C^\bullet_{i,n} \to D^\bullet_{i,n}$ be a morphism of strict finite-state descent towers, compatible with overlaps and sharing a certified eventual period. For the local cone towers let $T^{\mathrm{cone}}_{i,q}$ be the period monodromy. Then*

$$\mathbf{R}\varprojlim_n f_n : \mathbf{R}\varprojlim_n C^\bullet_n \longrightarrow \mathbf{R}\varprojlim_n D^\bullet_n$$

is a quasi-isomorphism if and only if

$$\mathrm{im}\big((T^{\mathrm{cone}}_{i,q})^{E_q}\big)\,(p) = 0$$

for every cover member U_i, every point $p \in U_i$, and every cohomological degree q, where E_q satisfies the corresponding stalkwise dimension bound of Theorem 15.2.

Proof. The global map is a quasi-isomorphism exactly when the derived inverse limit of its cone tower is acyclic. Cone formation commutes with restriction and strict descent. Apply Theorem 15.2 to the cone tower. □

Theorem 15.4 (UB-v21-DER-08: field-extension invariance of finite-state derived limits)**.** *Let K/k be any field extension and let $(C^\bullet_n)$ be a uniformly bounded finite-state tower of finite-dimensional k-complexes, or of such complexes in a fixed finite diagram category. Base change the complete finite-state presentation, including every bonding map and state equality. Then the canonical comparison map*

$$K \otimes^{\mathbf{L}}_k \mathbf{R}\varprojlim_n C^\bullet_n \longrightarrow \mathbf{R}\varprojlim_n (K \otimes_k C^\bullet_n)$$

is a quasi-isomorphism. Moreover, acyclicity and finite-state derived-limit quasi-isomorphisms are reflected by extension of scalars to K.

Proof. The field extension is faithfully flat, so derived tensor equals ordinary tensor and commutes with cohomology. For each q, VC7 gives natural Milnor identifications

$$H^q(\mathbf{R}\varprojlim C^\bullet_n) \cong \varprojlim H^q(C^\bullet_n) \cong W_q, \qquad \varprojlim{}^1 H^{q-1}(C^\bullet_n) = 0,$$

where $W_q = \mathrm{im}(T_q^{E_q})$ for a certified stabilizing exponent. Exactness of $K \otimes_k -$ gives

$$K \otimes_k W_q \cong \mathrm{im}((K \otimes_k T_q)^{E_q}),$$

and the base-changed finite-state tower has the same state cycle and again has vanishing $\varprojlim^1$. Naturality of the tensor-to-product comparison and of the Milnor map identifies the cohomology map induced by the displayed canonical comparison with this stable-image isomorphism. Hence the comparison is a quasi-isomorphism, even though tensor product need not commute with an uncontrolled countable product. Faithful flatness reflects zero objects; applying it to the derived-limit object or to the cone of a finite-state derived-limit morphism proves the reflection statements pointwise in a finite diagram category as well. $\square$

Warning 15.5 (Boundary of Theorem 15.4). The theorem uses the finite-state stable-image model. It does not assert that arbitrary scalar extension commutes with an uncontrolled countable product or an arbitrary derived inverse limit. Without finite-state reduction or another certified commutation theorem, the corresponding route remains `undefined`.

16 Unified regular–singular one-variable Iwasawa layer profile

Definition 16.1 (Exact regular–singular UB class)**.** Let $\Lambda = \mathcal{O}[[T]]$, where $\mathcal{O}$ is a complete DVR and $e = v_\varpi(p)$. An exact regular–singular UB object is an explicitly given direct sum

$$X = \bigoplus_{a=1}^{r} X_a^{\mathrm{reg}} \oplus \bigoplus_{b=1}^{s} \Lambda/(f_b),$$

where:

1. each X_a^{reg} has a certified VC10 square presentation with cyclotomic-free distinguished factor and exact tuple

$$(\mu_a, \lambda_a, C_a, n_a);$$

2. each f_b is a VC11 cyclic certificate with finite cyclotomic support S_b, support degree D_b, regular factor tuple, and exact reduced-torsion function $L_b^{\mathrm{sing}}(n)$ from UB-VC11-04;

3. the direct-sum decomposition is exact source data, not a pseudo-isomorphism or determinant-only surrogate.

For each singular summand set

$$N_b^{\mathrm{sing}} := \max(S_b \cup \{0, n_0(P_b)\}),$$

and define

$$N_X := \max(\{0\} \cup \{n_a\}_{a=1}^{r} \cup \{N_b^{\mathrm{sing}}\}_{b=1}^{s}).$$

Thus every empty support or empty summand family has an unambiguous threshold.

Theorem 16.2 (UB-v21-DER-09: exact mixed regular–singular finite-layer theorem)**.** *For every exact regular–singular UB object X and every $n \geq N_X$,*

$$\boxed{\operatorname{rank}_{\mathcal{O}}(X/\omega_n X) = \sum_{b=1}^{s} D_b}$$

and

$$\boxed{\begin{aligned}\operatorname{length}_{\mathcal{O}}(X/\omega_n X)_{\text{tors}} = &\sum_{a=1}^{r}(\mu_a p^n + \lambda_a en + C_a) \\ &+ \sum_{b=1}^{s} L_b^{\text{sing}}(n).\end{aligned}}$$

Thus the cyclotomic-free and cyclotomic-singular branches form one exact finite verifier on the declared decomposed class. The formula is an equality for every layer beyond one explicit finite threshold, not merely an asymptotic statement.

Proof. Quotient by ω_n commutes with finite direct sums. Each regular summand is finite and has the exact VC10 length formula. Each singular cyclic summand has free rank D_b and reduced torsion length $L_b^{\text{sing}}(n)$ by UB-VC11-04. Rank and torsion length are additive on the displayed direct sum. Taking the maximum threshold makes all component formulas simultaneous. □

Corollary 16.3 (UB-v21-DER-10: arithmetic mixed-layer Return)**.** *Suppose an external, source-bound control theorem gives actual isomorphisms*

$$A_n \xrightarrow{\sim} X/\omega_n X \qquad (n \geq N_{\text{ctrl}})$$

for an arithmetic layer A_n and an exact regular–singular UB object X. Then for every

$$n \geq \max(N_X, N_{\text{ctrl}})$$

the arithmetic rank and reduced torsion length are exactly the two formulas of Theorem 16.2. The control theorem, exact decomposition, and every characteristic/Fitting identification remain separate assumptions.

Proof. Transport rank and torsion length through the displayed isomorphism and apply Theorem 16.2. □

Theorem 16.4 (UB-v21-DER-11: branch-complete decision theorem on the exact one-variable class)**.** *For a one-variable finite-layer input carrying either:*

1. *a cyclotomic-free VC10 square-presentation certificate, or*
2. *an exact VC11 cyclic elementary-divisor certificate, or*
3. *an exact finite direct sum of such inputs,*

the UB verifier terminates after finite replay with an exact eventual layer profile consisting of finite length, or free rank plus reduced torsion length. For a specific arbitrary nonsplit square presentation containing a cyclotomic factor, absence of a transverse Bockstein certificate sufficient for the requested target and absence of exact elementary-divisor/higher-Fitting data gives atomic status `undefined` *and target result* `undefined` *at the missing reconstruction/rigidity leaf. Separately, any proposed uniform determinant-only reconstruction rule on the broader nonsplit* *singular class is refuted by the VC11 counterexample; that proposed stronger route is* `reject` *as theorem evidence and its requested stronger Return is* `obstructed`*. Determinant data alone do not license a stronger Return.*

Proof. The first branch is decided by the finite cyclotomic scan, logarithmic-norm certificate, and VC10 formula. The second is decided by the finite cyclotomic support, exact sequence, and resultant formula of VC11. The third follows by Theorem 16.2. A specific run with missing required reconstruction data is undefined by the Chapter 8 atomic-status rule. The VC11 determinant counterexample is a separate mathematical obstruction to a uniform determinant-only theorem on the broader class, so that proposed theorem is rejected and the corresponding stronger target is obstructed rather than inferred. □

17 Promotion assessment, deferred programs, and rejected upgrades

Classification 17.1 (Immediate v21 UB-local additions)**.** *Theorems 13.3–16.4 are admissible as Appendix-local v21 additions because their proofs use only already declared UB/Main/DG/HT inputs, their target classes are explicit, and their failure boundaries are preserved. They require no new external source theorem and no benchmark-derived premise.*

Classification 17.2 (Successor-Core promotion candidates)**.** *The following may be evaluated for a later Main/Core release, but are not promoted by this appendix:*

1. *the finite Pareto strength-frontier and minimal causal blocker calculus as a generic proof-object theorem;*
2. *strict descent of finite-state derived inverse limits as a reusable Core-DG/tail interaction theorem;*
3. *field-extension invariance of finite-state derived limits;*
4. *the exact mixed regular–singular one-variable Iwasawa kernel, provided its arithmetic terminology is parameterized away or retained in the Problem Interface layer.*

Promotion requires a distinct identity, Main-level proof and consumer audit, migration entry, and final governance action.

Classification 17.3 (Deferred research programs)**.** *The strongest mathematically credible UB continuations are:*

1. *higher Bockstein and higher-Fitting reconstruction for nonsplit repeated cyclotomic factors;*
2. *model-certified derived descent of non-strict finite-state towers with explicit higher coherence;*
3. *multivariable or noncommutative finite-layer profiles with pseudo-null and higher-codimension error control;*
4. *proof-carrying component optimization that minimizes assumptions or artifacts without changing the strength frontier;*

5. *a formal certificate-complexity theorem for the combined descent–inverse-limit verifier.*

Each remains a program until its missing mathematical edge is proved.

Non-claim 17.4 (Rejected upgrades). The following are not consequences of the present corpus:

1. twelve successful cases imply completeness of the universal Bridge library;
2. determinant or characteristic-ideal equality determines an arbitrary singular finite layer;
3. pairwise local inverse-limit agreement produces higher-coherent descent;
4. arbitrary scalar extension commutes with an uncontrolled derived inverse limit;
5. a finite tower prefix supplies a tail theorem without a finite-state, Noetherian, Mittag–Leffler, or other certified reduction;
6. exact one-variable formulas imply multivariable or noncommutative Iwasawa control;
7. operational replay, formal wiring, or Claim registration proves any missing mathematical premise.

Theorem 17.5 (UB-v21-DER-12: derived-theorem-pack closure)**.** *The Appendix-local additions above strengthen UB without changing any inherited theorem, canonical Main identity, Appendix DG descent theorem, Appendix HT module theorem, or source-specific validation result. Their new mathematical content is exactly:*

1. *an associative typed algebra of certified implementations and substitution;*
2. *finite Pareto strength and minimal-failure frontier theorems;*
3. *an independent product-category Return construction with no hidden cross-target obligation;*
4. *strict finite descent with stalkwise stabilization and field-extension transport for finite-state derived inverse limits;*
5. *an exact unified regular–singular finite-layer theorem on the declared one-variable decomposed class.*

Every stronger proposed reading remains either a separately named promotion candidate, a deferred research program, or an explicit non-claim.

Proof. Theorems 13.3–14.5 are formal consequences of the certified implementation and finite proof-DAG contracts. Theorems 15.2–15.4 compose strict finite descent with the finite-state stable-image/Roos–Milnor calculation and faithful field extension. Theorems 16.2–16.4 add the exact VC10 and VC11 layer formulas on an explicitly decomposed class. The admission and nonpromotion protocols prevent mutation or authority collision. □

Classification 17.6 (Second global re-audit disposition)**.** *The second full UB audit retains all twelve derived theorem locators and records the following exact dispositions:*

1. *DER-01 is strengthened by an explicit proof that interface equivalence is an equivalence relation and a congruence, so quotient composition is well defined;*
2. *DER-03 includes the empty-supported-set case and makes replacement monotonicity conditional on retained compatibility;*
3. *DER-04 now states inclusion-minimality and the stronger containment property precisely;*
4. *DER-05 is a product-category/product-record theorem and does not silently assume a categorical product or discharge cross-target coherence;*
5. *DER-06–DER-07 use explicit stalkwise stabilization exponents in the finite diagram category;*
6. *DER-08 proves the canonical base-change comparison through natural Milnor/stable-image identifications, without assuming tensor commutes with arbitrary countable products;*
7. *DER-09 has an unambiguous threshold for empty cyclotomic support and empty summand families;*
8. *DER-11 separates missing-evidence `undefined` from counterexample-certified method-level `obstructed`/`reject`.*

No predecessor theorem, validation case, Main/DG/HT owner, or source-specific Return ceiling is changed.

Part III

Retained v20 UB Predecessor, Executable Component Corpus, and Twelve Validation Cases

Remark 17.7 (Current-reading overlay). The following retained corpus preserves the complete predecessor theorem/proof and validation body. Generic items are consumed under the owner partition and migration rules above. UB locator strings remain stable component, proof, regression, and provenance identifiers; they are not competing Main theorem UIDs. Source-specific cases retain their exact printed strengths and non-claims.

Part IV

Retained Predecessor Sovereignty, Universal Bridge Data, and AK–HT Placement

18 Scope and final construction contract

18.1 Purpose

Appendix UB standardizes bridge components of high reuse across algebra, topology, geometry, arithmetic, representation theory, homological algebra, and finite-dimensional models. It is not a list of desired future arrows. Every theorem used by the UB closure theorem is proved in this appendix or is displayed as an explicitly named classical input with exact hypotheses and a proof sufficient for the stated use.

The intended global architecture is

$$\mathcal{C}_{\mathrm{source}} \xrightarrow{\mathfrak{B}_{\mathrm{in}}} \mathcal{A}_{\mathrm{AK}} \xrightarrow{\text{AK-Core/HT}} \mathcal{A}_{\mathrm{out}} \xrightarrow{\mathfrak{B}_{\mathrm{ret}}} \mathcal{C}_{\mathrm{target}}.$$

UB concentrates on the two bridge arrows and on finite reductions needed to make them certifiable.

Axiom 18.1 (Frozen-baseline sovereignty)**.** *The frozen v20.0.0 source set is an immutable predecessor dependency. For current v21 runs, Main Chapters 1–10, Foundation Appendices A–G, Appendix DG, and Appendix HT govern through the synchronization overlay above. A UB statement may connect to them only through an explicitly declared interface. No UB theorem changes hard threshold deletion, window-first evaluation, detector completeness, higher-edge eligibility, Type IV semantics, finite-to-infinite discipline, HT module strength, or Return status precedence.*

Principle 18.2 (Completion inside one appendix)**.** *A component is admitted to the canonical UB library only when its source and target types, hypotheses, positive theorem, proof, failure boundary, composition rule, and Return ceiling are all stated in this appendix. A domain-specific arithmetic, analytic, geometric, or homotopical theorem may remain external, but the point at which it is required must be uniquely exposed by the UB component contract.*

Principle 18.3 (New-theorem boundary)**.** *UB distinguishes:*

1. *classical constituent theorems, re-proved or specialized here for bridge use;*
2. *new AK component theorems, which package a constituent theorem into a typed reusable bridge;*
3. *new AK composition theorems, whose conclusion follows from a nontrivial conjunction of independently certified components;*
4. *external-domain novelty, which is claimed only after a new domain-specific input or a new application theorem is separately proved.*

The phrase "UB new theorem" refers to items (2) or (3), not automatically to a new theorem in the external mathematical literature.

18.2 What UB must and must not do

UB must provide:

- conservative realization or an exact promotion ceiling;
- a finite detector whenever a finite conclusion is asserted;
- coefficient and completion control;
- finite local-to-global or finite-to-infinite reduction when required;
- naturality or route-coherence verification;
- a rigidity theorem for every stronger Return;
- a finite replay object suitable for the AK verifier.

UB does not assert:

- that every mathematical category admits a finite conservative realization;
- that field coefficients retain integral or completed information;
- that finitely many slices classify arbitrary multiparameter persistence;
- that local equality implies global object descent without cocycle data;
- that a long finite tower prefix determines an arbitrary inverse limit;
- that an isomorphism after a nonfaithful observation functor returns an external isomorphism;
- that the UB closure theorem solves a named open problem without a complete domain adapter.

19 Universal bridge component and strength architecture

Definition 19.1 (Universal bridge component)**.** A universal bridge component is a tuple

$$\mathfrak{B} = (\mathcal{C}_{\mathrm{src}}, \mathcal{C}_{\mathrm{tgt}}, R, E, P, Q, S, \Sigma, \mathcal{E}, \mathcal{A}),$$

where:

1. $\mathcal{C}_{\mathrm{src}}$ and $\mathcal{C}_{\mathrm{tgt}}$ are the source and target categories or declared object classes;
2. R is the realization or comparison construction;
3. E is the finite AK-readable extraction;
4. P is the preservation or reflection theorem;
5. Q is the globalization, completion, descent, or finite-to-infinite theorem when invoked;
6. S is the external Return or rigidity theorem;
7. Σ is the requested Return strength;
8. $\mathcal{E}$ is the defect and evidence bundle;
9. $\mathcal{A}$ is the immutable source, proof, and artifact bundle.

A field may be marked NOT-INVOKED only through a scope-exclusion record.

Definition 19.2 (Retained bridge strengths under the v21 target-relative reading)**.** For a fixed target record, one reconstruction spine may contain

$$\Sigma_0 \preceq \Sigma_{\mathrm{pred}} \preceq \Sigma_{\mathrm{inv}} \preceq \Sigma_{\mathrm{obj}} \preceq \Sigma_{\mathrm{mor}} \preceq \Sigma_{\mathrm{ex}} \preceq \Sigma_{\mathrm{der}} \preceq \Sigma_{\mathrm{str}}.$$

The symbols mean no theorem-bearing Return, predicate Return, one specified invariant, object Return up to the declared equivalence, morphism Return, exact/Yoneda Return, bounded-derived Return, and structured Return. This is not a universal linear hierarchy. Determinants, ideals, graded pieces, stable classes, pseudo-isomorphism classes, and replacement targets remain distinct and incomparable unless a registered theorem relates their target records.

Definition 19.3 (Retained UB-local defect refinement)**.** The predecessor implementation-local stack

$$\begin{aligned}\mathfrak{D}^{\mathrm{loc}}_{\mathrm{UB}} = (&D_{\mathrm{src}}, D_{\mathrm{real}}, D_{\mathrm{gen}}, D_{\mathrm{amp}}, D_{\mathrm{coeff}},\\ &D_{\mathrm{multi}}, D_{\mathrm{desc}}, D_{\mathrm{tail}}, D_{\mathrm{coh}}, D_{\mathrm{rig}},\\ &D_{\mathrm{int}}, D_{\mathrm{return}}, D_{\mathrm{cert}})\end{aligned}$$

is retained only as a refinement of the canonical Main Definition 9.45 stack. Generator, amplitude, and multiparameter leaves map to the earliest applicable owner among D_{real}, D_P, D_{DG}, and D_{HT}. Interface leaves map to the exact Core, HT, tail, descent, coherence, or rigidity owner. No new top-level defect sort or double counting is created.

Protocol 19.4 (Retained status precedence under v21 typing)**.** *First evaluate every node and edge with the four atomic statuses* `pass`*,* `reject`*,* `undefined`*, and* *justified `not_invoked`. Then, for one fixed target record, emit exactly one Bridge result: `reject` for a false required assertion; otherwise `undefined` for a missing required object/theorem/map; otherwise `obstructed` for a valid existing comparison with a nonzero declared obstruction; otherwise `partial_return` for a certified weaker target; `regularized_return` only for a separately identified replacement target; and `return` when the exact requested target closes. A readiness token is neither an atomic status nor a Return result.*

Proposition 19.5 (No compensation)**.** *No metric margin, numerical score, AI confidence value, finite sample count, successful build, or manifest completeness compensates for a false or missing component of $\mathfrak{D}_{\mathrm{UB}}$.*

Proof. Each defect is a predicate on a different mathematical type. A real number cannot manufacture a functor, generator, coefficient theorem, cocycle, inverse-system tail theorem, naturality cell, conservative observation, or external comparison map. The conclusion follows from typed non-compensation. □

20 Retained predecessor interface with AK v20 and Appendix HT

Definition 20.1 (UB-to-AK admissibility)**.** A UB output is AK-admissible when it supplies an object in the declared AK-readable category together with the complete coefficient and variance policy and constructibility or finite-type evidence. When persistence is invoked, it also supplies monitored degree, window, threshold, and endpoint data, together with every required detector, representative, higher-edge, comparison, reduction, and artifact record.

Definition 20.2 (UB-to-HT interface)**.** A UB-to-HT interface certificate identifies:

1. the exact HT entry module among $\mathrm{HC}, \mathrm{PE}, \mathrm{SS}, \mathrm{RL}, \mathrm{IW}$;
2. the source object and the target object in the local HT theorem;
3. the comparison mode and requested strength;
4. the UB theorem that makes the entry conservative or purpose-faithful;
5. every coherence cell needed when parallel UB and HT routes coexist.

Theorem 20.3 (UB-CORE-01: Conservative UB–HT compatibility)**.** *Let a UB component terminate in the exact source type of an HT local theorem. If the UB component is certified at strength Σ_1, the HT theorem is certified at strength Σ_2, the interface cell is valid, and the target strength is no stronger than the certified composite strength, then their composite is a valid HT proof-DAG edge. If the UB observation is not conservative at the requested external strength, the composite Returns at most the corresponding weaker observation strength.*

Proof. The positive statement is ordinary composition in the declared target category, together with the HT interface contract. The ceiling statement follows because two non-equivalent source objects identified by the UB observation remain indistinguishable to every downstream functor that consumes only that observation. A stronger Return would contradict nonfaithfulness. □

Part V

Finite-Generator and Conservative Realization Kernel

21 Cone reduction and zero-object detection

Lemma 21.1 (Cone criterion). *Let $\mathcal{T}$ be a triangulated category and $f : X \to Y$ a morphism. Then*

$$f \text{ is an isomorphism} \quad \Longleftrightarrow \quad \mathrm{Cone}(f) \simeq 0.$$

Proof. Complete f to a distinguished triangle $X \to Y \to \mathrm{Cone}(f) \to X[1]$. If f is an isomorphism, the triangle is isomorphic to the trivial triangle and the cone is zero. Conversely, if the cone is zero, the long exact Hom sequences show that composition with f induces isomorphisms on all represented Hom functors; Yoneda gives that f is an isomorphism. □

Definition 21.2 (Finite probe family). For a finite family $\mathcal{G} = \{G_1, \dots, G_r\}$ in a triangulated category and a finite set $I \subset \mathbb{Z}$, define

$$\Pi_{\mathcal{G},I}(X) = \left(\mathrm{Hom}_{\mathcal{T}}(G_j, X[n])\right)_{1 \le j \le r,\ n \in I}.$$

The family is zero-reflecting on a class $\mathcal{C} \subseteq \mathcal{T}$ when vanishing of the displayed finite tuple implies $X \simeq 0$ for every $X \in \mathcal{C}$.

Proposition 21.3 (Isomorphism detection by a zero-reflecting probe). *If $\Pi_{\mathcal{G},I}$ is zero-reflecting on every cone of a morphism in $\mathcal{C}$, then a morphism $f : X \to Y$ in $\mathcal{C}$ is an isomorphism exactly when the induced probe maps certify $\Pi_{\mathcal{G},I}(\mathrm{Cone}(f)) = 0$.*

Proof. Apply the cone criterion and zero-reflection. □

Warning 21.4 (Finite observation is not zero-reflection). The finiteness of $\mathcal{G}$ and I does not prove zero-reflection. The next sections provide positive hypotheses under which it holds.

22 Bounded derived categories with a projective generator

Definition 22.1 (Projective-generator regime). Let $\mathcal{A}$ be an abelian category with a projective generator G: G is projective and

$$\mathrm{Hom}_{\mathcal{A}}(G, M) = 0 \Longrightarrow M = 0.$$

For integers $a \le b$, let $\mathrm{D}^{\mathrm{b}[a,b]}(\mathcal{A})$ denote objects with cohomology concentrated in $[a, b]$.

Theorem 22.2 (UB-GEN-01: Finite-amplitude generator detection). *Let $X \in \mathrm{D}^{\mathrm{b}[a,b]}(\mathcal{A})$. Then*

$$X \simeq 0 \quad \Longleftrightarrow \quad \mathrm{Hom}_{\mathrm{D}^{\mathrm{b}}(\mathcal{A})}(G, X[n]) = 0 \quad \textit{for every } n \in [a, b].$$

Thus the finite probe $\Pi_{\{G\},[a,b]}$ is zero-reflecting on $\mathrm{D}^{\mathrm{b}[a,b]}(\mathcal{A})$.

Proof. Because G is projective, $\mathrm{Hom}_{\mathcal{A}}(G,-)$ is exact. Hence

$$\mathrm{Hom}_{\mathrm{D^b}(\mathcal{A})}(G, X[n]) \cong H^n\,\mathrm{RHom}(G,X) \cong \mathrm{Hom}_{\mathcal{A}}(G, H^n(X)).$$

If the left side vanishes for $n \in [a,b]$, the generator property gives $H^n(X) = 0$ for every possible nonzero cohomological degree. Therefore $X \simeq 0$. The converse is immediate. ☐

Corollary 22.3 (Finite morphism Return)**.** *Let $f : X \to Y$ with $X, Y \in \mathrm{D}^{\mathrm{b}[a,b]}(\mathcal{A})$. Then f is an isomorphism if and only if*

$$\mathrm{Hom}(G, H^n(f)) : \mathrm{Hom}(G, H^n(X)) \longrightarrow \mathrm{Hom}(G, H^n(Y))$$

is an isomorphism for every $n \in [a,b]$.

Proof. The generator reflects whether each cohomology map is an isomorphism. A morphism in a derived category is an isomorphism exactly when it is a quasi-isomorphism. ☐

Theorem 22.4 (UB-GEN-02: Finite naturality-cell extension)**.** *Let $\mathcal{T}$ and $\mathcal{U}$ be idempotent-complete triangulated categories, let $\mathcal{G} = \{G_1, \dots, G_r\}$ satisfy $\mathcal{T} = \mathrm{thick}(\mathcal{G})$, and let $F, H : \mathcal{T} \to \mathcal{U}$ be exact functors. For an exact natural transformation $\eta : F \Rightarrow H$, compatible with the chosen suspension structures, if every η_{G_j} is an isomorphism, then η_X is an isomorphism for every $X \in \mathcal{T}$.*

Proof. Let $\mathcal{S}$ be the full subcategory of objects X for which η_X is an isomorphism. Naturality and exactness show that $\mathcal{S}$ is closed under shifts and cones; additivity and idempotent completeness show closure under finite direct sums and direct summands. Thus $\mathcal{S}$ is thick. Since it contains every G_j, it equals $\mathrm{thick}(\mathcal{G}) = \mathcal{T}$. ☐

Corollary 22.5 (Finite verification of base-change, projection-formula, and comparison cells)**.** *Any natural transformation between exact functors on a finitely thick-generated triangulated category is globally invertible once it is invertible on the declared finite generator family. This applies to a Beck–Chevalley map, projection-formula map, completion comparison, or realization comparison only after the functors and natural transformation are actually constructed.*

Counterexample 22.6 (Generator-free finite checking)**.** Checking a natural transformation on finitely many arbitrary objects does not imply global invertibility when those objects do not thick-generate the source. Take a product category $\mathcal{T} = \mathcal{T}_1 \times \mathcal{T}_2$, check only generators of $\mathcal{T}_1$, and let the transformation fail on $\mathcal{T}_2$.

23 Finite categories, finite posets, and Morita realization

Definition 23.1 (Finite-category representation)**.** Let $\mathcal{P}$ be a finite small category and k a field. Write

$$\mathrm{Rep}_k(\mathcal{P}) = \mathrm{Fun}(\mathcal{P}, \mathrm{Vect}_k^{\mathrm{fd}}).$$

For $p \in \mathcal{P}$, let $P_p = k[\mathrm{Hom}_{\mathcal{P}}(p,-)]$ be the linearized representable functor and set

$$G_{\mathcal{P}} = \bigoplus_{p \in \mathrm{Ob}(\mathcal{P})} P_p, \qquad A_{\mathcal{P}} = \mathrm{End}(G_{\mathcal{P}})^{\mathrm{op}}.$$

Theorem 23.2 (UB-MOR-01: Finite-category projective-generator realization)**.** *The object $G_{\mathcal{P}}$ is a finite projective generator of $\mathrm{Rep}_k(\mathcal{P})$, and*

$$\Phi_{\mathcal{P}} = \mathrm{Hom}(G_{\mathcal{P}}, -) : \mathrm{Rep}_k(\mathcal{P}) \longrightarrow A_{\mathcal{P}}\text{-}\mathrm{Mod}^{\mathrm{fd}}$$

is an exact equivalence. Consequently, objects, morphisms, kernels, cokernels, short exact sequences, bounded derived objects, and cones are Returnable through the finite algebra $A_{\mathcal{P}}$.

Proof. Each representable P_p is projective by Yoneda because evaluation at p is exact. If $M \neq 0$, then $M(p) \neq 0$ for some object p, and Yoneda gives $\mathrm{Hom}(P_p, M) \cong M(p) \neq 0$; hence the finite sum is a generator. The classical Morita argument for a projective generator constructs a quasi-inverse

$$G_{\mathcal{P}} \otimes_{A_{\mathcal{P}}} -.$$

The unit and counit are isomorphisms on the generator and therefore on all finite presentations. Since the category is finite-dimensional and every object has a finite presentation by representables, the equivalence follows. Exactness follows from projectivity of the generator. Exact and derived structures are transported by an exact equivalence. □

Corollary 23.3 (Finite-poset multiparameter Return)**.** *For a finite poset P, every finite-dimensional P-persistence module is exactly Returnable as a finite module over the incidence algebra A_P. No slicing is required. The Return strength is module, morphism, exact, and bounded-derived strength, not merely a rank invariant.*

Remark 23.4 (Relation to one-parameter PE). When P is a finite chain, the incidence algebra is a type-A quiver algebra, and Appendix HT's PE library may provide stronger barcode and Ext-signature reconstruction. For a general finite poset, UB Returns the full finite algebra module; it does not assert an interval decomposition or a barcode.

Counterexample 23.5 (Finite slices do not classify arbitrary multiparameter modules)**.** A finite collection of one-parameter restrictions can miss extension data supported away from the selected slices. Therefore slice equality is not substituted for the finite-category Morita realization unless a separate slice-completeness theorem is proved for the declared class.

24 UB finite realization theorem

Theorem 24.1 (UB-FIN-01: UB finite realization and isomorphism closure)**.** *Let $\mathcal{C}$ be one of the following:*

1. $\mathrm{D}^{\mathrm{b}[a,b]}(\mathcal{A})$ *for an abelian category with a finite projective generator;*
2. $\mathrm{D}^{\mathrm{b}}(\mathrm{Rep}_k(\mathcal{P}))$ *for a finite category $\mathcal{P}$, restricted to a bounded amplitude.*

Then there exists a finite algebra A, a finite generator G, and a finite shift interval I such that zero objects and isomorphisms in $\mathcal{C}$ are reflected by the finite algebraic probe

$$X \longmapsto (\mathrm{Hom}(G, X[n]))_{n \in I},$$

together with the induced module and morphism structure. The construction is functorial and compatible with cones.

Proof. In case (1), use finite-amplitude generator detection. In case (2), first apply finite-category Morita realization, then use the regular module A as projective generator of the module category and apply the same finite-amplitude theorem. Functoriality and cone compatibility follow from exactness of the realization. □

Remark 24.2 (UB novelty). The constituent generator and Morita statements are classical. The theorem is a UB component theorem because it gives a uniform finite, cone-compatible, Return-typed realization accepted by the AK–HT interface.

Part VI

Coefficient, Base-Change, and Completion Components

25 Faithfully flat conservativity

Theorem 25.1 (UB-FF-01: Faithfully flat derived zero-reflection)**.** *Let $R \to S$ be faithfully flat and let $X \in \mathrm{D}(R)$. Then*

$$X \simeq 0 \quad \Longleftrightarrow \quad X \otimes_R^{\mathbf{L}} S \simeq 0.$$

For a morphism $f : X \to Y$, f is an isomorphism if and only if $f \otimes_R^{\mathbf{L}} S$ is an isomorphism.

Proof. Flatness gives

$$H^n(X \otimes_R^{\mathbf{L}} S) \cong H^n(X) \otimes_R S.$$

Faithfulness implies $M \otimes_R S = 0 \Rightarrow M = 0$. Hence all cohomology modules of X vanish exactly when those of the base change vanish. Apply the same result to $\mathrm{Cone}(f)$ for the morphism statement. □

Corollary 25.2 (Finite product cover)**.** *If $S = \prod_{i=1}^{m} S_i$ is faithfully flat over R, then a derived zero or isomorphism can be verified on the finite family of fibers $X \otimes_R^{\mathbf{L}} S_i$.*

26 Derived Nakayama and finite residue-field probes

Theorem 26.1 (UB-NAK-01: Perfect derived Nakayama)**.** *Let $(R, \mathfrak{m}, k)$ be a local ring and $P \in \mathrm{Perf}(R)$. Then*

$$P \simeq 0 \quad \Longleftrightarrow \quad P \otimes_R^{\mathbf{L}} k \simeq 0.$$

A morphism $f : P \to Q$ in $\mathrm{Perf}(R)$ is an isomorphism if and only if its residue-field base change is an isomorphism.

Proof. Represent P by a bounded complex of finite free modules. By repeatedly splitting off a two-term direct summand on which a differential matrix contains a unit entry, obtain a homotopy-equivalent minimal finite free complex $P_{\min}$ whose differential matrices have entries in $\mathfrak{m}$. After tensoring with k, all differentials vanish. If the special fiber is acyclic, every graded vector space $P^n_{\min} \otimes_R k$ is zero. Nakayama gives $P^n_{\min} = 0$ for all n, so $P \simeq 0$. The converse is immediate. Apply the zero statement to $\mathrm{Cone}(f)$ for the morphism criterion. □

Theorem 26.2 (UB-NAK-02: Finite semilocal fiber atlas)**.** *Let R be semilocal with maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_s$, residue fields k_i, and let $P \in \mathrm{Perf}(R)$. Then*

$$P \simeq 0 \quad \Longleftrightarrow \quad P \otimes^{\mathbf{L}}_R k_i \simeq 0 \quad (1 \le i \le s).$$

The same finite family detects isomorphisms between perfect complexes.

Proof. If $P \not\simeq 0$, some finitely generated cohomology module $H^n(P)$ is nonzero. Its support contains a maximal ideal $\mathfrak{m}_i$. Localizing at $\mathfrak{m}_i$ gives a nonzero perfect complex over the local ring $R_{\mathfrak{m}_i}$. Perfect derived Nakayama implies the residue fiber at k_i is nonzero. The converse is immediate. Apply the result to cones for morphisms. □

Counterexample 26.3 (One residue field over a nonlocal ring)**.** A single residue field need not detect a perfect complex over a ring with several connected or maximal components. For $R = k \times k$, the perfect module $0 \times k$ has zero fiber at the first factor but is nonzero.

27 Ideal-adic completion component

Definition 27.1 (Completion-admissible perfect datum)**.** A completion-admissible datum consists of a Noetherian ring R, an ideal $I \subseteq \mathrm{Jac}(R)$, the completion $\widehat{R} = \varprojlim R/I^n$, and a perfect complex $P \in \mathrm{Perf}(R)$. The datum records whether Return is requested over R, over $\widehat{R}$, or only modulo I.

Theorem 27.2 (UB-COMP-01: Perfect completion zero-reflection)**.** *For a completion-admissible datum,*

$$P \simeq 0 \quad \Longleftrightarrow \quad P \otimes^{\mathbf{L}}_R R/I \simeq 0.$$

Moreover, if $R \to \widehat{R}$ is faithfully flat, then

$$P \simeq 0 \Longleftrightarrow P \otimes^{\mathbf{L}}_R \widehat{R} \simeq 0 \Longleftrightarrow P \otimes^{\mathbf{L}}_R R/I \simeq 0.$$

The analogous statement holds for isomorphisms of perfect complexes.

Proof. A perfect complex is represented by a bounded finite projective complex. Localize at each maximal ideal containing I. Since $I \subseteq \mathrm{Jac}(R)$, every maximal ideal contains I. If the reduction modulo I vanishes, then its further reduction to every residue field vanishes. The semilocal argument may be applied locally at each maximal ideal, giving that every localization of P is zero; hence $P = 0$. Faithfully flat zero-reflection gives the completion equivalence. Apply to cones for morphisms. □

Warning 27.3 (Derived completion outside the perfect regime). For arbitrary complexes, classical completion can vanish while derived completion contains a shifted $\varprojlim^1$- or Tor-origin class. UB therefore sends nonperfect completion problems to Appendix HT's RL derived-limit component rather than applying the perfect theorem.

28 Universal-coefficient boundary and Bockstein control

Theorem 28.1 (Two-row universal-coefficient boundary)**.** *Let R be a principal ideal domain, let C be a bounded complex of projective R-modules, and let $0 \neq t \in R$ and $S = R/(t)$. For every n there is a natural short exact sequence*

$$0 \longrightarrow H^n(C) \otimes_R S \longrightarrow H^n(C \otimes_R S) \longrightarrow \mathrm{Tor}_1^R(H^{n+1}(C), S) \longrightarrow 0.$$

The sequence is not canonically split in general.

Proof. Use the derived tensor spectral sequence

$$E_2^{-p,q} = \mathrm{Tor}_p^R(H^q(C), S) \Longrightarrow H^{q-p}(C \otimes_R^{\mathbf{L}} S).$$

Since S has projective dimension one, only rows $p = 0, 1$ survive. The two filtration pieces in total degree n give the short exact sequence. □

Definition 28.2 (Coefficient boundary defect)**.** The coefficient boundary defect in degree n is

$$D^n_{\mathrm{Tor}}(C; S) = \mathrm{Tor}_1^R(H^{n+1}(C), S).$$

A naive cohomology base-change Return is licensed only when this defect vanishes in every consumed degree.

Proposition 28.3 (Bockstein interpretation)**.** *For the short exact coefficient sequence $0 \to R \xrightarrow{t} R \to S \to 0$, the connecting morphism in cohomology detects the same failure of naive specialization represented by the Tor boundary. A nonzero Bockstein is therefore a coefficient defect, not a failure of the original source theorem.*

29 UB coefficient-completion composition theorem

Theorem 29.1 (UB-COEFF-01: Finite coefficient-completion Return)**.** *Let R be a Noetherian semilocal ring with maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_s$, let $I \subseteq \mathrm{Jac}(R)$, and let $f : P \to Q$ be a morphism of perfect complexes. The following are equivalent:*

1. *f is an isomorphism over R;*
2. *every residue-fiber map $f \otimes_R^{\mathbf{L}} k_i$ is an isomorphism;*
3. *the completion map $f \otimes_R^{\mathbf{L}} \widehat{R}$ is an isomorphism, provided $R \to \widehat{R}$ is faithfully flat;*
4. *the reduction $f \otimes_R^{\mathbf{L}} R/I$ is an isomorphism.*

Thus a finite coefficient atlas, completion Return, and source isomorphism Return coincide in this regime.

Proof. Apply the finite semilocal fiber atlas and perfect completion zero-reflection to the cone of f. □

Remark 29.2 (UB novelty). This is a new UB composite theorem. It combines classical conservativity principles into one finite certificate that can be passed directly to HT PE-10, RL completion, or IW Return.

Part VII

Noetherian, Finite-State, and Finite-to-Infinite Globalization

30 Finitely generated graded obstruction modules

Definition 30.1 (Graded obstruction package)**.** Let $A = \bigoplus_{n \geq 0} A_n$ be a nonnegatively graded ring and $M = \bigoplus_{n \geq 0} M_n$ a graded A-module. A degree bound d is a certified generation bound when M is generated as an A-module by $\bigoplus_{0 \leq n \leq d} M_n$.

Theorem 30.2 (UB-NOETH-01: Noetherian finite-window zero theorem)**.** *If M has certified generation bound d, then*

$$M = 0 \quad \Longleftrightarrow \quad M_n = 0 \quad (0 \leq n \leq d).$$

Proof. Every homogeneous element of M is an A-linear combination of homogeneous generators lying in degrees at most d. If all possible generator degrees vanish, every element vanishes. □

Theorem 30.3 (UB-NOETH-02: Finite-window morphism theorem)**.** *Let $f : M \to N$ be a graded morphism. Suppose* ker f *and* coker f *have certified generation bound d. Then*

$$f \text{ is an isomorphism} \quad \Longleftrightarrow \quad f_n : M_n \to N_n \text{ is an isomorphism for } 0 \leq n \leq d.$$

Proof. The finite degree condition says that the kernel and cokernel vanish in all degrees in which their generators may occur. The previous theorem gives global vanishing. □

Warning 30.4 (Unknown generation bound)*.* Finite generation without a certified degree bound does not license a predetermined finite window. The bound is a non-scalar theorem input.

31 Finite-state tail certification

Definition 31.1 (Deterministic tail system)**.** A deterministic tail system is a finite set S, a transition map $T : S \to S$, an initial state s_0, and a terminal subset $Z \subseteq S$ satisfying $T(Z) \subseteq Z$. The orbit is $s_n = T^n(s_0)$.

Theorem 31.2 (UB-STATE-01: Finite-state terminality theorem)**.** *Let $|S| = N$. Exactly one of the following occurs within the prefix $s_0, \dots, s_N$:*

1. *some $s_n \in Z$, in which case every later state lies in Z;*
2. *two states $s_i = s_j \notin Z$ occur with $0 \leq i < j \leq N$, in which case the orbit is eventually periodic outside Z and never reaches Z.*

Hence eventual terminality is decidable by a finite prefix of length at most $N + 1$.

Proof. If no state in the prefix lies in Z, the pigeonhole principle gives a repeated state. Determinism implies the subsequent orbit repeats with period $j - i$. Since the repeated cycle contains no terminal state and Z is forward invariant, the orbit never reaches Z. If a terminal state occurs, forward invariance gives permanent terminality. □

Corollary 31.3 (Finite-state obstruction vanishing). *If each state carries an obstruction object $O(s)$, $O(s) = 0$ for $s \in Z$, and zero is reflected by the state encoding, then the infinite obstruction tail vanishes exactly when the finite terminality certificate passes.*

Counterexample 31.4 (Arbitrarily long finite prefixes). Without a finite state space or another quantified recurrence theorem, two towers may agree for an arbitrarily long finite prefix and diverge later. Therefore prefix length alone is not a finite-to-infinite theorem.

32 Finite length and eventual stabilization

Theorem 32.1 (Finite-length chain stabilization). *Let M be a module of finite composition length $\ell(M)$. Every strictly increasing or strictly decreasing chain of submodules has at most $\ell(M)$ strict inclusions. Consequently every monotone chain stabilizes.*

Proof. A strict inclusion changes composition length by at least one. The length is bounded between zero and $\ell(M)$. □

Theorem 32.2 (UB-RL-01: Eventually constant inverse-system Return). *Let*

$$A_0 \leftarrow A_1 \leftarrow A_2 \leftarrow \cdots$$

be an inverse system in an abelian category. Suppose there exists N such that every transition $A_{n+1} \to A_n$ is an isomorphism for $n \geq N$. Then

$$\varprojlim A_n \cong A_N, \qquad \varprojlim{}^1 A_n = 0.$$

Any declared apex A_∞ equipped with an isomorphism $A_N \cong A_\infty$ is exactly Returned from the tower.

Proof. Compatible sequences are uniquely determined by their stabilized value in A_N, giving the inverse-limit isomorphism. The inverse-limit functor is exact on eventually constant systems, so its first derived functor vanishes. Compose with the declared apex isomorphism. □

Theorem 32.3 (UB-RL-02: Finite-state derived-limit closure). *Suppose an inverse system is generated by a deterministic finite-state rule, and terminal states are exactly those after which every transition is an isomorphism. Then the first $|S| + 1$ states provide a complete certificate for one of two conclusions:*

1. *the tower is eventually constant, $\varprojlim^1 = 0$, and the stable apex is Returnable;*
2. *the state orbit enters a nonterminal cycle, so the eventual-constancy Return route is rejected.*

Proof. Apply the finite-state terminality theorem. In the terminal case apply eventual-constant inverse-system Return. In the cyclic case the declared terminal criterion never holds, so that particular Return route is false rather than merely missing. □

33 Noetherian globalization component

Definition 33.1 (Finite-generation globalization datum)**.** A finite-generation globalization datum consists of an obstruction module M, a certified finite generation degree d, a finite AK-readable realization of $M_0, \dots, M_d$, and a Return theorem identifying $M = 0$ with the external target predicate.

Theorem 33.2 (UB Noetherian globalization)**.** *For a finite-generation globalization datum, a complete finite zero certificate for the degrees $0, \dots, d$ Returns the global external vanishing predicate. If the finite certificate is incomplete, the global status is UNDEFINED; if a certified finite degree is nonzero, the global vanishing claim is REJECT.*

Proof. The mathematical implication is the Noetherian finite-window zero theorem. The status clauses follow from the UB precedence rule and detector completeness separation. □

Part VIII

Descent and Local-to-Global Components

34 Finite faithfully flat module descent

Definition 34.1 (Descent datum)**.** Let $R \to S$ be faithfully flat. A descent datum on an S-module N is an isomorphism

$$\theta : S \otimes_R N \xrightarrow{\sim} N \otimes_R S$$

of $S \otimes_R S$-modules satisfying the usual cocycle condition after tensoring to $S \otimes_R S \otimes_R S$.

Theorem 34.2 (UB-DESC-01: Effective faithfully flat descent for modules)**.** *The functor*

$$M \longmapsto (S \otimes_R M, \theta_{\text{can}})$$

from R-modules to S-modules with descent datum is an equivalence. A quasi-inverse sends (N, θ) to the equalizer

$$M = \mathrm{Eq}\Big(N \rightrightarrows S \otimes_R N\Big),$$

where the two arrows are induced by the unit and the descent datum.

Proof. The Amitsur sequence

$$0 \longrightarrow M \longrightarrow S \otimes_R M \rightrightarrows S \otimes_R S \otimes_R M$$

is exact because faithful flatness can be checked after tensoring with S, where it becomes split exact by insertion of units. This proves that the equalizer recovers every extended module. Conversely, the cocycle condition makes the canonical map $S \otimes_R M \to N$ compatible with the two descent arrows; after faithfully flat base change it becomes an isomorphism, hence it is an isomorphism. The constructions are functorial and mutually inverse. □

Corollary 34.3 (Finite product descent). *If $S = \prod_{i=1}^{m} S_i$, a global module is Returnable from finitely many local modules, pairwise overlap isomorphisms, and the finite cocycle condition.*

Warning 34.4 (Pairwise agreement is not enough). Local modules and pairwise isomorphisms without the cocycle condition need not descend. Missing triple-overlap coherence gives UNDEFINED, not an obstruction number and not a global object.

35 Perfect-complex descent

Theorem 35.1 (UB-DESC-02: Finite faithfully flat descent for perfect complexes). *Let $R \to S$ be faithfully flat. A bounded complex of finite projective S-modules equipped degreewise with compatible descent data, with differential commuting with those data, descends to a bounded complex of finite projective R-modules. The descended complex is unique up to unique isomorphism compatible with the descent datum.*

Proof. Apply module descent degreewise. Finite projectivity descends under faithfully flat maps. Compatibility of the differential with descent data implies that the degreewise descended maps form a complex. The relation $d^2 = 0$ can be checked after faithfully flat base change and therefore holds over R. Uniqueness follows from full faithfulness of module descent in each degree. □

Corollary 35.2 (Local cone Return). *A morphism of perfect complexes over R is an isomorphism if and only if its faithfully flat localizations are isomorphisms and the local morphisms form a valid descent datum.*

36 Finite acyclic-cover hypercohomology

Definition 36.1 (UB-acyclic finite cover). Let X be a topological space, $\mathcal{U} = \{U_1, \ldots, U_m\}$ a finite open cover, and $\mathcal{F}^\bullet$ a bounded-below complex of sheaves. The cover is UB-acyclic for $\mathcal{F}^\bullet$ when, for every nonempty finite intersection $U_{i_0\cdots i_p}$ and every degree q,

$$H^r(U_{i_0\cdots i_p}, \mathcal{F}^q) = 0 \qquad (r > 0).$$

Theorem 36.2 (UB-DESC-03: Finite Čech-to-derived Return). *For a UB-acyclic finite cover, the total Čech complex computes derived global sections:*

$$\operatorname{Tot} \check{C}^\bullet(\mathcal{U}, \mathcal{F}^\bullet) \simeq R\Gamma(X, \mathcal{F}^\bullet).$$

Consequently, global hypercohomology and every morphism induced on it are Returnable from finitely many intersections and finite Čech differentials.

Proof. Resolve $\mathcal{F}^\bullet$ by an injective double complex and form the Čech–resolution triple complex. Filtering first by sheaf-cohomological degree gives a spectral sequence whose higher rows vanish on every finite intersection by UB-acyclicity. The spectral sequence collapses to the Čech total complex. Filtering in the other direction computes derived global sections because the augmented Čech complex of an injective sheaf is exact. The two totalizations are therefore quasi-isomorphic. □

Corollary 36.3 (Finite local-to-global zero certificate)**.** *Under the same hypotheses, if the finite total Čech complex is certified acyclic, then $R\Gamma(X, \mathcal{F}^\bullet) \simeq 0$. If the complex is bounded and its terms are finite-dimensional, the finite-generator kernel of UB gives a finite proof certificate.*

Remark 36.4 (SS interface). The Čech double complex may instead be passed to Appendix HT's SS module when pagewise information, nonzero differentials, filtration, or extension data are part of the requested Return. UB's theorem supplies the convergence and abutment identification under the displayed acyclicity hypothesis.

37 Finite descent closure theorem

Theorem 37.1 (UB-DESC-04: UB finite descent closure)**.** *Suppose an external object is represented by a finite faithfully flat cover or a finite UB-acyclic open cover. Assume:*

1. *every local object is AK-readable;*
2. *all overlap maps and cocycle cells are supplied;*
3. *local comparison maps are certified at the requested local strength;*
4. *the applicable module or perfect-complex descent theorem, or Čech-to-derived theorem, is invoked;*
5. *the global Return strength does not exceed the descent theorem's strength.*

Then the local finite certificate Returns a unique global object or global derived object at that strength. A failed cocycle is REJECT*; a missing cocycle is* UNDEFINED*; a valid nonzero comparison cone is* OBSTRUCTED*.*

Proof. Effectivity gives existence and uniqueness of the global object in the algebraic descent cases. The Čech theorem gives the global derived object in the sheaf case. The local comparison maps descend because their compatibility is part of the datum. The status clauses follow from the typed meaning of false, missing, and nonzero terminal comparison data. □

Part IX

Monoidal, Duality, Coherence, and Rigidity Components

38 Kunneth components

Theorem 38.1 (UB-KUN-01: Field Kunneth Return)**.** *Let k be a field and let C, D be bounded complexes of finite-dimensional k-vector spaces. Then*

$$H^n(C \otimes_k D) \cong \bigoplus_{p+q=n} H^p(C) \otimes_k H^q(D)$$

naturally in C and D.

Proof. Every short exact sequence of vector spaces splits. Replace each complex by the direct sum of its cohomology with zero differential and contractible two-term summands. Tensoring preserves this decomposition; contractible summands remain contractible, and the cohomology summands give the displayed formula. □

Theorem 38.2 (PID Kunneth boundary)**.** *Let R be a principal ideal domain and let C, D be bounded complexes of projective modules with finitely generated cohomology. There is a natural short exact sequence*

$$0 \to \bigoplus_{p+q=n} H^p(C) \otimes_R H^q(D) \to H^n(C \otimes_R D) \to \bigoplus_{p+q=n+1} \mathrm{Tor}_1^R(H^p(C), H^q(D)) \to 0.$$

A tensor-product Return that omits the Tor term is valid only when the displayed Tor groups vanish.

Proof. Apply the standard tensor-product spectral sequence. Over a PID, the relevant modules have projective resolutions of length at most one, leaving two rows and hence the short exact sequence. □

39 Perfect duality and invertible tensor transport

Theorem 39.1 (UB-DUAL-01: Perfect duality zero-reflection)**.** *For $P \in \mathrm{Perf}(R)$, set $P^\vee = \mathrm{RHom}_R(P, R)$. The bidual map*

$$P \xrightarrow{\sim} \mathrm{RHom}_R(P^\vee, R)$$

is an isomorphism. Consequently,

$$P \simeq 0 \iff P^\vee \simeq 0,$$

and a morphism of perfect complexes is an isomorphism exactly when its dual is an isomorphism.

Proof. The statement is immediate for a finite projective module and hence for bounded finite projective complexes, term by term, with the usual sign convention. The morphism result follows by dualizing its cone and using biduality. □

Theorem 39.2 (Invertible tensor conservativity)**.** *Let* $(\mathcal{T}, \otimes, \mathbf{1})$ *be a monoidal category and let* L *be invertible with inverse* L^{-1}*. Then*

$$L \otimes - : \mathcal{T} \longrightarrow \mathcal{T}$$

is an equivalence. It reflects zero objects, isomorphisms, exact triangles when the tensor structure is exact, and every Return strength expressed in the underlying category.

Proof. The functor $L^{-1} \otimes -$ is a quasi-inverse using the evaluation isomorphisms $L^{-1} \otimes L \cong \mathbf{1}$ and $L \otimes L^{-1} \cong \mathbf{1}$. □

Warning 39.3 (Dualizable is not invertible)*.* A dualizable object need not make tensoring conservative. For example, tensoring with the zero object is not conservative, and finite projective modules of rank greater than one are dualizable but not generally invertible. UB requires invertibility or an independent conservativity theorem.

40 Finite-generator coherence and rigidity

Definition 40.1 (Rigidity class)**.** Let $O : \mathcal{T} \to \mathcal{A}$ be an observation functor. A class $\mathcal{C} \subseteq \mathcal{T}$ is O-rigid when

$$O(f) \text{ is an isomorphism} \Longrightarrow f \text{ is an isomorphism}$$

for every morphism in $\mathcal{C}$.

Theorem 40.2 (UB-RIG-01: Generator criterion for rigidity)**.** *Let*

$$\mathcal{T} = \operatorname{thick}(G_1, \dots, G_r).$$

Let $\mathcal{A}$ *be triangulated, and let*

$$O : \mathcal{T} \longrightarrow \mathcal{A}$$

be exact. Assume that, for every i, j *and every degree* n *in which the source Hom-group may be nonzero, the comparison*

$$\operatorname{Hom}_{\mathcal{T}}(G_i, G_j[n]) \longrightarrow \operatorname{Hom}_{\mathcal{A}}(OG_i, OG_j[n])$$

is an isomorphism. Assume also that $O(G_1), \dots, O(G_r)$ *generate the essential image. Then* O *is fully faithful on* $\mathcal{T}$ *and conservative there.*

Proof. For a fixed generator G_i, let $\mathcal{S}_i$ be the class of objects X for which the Hom comparison from G_i to every shift of X is an isomorphism. Exactness shows $\mathcal{S}_i$ is thick and the hypotheses place every generator in it; hence $\mathcal{S}_i = \mathcal{T}$. Fixing the target and repeating the argument in the first variable proves the Hom comparison for every pair of objects. Thus O is fully faithful. If $O(X) = 0$, full faithfulness gives $\mathrm{Id}_X = 0$, so $X = 0$; applying this to cones proves conservativity. □

Remark 40.3 (Finite check)*.* The Hom-degree check is finite when the generator algebra has certified finite cohomological amplitude. Without such an amplitude bound, the generator family is finite but the shift verification may remain infinite.

Theorem 40.4 (UB-RIG-02: Rigidity promotion theorem)**.** *Suppose a UB–HT route Returns an object or morphism after applying an observation functor O. If the source and target lie in a certified O-rigid class, then the observed Return promotes to the corresponding external object or morphism Return. If rigidity is absent, the observed Return remains at the weaker observation strength.*

Proof. Conservativity reflects isomorphisms, and full faithfulness Returns unique morphisms compatible with the observed morphisms. The ceiling statement follows from the existence of non-equivalent objects with identical observations when rigidity fails. □

41 Universal coherence-cell theorem

Definition 41.1 (Generator-determined transformation regime)**.** A family of enhanced exact functors out of a thick-generated category is generator-determined when restriction to the declared finite generator family is faithful on the natural transformations consumed by the bridge. Concretely, this holds for derived tensor functors on $\mathrm{Perf}(A)$ when transformations are represented by actual bimodule maps: evaluation on the regular generator A recovers the representing map.

Theorem 41.2 (UB-COH-01: Finite coherence-cell certification)**.** *Let a proposed bridge contain finitely many exact functors between idempotent-complete triangulated categories, each source thick-generated by a declared finite family. Suppose every required naturality, base-change, projection-formula, associator, or route-comparison cell is an isomorphism on the corresponding source generators. Then every such cell is an isomorphism on the full generated subcategory. If, in addition, the consumed transformations form a generator-determined transformation regime, every finite diagram built from them commutes globally whenever its two composite transformations agree on the generators.*

Proof. Invertibility follows from finite naturality-cell extension. For commutativity, the two routes define natural transformations with the same source and target. In a generator-determined regime, equality after restriction to the generator family implies equality of the transformations themselves. Therefore the diagram commutes on every object. The additional generator-determination hypothesis is essential: arbitrary triangulated categories may admit phantom phenomena that are invisible on a chosen generator-level object test. □

Remark 41.3 (GPT completion value)*.* The theorem converts a potentially unbounded coherence review into finite generator-level verification in enhanced regimes where transformations are actually determined by the generator data. It does not infer global equality of arbitrary triangulated natural transformations from objectwise generator checks alone.

Part X

Universal Composition with Appendix HT

42 Standard UB component library

Component	Positive regime	Returned strength	Principal ceiling
UB-GEN	bounded derived category with finite projective generator	zero, isomorphism, bounded derived	no amplitude/generator, no finite proof
UB-MOR	finite category or finite poset	module; morphism; exact; derived	no barcode for a general poset
UB-FF	faithfully flat base change	derived zero and isomorphism	not a reconstruction of integral structure
UB-NAK	perfect complex over local or semilocal ring	finite-fiber zero and isomorphism	nonperfect derived completion excluded
UB-COMP	perfect completion datum, ideal in Jacobson radical	source–completion–special-fiber equivalence	derived-limit defect outside perfect regime
UB-NOETH	certified graded generation degree	all-degree vanishing and isomorphism	unknown bound gives undefined
UB-STATE	deterministic finite-state transition	exact tail decision	arbitrary tower prefixes excluded
UB-DESC	finite faithfully flat or acyclic-cover descent	global object or derived sections	missing cocycle blocks Return
UB-KUN	field or PID tensor regime	tensor cohomology with displayed Tor correction	ring-level multiplication not automatic
UB-DUAL	perfect duality or invertible tensor	zero, isomorphism, dual Return	dualizable alone not conservative
UB-COH	finite thick generator and exact functors	global coherence cells	no generator, no finite extension
UB-RIG	fully faithful and conservative observation on a rigid class	promotion to stronger external Return	nonfaithfulness imposes ceiling

43 Interfaces to HC, PE, SS, RL, and IW/Return

Proposition 43.1 (UB to HC)**.** *A finite complex produced by UB-GEN, UB-MOR, UB-DESC, or UB-COMP may enter HT-HC when the filtration, basis, pivot, zero-shift, protected-degree, and replay hypotheses of HC are independently supplied. UB finite generation does not authorize a filtered cancellation.*

Proposition 43.2 (UB to PE)**.** *A finite chain or finite-poset object may enter PE through its exact finite algebra realization. PE's barcode, Ext-signature, morphism,* *Yoneda, derived, A_∞, deformation, descent, and boundary strengths remain governed by the corresponding PE theorem. UB-MOR alone Returns a module over a finite algebra, not a PE-2 interval barcode unless the algebra is in the declared type-A regime.*

Proposition 43.3 (UB to SS)**.** *UB-DESC supplies finite Čech and double complexes together with an abutment identification under acyclicity. UB-KUN and UB-COH may supply multiplicative or naturality inputs. Unknown differentials, convergence, filtration, and extension problems remain SS defects.*

Proposition 43.4 (UB to RL)**.** *UB-STATE and UB-NOETH may provide finite-to-infinite reduction. UB-COMP supplies perfect completion zero-reflection. Outside those regimes, RL must compute or control $\varprojlim^1$, derived completion, and apex mismatch directly.*

Proposition 43.5 (UB to IW/Return)**.** *UB-RIG supplies the promotion theorem required by IW's "no promotion without rigidity" boundary. UB-COMP, UB-DUAL, and UB-DESC may supply native target comparisons. Characteristic, determinant, pseudo-isomorphism, or observation-level Return does not become exact object Return unless the rigidity hypotheses are explicitly met.*

44 Composable UB–HT proof DAG

Definition 44.1 (Composable UB–HT proof DAG)**.** A composable UB–HT proof DAG is a finite directed acyclic graph whose vertices are certified UB components, AK Core clauses, or HT modules. It contains:

1. a local theorem certificate for every vertex;
2. a type-correct interface for every edge;
3. a finite generator-level or explicit coherence certificate for every parallel route;
4. every required coefficient, descent, tail, and rigidity record;
5. a requested Return strength and native target comparison;
6. one complete dependency and replay object.

Theorem 44.2 (UB-HT-01: UB–HT component composition)**.** *Let $\mathcal{B}$ be a composable UB–HT proof DAG from an external source X to an external target Y at strength Σ. If every local theorem hypothesis, interface, coherence cell, finite detector, coefficient condition, descent condition, tail theorem, rigidity condition, and Return obligation is discharged, then*

$$\mathrm{Return}_\Sigma^{\mathrm{UBHT}}(X, Y)$$

is certified with primary status RETURN*. If a stronger rigidity condition is missing but a weaker observation Return is certified, the primary status at Σ is* PARTIAL-RETURN*.*

Proof. Compose the UB realization and globalization arrows with the AK/HT proof DAG. Coherence is global by the finite coherence-cell theorem. The HT composition theorem Returns the internal target at its certified strength. UB-RIG promotes the internal or observed Return to the external target exactly when the rigidity class is certified. Status precedence handles false, missing, nonzero-defect, and weaker-strength cases. □

45 Universal Finite Bridge Closure Theorem

Definition 45.1 (UB-finitely closable route)**.** A route is UB-finitely closable when it satisfies all applicable clauses:

1. **Finite realization:** the source lies in a bounded generated derived class or a finite-category representation class covered by UB-GEN or UB-MOR;
2. **Finite coefficient atlas:** coefficient passage is covered by UB-FF, UB-NAK, UB-COMP, or a finite semilocal residue atlas;
3. **Finite globalization:** all-degree or infinite-stage claims are covered by UB-NOETH, UB-STATE, eventual constancy, or an independently certified finite-to-infinite theorem;
4. **Finite descent:** every local-to-global step is covered by UB-DESC or an independently certified effective descent theorem;
5. **Finite coherence:** all naturality and route cells are covered by UB-COH;
6. **Rigid Return:** the target strength is covered by UB-RIG or a stronger domain-specific rigidity theorem;
7. **AK–HT admissibility:** every invoked Core and HT local theorem is consumed at exact local strength.

Theorem 45.2 (UB-CLOSE-01: Universal Finite Bridge Closure)**.** *Every UB-finitely closable route admits a finite proof certificate whose successful verification Returns the declared external predicate, object, morphism, exact sequence, derived object, or structured target at the requested certified strength. The certificate consists of finitely many:*

- *generator and shift probes;*
- *coefficient fibers or faithfully flat localizations;*
- *generation-bound or finite-state tail checks;*
- *overlap maps and cocycle cells;*
- *naturality checks on finite generators;*
- *native comparison and rigidity checks;*
- *AK/HT local certificates and interface records.*

No unrecorded infinite quantifier remains in the selected proof route.

Proof. Finite realization reduces zero and isomorphism questions to finitely many generator and amplitude probes. The coefficient component reflects the relevant predicates through finitely many fibers or one faithfully flat comparison. Noetherian or finite-state globalization replaces every invoked all-degree or tail quantifier by a certified finite window. Effective finite descent reconstructs the global object from finitely many local objects and coherence cells. Finite-generator coherence extends the checked comparison cells to the full generated classes. The AK/HT composition theorem transports the finite model through the selected internal modules. Finally, rigidity promotes the observed or internal target to the declared external strength. Every set appearing in the resulting certificate is finite by hypothesis; composing the component implications proves the Return. □

Corollary 45.3 (Finite contradiction localization)**.** *If the Universal Finite Bridge Closure certificate fails, the first failure lies in exactly one typed component of* $\mathfrak{D}_{\mathrm{UB}}$ *or the inherited AK–HT defect stack. Therefore the unresolved mathematical core is exposed as a finite list of missing or false bridge obligations rather than an undifferentiated failure of the whole problem.*

Remark 45.4 (Principal new theorem). In the frozen v20 predecessor, the Universal Finite Bridge Closure Theorem was the principal integrated UB closure statement. For v21, canonical generic closure authority is Main Theorem 9.47; UB-CLOSE-01 is retained as a predecessor proof/implementation and regression locator at no stronger strength. Its mathematical content is not the claim that every problem satisfies the hypotheses; its force is that, once the hypotheses are met, GPT-assisted construction and verification can terminate without a hidden local, coefficient, infinite, coherence, or Return gap.

Corollary 45.5 (UB-CLOSE-02: Certificate complexity bound)**.** *Fix one UB-finitely closable route. Let* q *be the number of local or overlap objects,* r *the largest generator-family size,* L *the largest number of monitored shifts,* s *the largest finite coefficient-atlas size,* N *the largest finite-state count,* c *the number of cocycle and coherence cells, and* h *the number of invoked AK/HT local certificates. The route admits a certificate with at most*

$$qrLs + (N+1) + c + h$$

primary verification blocks, in addition to the finitely encoded source and target objects. Each block may contain a finite matrix, module, chain, or proof object, but no additional family indexed by an unbounded parameter is required.

Proof. For each local or overlap object, at most rL generator-shift probes are evaluated on at most s coefficient fibers, giving $qrLs$ blocks. The deterministic tail check uses at most $N+1$ state records. The descent and route comparison contribute c cells, and the internal AK/HT path contributes h local certificates. Summing these finite families gives the displayed upper bound. Shared probes may reduce the actual size. □

Part XI

Validated Success Case: Exact Minimal Rank-Probe Reconstruction

46 Validation target and exact success criterion

This part validates the universal bridge architecture on one nontrivial external mathematical class without leaving a problem-specific theorem as an unresolved hypothesis. The source objects are finite multiparameter persistence modules that are declared to be finite direct sums of box modules. The external conclusion is exact object reconstruction and isomorphism classification from a finite scalar certificate. The result is not merely a conditional adapter: the probe family, inverse transform, lower bound, verifier, and Return are all proved below.

Definition 46.1 (Finite product grid)**.** Fix an integer $d \geq 1$ and positive integers

$$\mathbf{n} = (n_1, \dots, n_d).$$

Let

$$P_{\mathbf{n}} := [n_1] \times \cdots \times [n_d], \qquad [n_r] = \{1, \dots, n_r\},$$

with the product order. For $\mathbf{a}, \mathbf{b} \in P_{\mathbf{n}}$ with $\mathbf{a} \leq \mathbf{b}$, define the box

$$[\mathbf{a}, \mathbf{b}] := \{\mathbf{p} \in P_{\mathbf{n}} : \mathbf{a} \leq \mathbf{p} \leq \mathbf{b}\}.$$

The number of boxes is

$$R(\mathbf{n}) = \prod_{r=1}^{d} \frac{n_r(n_r+1)}{2}.$$

Definition 46.2 (Box module and box-decomposable module)**.** For a box $B = [\mathbf{a}, \mathbf{b}]$, let $k_B \in \mathrm{Rep}_k(P_{\mathbf{n}})$ be the representation with value k at points of B, value 0 outside B, identity maps between comparable points lying in B, and zero maps otherwise.

A module $M \in \mathrm{Rep}_k(P_{\mathbf{n}})$ is *box-decomposable* when

$$M \cong \bigoplus_{B \in \mathcal{B}_{\mathbf{n}}} k_B^{m_B}$$

for uniquely specified nonnegative multiplicities $m_B \in \mathbb{N}$, where $\mathcal{B}_{\mathbf{n}}$ is the finite box set. The declared source class is

$$\mathrm{BoxDec}_k(P_{\mathbf{n}}) := \left\{ \bigoplus_{B \in \mathcal{B}_{\mathbf{n}}} k_B^{m_B} \right\}.$$

No assertion is made that every representation of $P_{\mathbf{n}}$ is box-decomposable.

Lemma 46.3 (Krull–Schmidt uniqueness of box multiplicities)**.** *Every box module k_B is indecomposable, and $k_B \cong k_C$ implies $B = C$. Consequently every finite box decomposition has unique multiplicities.*

Proof. The support of k_B is exactly B, so an isomorphism $k_B \cong k_C$ forces $B = C$. A box is connected in the underlying Hasse graph. Any endomorphism of k_B is therefore multiplication by one common scalar on every nonzero vertex space, and

$$\mathrm{End}(k_B) \cong k.$$

Hence k_B is indecomposable. The category $\mathrm{Rep}_k(P_{\mathbf{n}})$ is the category of finite-dimensional modules over the finite-dimensional incidence algebra $kP_{\mathbf{n}}$, and is Krull–Schmidt. The multiplicities of the pairwise nonisomorphic indecomposable box summands are therefore unique. □

Definition 46.4 (Comparable-pair rank probe)**.** For $M \in \mathrm{Rep}_k(P_{\mathbf{n}})$ and $\mathbf{u} \leq \mathbf{v}$, define

$$\rho_M(\mathbf{u}, \mathbf{v}) := \mathrm{rank}_k(M(\mathbf{u} \leq \mathbf{v})).$$

The comparable pair $(\mathbf{u}, \mathbf{v})$ is identified with its query box $Q = [\mathbf{u}, \mathbf{v}]$. Thus the complete rank-probe vector is

$$\rho_M = \left(\rho_M(Q)\right)_{Q \in \mathcal{B}_{\mathbf{n}}} \in \mathbb{N}^{R(\mathbf{n})}.$$

Definition 46.5 (Exact success criterion)**.** The UB validation succeeds when all of the following are proved:

1. a closed formula reconstructs every m_B from ρ_M;
2. equality of rank-probe vectors is equivalent to isomorphism inside the declared box-decomposable source class;
3. the probe number $R(\mathbf{n})$ is minimal among all additive integer-valued scalar probe families;
4. a finite verifier checks a proposed reconstruction and replays it;
5. the result holds for arbitrary d and arbitrary finite product grids.

47 Rank probes as an integral zeta transform

Lemma 47.1 (Contribution of one box)**.** *Let* $B, Q \in \mathcal{B}_{\mathbf{n}}$*. Then*

$$\rho_{k_B}(Q) = \begin{cases} 1, & Q \subseteq B, \\ 0, & Q \not\subseteq B. \end{cases}$$

Proof. Write $Q = [\mathbf{u}, \mathbf{v}]$. If $Q \subseteq B$, both endpoint spaces of $k_B(\mathbf{u} \le \mathbf{v})$ are k and the structure map is the identity. Otherwise at least one endpoint lies outside B, or the required interval is not fully supported, and the structure map has rank zero. □

Theorem 47.2 (UB-SC-01: Integral zeta-transform theorem)**.** *Let*

$$M \cong \bigoplus_{B \in \mathcal{B}_{\mathbf{n}}} k_B^{m_B} \in \mathrm{BoxDec}_k(P_{\mathbf{n}}).$$

Then for every query box Q*,*

$$\rho_M(Q) = \sum_{B \supseteq Q} m_B.$$

Equivalently, if $m = (m_B)_B$ *and* $\rho = (\rho(Q))_Q$*, then*

$$\rho = Z_{\mathbf{n}} m, \qquad (Z_{\mathbf{n}})_{Q,B} = \mathbf{1}_{\{Q \subseteq B\}}.$$

After ordering boxes by nonincreasing inclusion size, $Z_{\mathbf{n}}$ *is unitriangular over* $\mathbb{Z}$*. In particular,*

$$\det Z_{\mathbf{n}} = 1.$$

Proof. Rank is additive under direct sum, and the preceding lemma gives the contribution of each summand. If boxes are ordered so that a strict container precedes every contained box, a nonzero entry can occur only on or before the diagonal, while each diagonal entry is one. Hence the matrix is unitriangular and unimodular. □

48 Explicit Möbius inversion and exact object Return

For vectors $\epsilon, \eta \in \{0,1\}^d$, write

$$|\epsilon| = \sum_r \epsilon_r, \qquad |\eta| = \sum_r \eta_r.$$

Extend a rank function by the convention

$$\rho(\mathbf{u}, \mathbf{v}) = 0$$

whenever some coordinate of $\mathbf{u}$ is below 1, some coordinate of $\mathbf{v}$ is above n_r, or $\mathbf{u} \not\leq \mathbf{v}$.

Theorem 48.1 (UB-SC-02: Explicit $2d$-difference inversion)**.** *For every box $B = [\mathbf{a}, \mathbf{b}]$,*

$$\boxed{m_{[\mathbf{a},\mathbf{b}]} = \sum_{\epsilon,\eta \in \{0,1\}^d} (-1)^{|\epsilon|+|\eta|} \rho_M(\mathbf{a} - \epsilon, \mathbf{b} + \eta).}$$

Thus every box multiplicity is reconstructed from at most 2^{2d} rank entries, with out-of-range terms interpreted as zero.

Proof. In one coordinate, the cumulative relation

$$\rho(a, b) = \sum_{c \leq a,\ d \geq b} m(c, d)$$

is inverted by applying the two finite differences

$$(\Delta_a^- f)(a, b) = f(a, b) - f(a-1, b), \qquad (\Delta_b^+ f)(a, b) = f(a, b) - f(a, b+1).$$

The product grid and box-containment relation factor coordinatewise. Applying $\Delta_{a_r}^- \Delta_{b_r}^+$ for every $r = 1, \dots, d$ gives the displayed $2d$-fold inclusion–exclusion formula. This is the Möbius inverse of the unimodular zeta matrix $Z_{\mathbf{n}}$. □

Corollary 48.2 (Exact object-isomorphism Return)**.** *For $M, N \in \mathrm{BoxDec}_k(P_{\mathbf{n}})$,*

$$\rho_M = \rho_N \quad \Longleftrightarrow \quad M \cong N.$$

Consequently the complete comparable-pair rank vector is a finite, exact, zero-loss object classifier on the declared source class.

Proof. Equality of rank vectors gives equality of every multiplicity by the inversion formula. Conversely, isomorphic box decompositions have identical structure-map ranks. □

Corollary 48.3 (Existence criterion for abstract box-rank data)**.** *Let*

$$\rho : \{(\mathbf{u}, \mathbf{v}) : \mathbf{u} \leq \mathbf{v}\} \longrightarrow \mathbb{Z}$$

be an integer-valued function. Define m_B by the explicit inversion formula. Then there exists a box-decomposable module M_ρ satisfying $\rho_{M_\rho} = \rho$ if and only if

$$m_B \in \mathbb{N} \qquad \text{for every } B.$$

When it exists, M_ρ is unique up to isomorphism inside $\mathrm{BoxDec}_k(P_{\mathbf{n}})$.

Proof. Necessity follows from the multiplicity interpretation. For sufficiency, form

$$M_\rho := \bigoplus_B k_B^{m_B}.$$

The zeta-transform theorem and unimodularity show that its rank vector is exactly ρ. Uniqueness follows from the preceding corollary. □

Warning 48.4 (Membership boundary). For an arbitrary $M \in \mathrm{Rep}_k(P_{\mathbf{n}})$, non-negative inversion of ρ_M proves that the same rank function is realized by a box-decomposable module. It does not by itself prove that the original M is box-decomposable. The declared source-class membership or an independent decomposition theorem is a non-scalar obligation.

49 Sharp minimality among additive scalar probes

Definition 49.1 (Additive integer-valued scalar probe)**.** An additive scalar probe on $\mathrm{BoxDec}_k(P_{\mathbf{n}})$ is a function

$$\psi : \mathrm{BoxDec}_k(P_{\mathbf{n}}) \longrightarrow \mathbb{Z}$$

such that

$$\psi(0) = 0, \qquad \psi(M \oplus N) = \psi(M) + \psi(N).$$

A finite family $\Psi = (\psi_1, \dots, \psi_s)$ is *object-separating* if

$$\Psi(M) = \Psi(N) \Longrightarrow M \cong N$$

for all objects in the declared source class.

Theorem 49.2 (UB-SC-03: Exact minimal additive-probe theorem)**.** *Let*

$$R = R(\mathbf{n}) = \prod_{r=1}^{d} \frac{n_r(n_r+1)}{2}.$$

Every object-separating family of additive integer-valued scalar probes on $\mathrm{BoxDec}_k(P_{\mathbf{n}})$ *has at least* R *members. The complete comparable-pair rank family has exactly* R *members and is object-separating. Therefore*

$$\boxed{\mathrm{MinCert}_{\text{additive scalar}}(\mathrm{BoxDec}_k(P_{\mathbf{n}})) = R(\mathbf{n}).}$$

Proof. The isomorphism classes under direct sum form the free commutative monoid

$$\mathbb{N}^R$$

on the R box indecomposables. Every additive probe extends uniquely to a group homomorphism from the group completion $\mathbb{Z}^R$ to $\mathbb{Z}$. Thus a family of s probes defines an integer matrix

$$A_\Psi : \mathbb{Z}^R \longrightarrow \mathbb{Z}^s.$$

If $s < R$, then $A_\Psi \otimes_{\mathbb{Z}} \mathbb{Q}$ has a nonzero rational kernel. Clearing denominators gives $0 \neq z \in \mathbb{Z}^R$ with $A_\Psi z = 0$. Write

$$z = z^+ - z^-, \qquad z^+, z^- \in \mathbb{N}^R,$$

with disjoint supports. The two distinct box-decomposable modules represented by z^+ and z^- have identical probe vectors, contradicting separation. Hence $s \geq R$.

There are exactly R comparable pairs, because a comparable pair $\mathbf{u} \leq \mathbf{v}$ is the same datum as a box $[\mathbf{u}, \mathbf{v}]$. Their rank probes are additive and separate objects by exact Möbius inversion. Hence the lower bound is attained. □

Remark 49.3 (Strength of the lower bound). The theorem is stronger than minimality among subsets of coordinate rank probes. It excludes every smaller family of additive integer-valued scalar invariants, regardless of how those invariants are formed. Nonadditive encodings with unbounded ad hoc information capacity are outside the MinCert information class.

50 Finite verifier and certificate complexity

Definition 50.1 (Rank-reconstruction certificate). A rank-reconstruction certificate is

$$\mathcal{C}_{\text{rank}} = (\mathbf{n}, \rho, m, \text{source\_class\_witness}),$$

where $\rho \in \mathbb{N}^R$, $m \in \mathbb{N}^R$, and the final field records the declared box-decomposable source scope or its independent proof.

Verifier contract 50.2 (UB-SC-04: Exact finite replay verifier). The verifier:

1. checks that the grid and box index each have size $R(\mathbf{n})$;
2. checks integrality and nonnegativity of every claimed m_B;
3. recomputes every m_B by the $2d$-difference formula;
4. replays the forward relation
$$\rho(Q) = \sum_{B \supseteq Q} m_B$$
for every query box Q;
5. compares the source module's finite structure-map ranks with the declared ρ, when explicit source matrices are supplied;
6. Returns the unique isomorphism class
$$\bigoplus_B k_B^{m_B}$$
inside the declared source class.

Any failed arithmetic equality gives REJECT. Missing source-class evidence gives UNDEFINED, not a positive reconstruction verdict for an arbitrary representation.

Theorem 50.3 (Certificate-size and runtime bound). *The certificate uses R rank scalars and R multiplicities. Direct inversion requires at most*

$$2^{2d} R$$

signed scalar accesses. Forward replay requires at most R^2 additions in the naive implementation and can be accelerated by multidimensional cumulative sums. For fixed d, the inversion cost is linear in the number of boxes.

Proof. There is one certificate field per box and one inversion stencil of size at most 2^{2d} per multiplicity. The forward relation has at most R summands for each of R queries. $\square$

51 Explicit two-by-two computation

Let
$$P = [2] \times [2].$$
Order the nine boxes as
$$\begin{array}{lll} B_1 = [(1,1),(2,2)], & B_2 = [(1,1),(1,2)], & B_3 = [(2,1),(2,2)], \\ B_4 = [(1,1),(2,1)], & B_5 = [(1,2),(2,2)], & B_6 = \{(1,1)\}, \\ B_7 = \{(1,2)\}, & B_8 = \{(2,1)\}, & B_9 = \{(2,2)\}. \end{array}$$
Use the same order for the nine query boxes. Write
$$x_i = m_{B_i}, \qquad r_i = \rho_M(B_i).$$

Theorem 51.1 (UB-SC-05: 2×2 exact matrix and inverse)**.** *The rank and multiplicity vectors satisfy*
$$\begin{pmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \\ r_7 \\ r_8 \\ r_9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \\ x_9 \end{pmatrix}.$$
The determinant is 1, and the exact inverse is
$$\begin{array}{llll} x_1 = r_1, & & & \\ x_2 = r_2 - r_1, & x_3 = r_3 - r_1, & x_4 = r_4 - r_1, & x_5 = r_5 - r_1, \\ x_6 = r_6 - r_2 - r_4 + r_1, & x_7 = r_7 - r_2 - r_5 + r_1, & & \\ x_8 = r_8 - r_3 - r_4 + r_1, & x_9 = r_9 - r_3 - r_5 + r_1. & & \end{array}$$
Exactly nine additive integer-valued scalar probes are necessary and sufficient.

Proof. The matrix entry is one precisely when the column box contains the row query. The displayed inverse is direct substitution, or the two-dimensional four-boundary difference formula. The determinant and sharp minimality follow from the general theorems. $\square$

Example 51.2 (Complete numerical replay)**.** Take
$$(x_1, \dots, x_9) = (2,1,0,3,1,4,2,0,5).$$
The forward rank vector is
$$(r_1, \dots, r_9) = (2,3,2,5,3,10,6,5,8).$$
Substitution into the inverse equations returns exactly
$$(2,1,0,3,1,4,2,0,5),$$
with no ambiguity and no external oracle.

52 UB closure and expansion theorem

Theorem 52.1 (UB-SC-06: UB exact rank-probe success-case closure)**.** *For every field k, every finite product grid $P_{\mathbf{n}}$, and the declared source category* $\mathrm{BoxDec}_k(P_{\mathbf{n}})$*, the following external theorem is completely closed:*

> *Every object is uniquely reconstructed up to isomorphism from the finite family of all comparable-pair rank probes. The reconstruction is given by an explicit integral Möbius inversion. The family has the smallest possible cardinality among all additive integer-valued scalar object-separating probe families.*

The realization, finite observation family, completeness theorem, MinCert lower bound, reconstruction algorithm, verifier, and external object Return are all contained in this appendix. No finite-to-infinite reduction, metric threshold transport, spectral-sequence degeneration, or unproved rigidity theorem is required.

Proof. Finite-grid realization is exact by the finite-poset incidence-algebra component. The zeta-transform and inversion theorems prove completeness and object Return. The additive-probe theorem proves the sharp lower bound. The verifier contract supplies finite replay. Every invoked arrow is exact and finite, so the Universal Finite Bridge Closure conditions reduce to the proved UB-MOR and UB-FIN components plus the present explicit inverse. □

Corollary 52.2 (Dimension-independent extension)**.** *The success case holds unchanged for arbitrary multiparameter dimension d. Only the local inversion stencil changes from four boundary differences in one dimension to $2d$ boundary differences, with at most 2^{2d} terms per box.*

Non-claim 52.3 (Novelty and strength boundary). This part establishes a fully executed AK–UB success case and a sharp additive MinCert theorem. It does not claim:

1. external-literature novelty for the classical rank-invariant or Möbius inversion principle;
2. classification of arbitrary finite-grid persistence modules;
3. reconstruction of a particular morphism, extension class, or A_∞-structure from scalar ranks;
4. that fewer nonadditive, unbounded, encoding-style outputs are impossible;
5. a solution of a named open conjecture.

Its value is exact completion: unlike a conditional application schema, no mathematical obligation inside the declared theorem remains open.

Part XII

Validated Success Case II: Minimal Enriched Projective-Probe Return

53 Validation target and exact closure criterion

The first validated success case closed exact object reconstruction for the declared box-decomposable source class by scalar rank probes. Scalar ranks cannot encode a particular morphism or a Yoneda extension class. The present success case closes that missing level by replacing scalar observations with an enriched projective probe whose module action is retained.

Throughout this part, let P be a finite poset, regarded as a finite category, let k be a field, and let

$$\mathrm{Rep}_k(P) = \mathrm{Fun}(P, \mathrm{Vect}_k^{\mathrm{fd}}).$$

Write $q = |P|$.

Definition 53.1 (Atomic representable probes)**.** For $x \in P$, define the covariant representable projective

$$P_x := k[\mathrm{Hom}_P(x, -)].$$

Set

$$G_P := \bigoplus_{x \in P} P_x, \qquad A_P := \mathrm{End}_{\mathrm{Rep}_k(P)}(G_P)^{\mathrm{op}},$$

and define the enriched projective-probe functor

$$\mathcal{E}_P := \mathrm{Hom}(G_P, -) : \mathrm{Rep}_k(P) \longrightarrow A_P\text{-}\mathrm{Mod}^{\mathrm{fd}}.$$

The word *enriched* means that $\mathcal{E}_P(M)$ is retained as an A_P-module. The scalar tuple $(\dim_k \mathrm{Hom}(P_x, M))_{x \in P}$ is not substituted for this module.

Definition 53.2 (VC2 exact closure criterion)**.** The second validation succeeds when all of the following are proved inside this appendix:

1. $\mathcal{E}_P$ reconstructs every object and every morphism;
2. it preserves and reflects kernels, cokernels, monomorphisms, epimorphisms, and short exact sequences;
3. it returns every Yoneda Ext^n-class and every bounded derived triangle;
4. the atomic projective family $\{P_x\}_{x \in P}$ is minimal among projective-generator probe families, in the precise indecomposable-summand sense stated below;
5. all claims admit a finite matrix verifier;
6. the result composes with the scalar rank reconstruction of Success Case I without claiming that scalar ranks determine morphisms or extensions.

54 Exact enriched realization and explicit inverse

Lemma 54.1 (Yoneda evaluation)**.** *For every $x \in P$ and $M \in \mathrm{Rep}_k(P)$, there is a natural isomorphism*

$$\mathrm{Hom}(P_x, M) \cong M(x).$$

In particular, every P_x is projective.

Proof. This is the linear Yoneda lemma. Evaluation at a vertex is exact because kernels and cokernels in a functor category are computed pointwise. Hence $\mathrm{Hom}(P_x, -) \cong (-)(x)$ is exact, proving projectivity. □

Theorem 54.2 (UB-VC2-01: Exact enriched projective-probe equivalence)**.** *The functor*

$$\mathcal{E}_P = \mathrm{Hom}(G_P, -)$$

is an exact equivalence of abelian categories. A quasi-inverse is

$$\mathcal{R}_P(V) := G_P \otimes_{A_P} V.$$

The unit and counit

$$V \longrightarrow \mathcal{E}_P\mathcal{R}_P(V), \qquad \mathcal{R}_P\mathcal{E}_P(M) \longrightarrow M$$

are natural isomorphisms.

Proof. The object G_P is projective by the preceding lemma. If $M \neq 0$, then $M(x) \neq 0$ for some x, and therefore $\mathrm{Hom}(P_x, M) \neq 0$; hence G_P is a generator. Every object of $\mathrm{Rep}_k(P)$ is finite-dimensional and has a finite presentation by finite sums of representables. The standard projective-generator Morita construction gives $\mathcal{R}_P$ and shows that the unit and counit are isomorphisms on G_P, on finite sums of G_P, and then on cokernels of maps between such sums. This covers every object. Exactness of $\mathcal{E}_P$ follows from projectivity of G_P, and exactness of the quasi-inverse follows because it is quasi-inverse to an exact equivalence. □

Corollary 54.3 (Object and morphism Return)**.** *For all $M, N \in \mathrm{Rep}_k(P)$, the canonical map*

$$\mathrm{Hom}_{\mathrm{Rep}_k(P)}(M, N) \xrightarrow{\sim} \mathrm{Hom}_{A_P}(\mathcal{E}_P(M), \mathcal{E}_P(N))$$

is an isomorphism. Consequently:

$$\mathcal{E}_P(M) \cong \mathcal{E}_P(N) \quad \Longleftrightarrow \quad M \cong N,$$

and every A_P-linear morphism between enriched probe modules Returns a unique morphism of P-representations.

Proof. An equivalence is fully faithful and reflects isomorphisms. Apply the quasi-inverse to an A_P-linear morphism to obtain the unique returned natural transformation. □

Proposition 54.4 (Finite matrix form of the enriched probe)**.** *Choose bases of the vertex spaces of M. Then $\mathcal{E}_P(M)$ is specified by the finite data*

$$\left(M(x), M(x \leq y)\right)_{x \leq y},$$

subject to identity and composition equations. It is enough to store matrices on the Hasse arrows and impose equality of composites along all parallel saturated chains. A morphism $f : M \to N$ is specified by matrices $f_x : M(x) \to N(x)$ satisfying

$$N(x \le y) f_x = f_y M(x \le y) \qquad (x \le y).$$

All conditions are finite systems of polynomial, in fact linear for morphisms, matrix equations over k.

Proof. A finite-poset representation is a functor, so it is exactly vertex-space data plus structure maps satisfying functoriality. The action of the incidence basis element corresponding to $x \le y$ is the structure map $M(x \le y)$. Naturality is the displayed commutative-square equation. Finiteness of P makes every list finite. □

55 Exactness, kernels, cokernels, and cone Return

Theorem 55.1 (UB-VC2-02: Exact structural Return)**.** *Let $f : M \to N$ be a morphism in $\mathrm{Rep}_k(P)$. Then there are natural isomorphisms*

$$\mathcal{E}_P(\ker f) \cong \ker \mathcal{E}_P(f), \qquad \mathcal{E}_P(\operatorname{coker} f) \cong \operatorname{coker} \mathcal{E}_P(f).$$

The functor $\mathcal{E}_P$ preserves and reflects monomorphisms, epimorphisms, isomorphisms, and short exact sequences. In particular,

$$0 \to L \xrightarrow{u} M \xrightarrow{v} N \to 0$$

is exact if and only if its enriched probe image is exact.

Proof. An exact equivalence preserves kernels and cokernels up to their unique natural isomorphisms. Reflection follows by applying the exact quasi-inverse. The stated morphism properties and short exactness are characterized by kernels and cokernels. □

Corollary 55.2 (Pointwise finite exactness verifier)**.** *A composable pair*

$$L \xrightarrow{u} M \xrightarrow{v} N$$

is short exact if and only if, for every $x \in P$,

$$0 \to L(x) \xrightarrow{u_x} M(x) \xrightarrow{v_x} N(x) \to 0$$

is an exact sequence of finite-dimensional vector spaces. Equivalently, a verifier checks

$$\begin{aligned} v_x u_x &= 0, \\ \operatorname{rank}(u_x) &= \dim L(x), \qquad \operatorname{rank}(v_x) = \dim N(x), \\ \dim M(x) &= \dim L(x) + \dim N(x). \end{aligned}$$

for every vertex, together with naturality on the Hasse arrows.

Proof. Limits and colimits in the functor category are pointwise. The rank conditions are the usual finite-dimensional criterion for exactness. □

Theorem 55.3 (Bounded derived and cone Return). *The termwise extension of $\mathcal{E}_P$ induces an exact equivalence*

$$\mathrm{D}^{\mathrm{b}}(\mathrm{Rep}_k(P)) \xrightarrow{\sim} \mathrm{D}^{\mathrm{b}}(A_P\text{-}\mathrm{Mod}^{\mathrm{fd}}).$$

For every chain map $f^\bullet : M^\bullet \to N^\bullet$,

$$\mathcal{E}_P(\mathrm{Cone}(f^\bullet)) \cong \mathrm{Cone}(\mathcal{E}_P(f^\bullet)),$$

and a bounded triangle is distinguished if and only if its enriched probe image is distinguished.

Proof. An exact equivalence of abelian categories extends termwise to an equivalence of bounded complex categories, preserves quasi-isomorphisms, and descends to bounded derived categories. Cones are defined by finite direct sums, signs, and the given differentials, all preserved by the additive exact functor. Distinguished triangles are those isomorphic to cone triangles and are therefore preserved and reflected. ☐

56 Yoneda extension and all-degree Ext Return

Theorem 56.1 (UB-VC2-03: Yoneda-class Return). *For every $n \geq 0$ and $M, N \in \mathrm{Rep}_k(P)$, the exact equivalence induces a natural isomorphism*

$$\mathrm{Ext}^n_{\mathrm{Rep}_k(P)}(M, N) \xrightarrow{\sim} \mathrm{Ext}^n_{A_P}(\mathcal{E}_P(M), \mathcal{E}_P(N)).$$

For $n = 1$, the isomorphism sends the Yoneda class of

$$0 \to N \to E \to M \to 0$$

to the class of its enriched probe image, and the inverse is obtained by applying $\mathcal{R}_P$ to the entire short exact sequence. Thus extension classes, not only Ext dimensions, are exactly Returnable.

Proof. An exact equivalence maps n-fold exact extensions to n-fold exact extensions and maps equivalences of extensions to equivalences. Applying the exact quasi-inverse defines the inverse map on Yoneda classes. The two operations are inverse because the unit and counit are natural isomorphisms. Equivalently, use the bounded derived equivalence and the standard identification

$$\mathrm{Ext}^n(M, N) \cong \mathrm{Hom}_{\mathrm{D}^{\mathrm{b}}}(M, N[n]).$$

☐

Corollary 56.2 (Finite verifier for a represented Yoneda class). *A represented Ext^1-class is certified by finite matrices for $N \to E \to M$, the vertexwise exactness checks, and the naturality equations. Two represented extensions define the same Yoneda class if and only if there exists an isomorphism between their middle terms commuting with the endpoint maps; the existence of such an isomorphism is a finite system of linear equations together with a nonzero determinant condition. The same certificate may be checked after applying $\mathcal{E}_P$, with exactly the same answer.*

Proof. The Yoneda equivalence relation for one-extensions is precisely isomorphism of short exact sequences fixing the endpoints. Every object and map is finite-dimensional, so all commutation conditions are finite matrix equations. The preceding theorem proves invariance and exact Return under $\mathcal{E}_P$. □

Remark 56.3 (Computing rather than merely transporting Ext). For a fixed degree n, a projective resolution by finite sums of the representables P_x reduces Ext^n to the cohomology of a finite matrix complex. VC2 requires no scalar-dimension shortcut: the cocycles, coboundaries, and resulting extension classes are retained.

57 Sharp atomic minimality of the enriched probe

Definition 57.1 (Vertex simple)**.** For $x \in P$, let S_x be the representation with value k at x, value 0 at every other vertex, and all nonidentity structure maps zero.

Lemma 57.2 (Atomic separation matrix)**.** *For $x, y \in P$,*

$$\mathrm{Hom}(P_y, S_x) \cong \begin{cases} k, & x = y, \\ 0, & x \neq y. \end{cases}$$

Proof. By Yoneda, $\mathrm{Hom}(P_y, S_x) \cong S_x(y)$, which is k exactly when $y = x$. □

Theorem 57.3 (UB-VC2-04: Minimal basic projective-generator theorem)**.** *Let Q be a finite projective generator of $\mathrm{Rep}_k(P)$. In its Krull–Schmidt decomposition, Q contains every P_x as a direct summand. Consequently:*

1. *every basic projective generator has exactly the q pairwise nonisomorphic indecomposable summands $\{P_x : x \in P\}$;*
2. *omitting any one atomic representable probe destroys zero-reflection;*
3. *$G_P = \bigoplus_x P_x$ has the smallest possible number q of indecomposable projective probe types among all projective generators.*

The count is a count of indecomposable probe types; packaging their direct sum as one object does not reduce this atomic complexity.

Proof. A projective generator Q must detect the nonzero simple S_x, so $\mathrm{Hom}(Q, S_x) \neq 0$ for every x. Every indecomposable projective of the finite-dimensional incidence algebra is isomorphic to one of the P_y. Decompose Q into indecomposable projectives. The atomic separation lemma shows that a summand contributes a nonzero map to S_x only when that summand is P_x. Thus P_x occurs for every x. The three conclusions follow. □

Corollary 57.4 (Explicit failure certificate for every omitted probe)**.** *For a proper subset $J \subsetneq P$, choose $x \in P \setminus J$. Then*

$$\mathrm{Hom}(P_y, S_x) = 0 \qquad (y \in J),$$

although $S_x \neq 0$. Hence the reduced enriched probe $\bigoplus_{y \in J} P_y$ is not conservative and cannot provide exact, Yoneda, or derived Return on all of $\mathrm{Rep}_k(P)$.

58 Explicit execution on the two-by-two grid

Let

$$P = [2] \times [2] = \{00, 10, 01, 11\}$$

with cover arrows

$$a : 00 \to 10, \qquad b : 00 \to 01, \qquad c : 10 \to 11, \qquad d : 01 \to 11.$$

The unique square relation is

$$ca = db.$$

Proposition 58.1 (Concrete incidence-algebra model)**.** *The enriched algebra is the finite bound-quiver algebra*

$$A_P \cong kQ/(ca - db).$$

A representation is exactly a tuple

$$(M_{00}, M_{10}, M_{01}, M_{11}; A, B, C, D)$$

of finite-dimensional vector spaces and matrices satisfying

$$CA = DB.$$

A morphism $f : M \to N$ *is exactly a four-tuple of matrices* $(f_{00}, f_{10}, f_{01}, f_{11})$ *satisfying*

$$f_{10}A_M = A_N f_{00}, \quad f_{01}B_M = B_N f_{00},$$
$$f_{11}C_M = C_N f_{10}, \quad f_{11}D_M = D_N f_{01}.$$

The four atomic projective probes are

$$P_{00}, P_{10}, P_{01}, P_{11},$$

and each is necessary.

Proof. The Hasse quiver and its commutativity relation present the incidence algebra. The module and morphism descriptions are the functoriality and naturality equations. Necessity of each projective follows from the four simples $S_{00}, S_{10}, S_{01}, S_{11}$. □

Example 58.2 (UB-VC2-05: Finite exactness and extension replay)**.** Given matrices for a candidate sequence

$$0 \to L \xrightarrow{u} E \xrightarrow{v} M \to 0,$$

a verifier performs the following finite operations:

1. check the square relation for L, E, M;
2. check the eight naturality equations for u and v;
3. check $v_x u_x = 0$ and the rank equalities at the four vertices;
4. serialize the resulting A_P-module extension;
5. Return the same sequence by $G_P \otimes_{A_P} -$.

No slice completeness, barcode hypothesis, or unproved rigidity statement is used.

59 Two-tier composition with scalar rank reconstruction

Definition 59.1 (Two-tier finite certificate)**.** For a box-decomposable object M on a finite product grid, define:

1. the *object tier* to be the complete comparable-pair rank vector ρ_M of Success Case I;
2. the *structural tier* to be the enriched incidence-algebra module $\mathcal{E}_P(M)$, together with the actual A_P-linear map, exact sequence, extension, or complex data requested by the claim.

Theorem 59.2 (UB-VC2-06: Scalar-to-enriched UB composition)**.** *Let P be a finite product grid.*

1. *On $\mathrm{BoxDec}_k(P)$, the object tier reconstructs every object up to isomorphism by the explicit Möbius inversion of Success Case I.*
2. *On all of $\mathrm{Rep}_k(P)$, the structural tier reconstructs every morphism, kernel, cokernel, short exact sequence, Yoneda class, bounded complex, and distinguished triangle by the exact enriched equivalence of Success Case II.*
3. *The two tiers compose without strength inflation: scalar ranks are used only for object classification in the box-decomposable class, while actual enriched module action is required for morphism and extension Return.*

Thus the previous object-level success case extends to an exact structural Return pipeline, but no morphism is inferred from scalar ranks alone.

Proof. Part (1) is the exact rank-probe reconstruction theorem. Part (2) is the enriched projective-probe equivalence and its exact, Yoneda, and derived consequences. The source classes and Return strengths are explicitly separated, proving Part (3). □

60 VC2 verifier contract and closure theorem

Verifier contract 60.1 (Minimal enriched probe certificate)**.** A VC2 certificate contains:

1. the finite poset P, its Hasse arrows, and all path-equality relations;
2. the field k and the vertex dimensions;
3. structure matrices for every represented object;
4. matrices for every represented morphism or differential;
5. functoriality and naturality equation checks;
6. vertexwise kernel, cokernel, rank, and exactness checks as required;
7. for a Yoneda class, the complete finite extension sequence and endpoint maps;
8. for minimality, the omitted vertex and the corresponding simple-module failure certificate;

9. the requested Return strength and the statement that scalar dimensions alone are not used for structural Return.

All fields are finite. Replay consists entirely of finite matrix arithmetic and the explicit tensor quasi-inverse.

Theorem 60.2 (UB-VC2-07: UB enriched projective-probe success-case closure)**.** *For every field k and every finite poset P, the following external theorem is completely closed:*

> *The enriched probe formed by all vertex representable projectives is an exact, fully faithful, essentially surjective finite realization. It reconstructs all objects and morphisms, preserves and reflects exact sequences, returns every Yoneda extension class and every bounded derived triangle, and has the smallest possible set of indecomposable projective probe types among projective-generator realizations.*

The source category, probe family, endomorphism algebra, quasi-inverse, structural Return, minimality countercertificates, and finite verifier are all contained in this appendix. No unresolved mathematical hypothesis remains inside the declared theorem.

Proof. Exact equivalence and its inverse are proved by the projective-generator theorem. Full morphism Return, exactness, Yoneda Return, and derived Return follow from the exact equivalence. Atomic minimality follows from the simple-module separation matrix. The verifier contract reduces every claim to finite matrix equations. The two-tier theorem links this structural result to the earlier scalar object reconstruction without overclaiming scalar strength. □

Corollary 60.3 (HT PE-3–PE-5 realization)**.** *On finite-poset representations, the enriched projective-probe component supplies a fully explicit realization of the object, morphism, exact/Yoneda, and bounded-derived Return strengths corresponding to the PE-2 through PE-5 hierarchy. Appendix HT may transport or compose these data, but no additional completeness hypothesis is needed for the present finite source category.*

Non-claim 60.4 (VC2 novelty and strength boundary)*.* This success case does not assert external-literature novelty for Morita theory, Yoneda's lemma, or exact-equivalence invariance of Ext. It establishes an executed AK–UB validation with a sharp atomic projective-generator minimum. It does not claim that:

1. scalar dimensions of Hom or Ext recover their module action or classes;
2. a proper subset of vertex probes is insufficient on every restricted source subclass;
3. box-decomposable modules form an extension-closed subcategory;
4. arbitrary infinite posets admit the same finite certificate;
5. the enriched probe solves a named open conjecture.

Part XIII

Validated Success Case III: Explicit Finite-Window Tachikawa Closure

61 Validation target and exact success criterion

This success case moves the finite-state Tachikawa bridge from a conditional package to a fully executed external theorem on an infinite family of self-injective algebras. Fix a field k, integers $m \geq 1$ and $L \geq 2$, and let Q_m be the oriented cycle

$$0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow m-1 \longrightarrow 0,$$

with vertices read in $\mathbb{Z}/m\mathbb{Z}$. If J is the arrow ideal, set

$$A_{m,L} := kQ_m/J^L.$$

All modules in this part are finitely generated right $A_{m,L}$-modules.

Definition 61.1 (VC3 exact closure criterion)**.** VC3 is closed only when the following are all proved inside this appendix:

1. $A_{m,L}$ is self-injective for every declared (m, L);
2. all indecomposable nonprojective modules are explicitly classified;
3. the syzygy operator is computed on every indecomposable type;
4. a uniform, computable positive degree $p_{m,L}$ is obtained with $\Omega^{p_{m,L}} M \cong M$ in the stable category for every module;
5. nonprojectivity is converted into a nonzero self-extension in that degree;
6. Tachikawa's second conjecture is therefore proved for the whole family;
7. the calculation is replayable from finite quiver, relation, decomposition, syzygy-permutation, and stable-Hom data;
8. no assertion is promoted to arbitrary self-injective algebras.

Proposition 61.2 (Self-injectivity of the truncated cycle)**.** *For every field k, $m \geq 1$, and $L \geq 2$, the algebra $A_{m,L}$ is finite-dimensional and self-injective.*

Proof. Let e_i be the primitive idempotent at vertex i, and put $P_i = e_i A_{m,L}$. The radical chain

$$P_i \supset e_i J \supset \cdots \supset e_i J^{L-1} \supset 0$$

has one-dimensional successive factors

$$S_i, S_{i+1}, \ldots, S_{i+L-1},$$

so P_i is uniserial of length L, with top S_i and socle S_{i+L-1}. The indecomposable injective right module with socle S_j is $I_j = D(A_{m,L} e_j)$. Its composition series has length L, top S_{j-L+1}, and socle S_j. Hence I_{i+L-1} and P_i are uniserial modules with the same top and length, and therefore

$$P_i \cong I_{i+L-1}.$$

Every indecomposable projective is injective. Thus $A_{m,L}$ is self-injective. □

62 Indecomposable modules and the exact syzygy permutation

Definition 62.1 (Uniserial type)**.** For $i \in \mathbb{Z}/m\mathbb{Z}$ and $1 \le \ell \le L$, define

$$M(i,\ell) := e_i A_{m,L}/e_i J^\ell.$$

It has length ℓ and composition factors $S_i, S_{i+1}, \dots, S_{i+\ell-1}$. The case $\ell = L$ is the projective module P_i.

Proposition 62.2 (Complete indecomposable list)**.** *Every indecomposable finitely generated right $A_{m,L}$-module is uniquely isomorphic to one $M(i,\ell)$ with $i \in \mathbb{Z}/m\mathbb{Z}$ and $1 \le \ell \le L$. Consequently the nonprojective indecomposable types are exactly*

$$\mathcal{I}_{m,L} := \{(i,\ell) : i \in \mathbb{Z}/m\mathbb{Z}, 1 \le \ell \le L-1\}.$$

Proof. The indecomposable projectives are uniserial, so $A_{m,L}$ is a Nakayama algebra. An indecomposable module over a Nakayama algebra is uniserial and is a quotient of its indecomposable projective cover. If its top is S_i and its length is ℓ, the unique such quotient is $e_i A_{m,L}/e_i J^\ell = M(i,\ell)$. Top and length determine the displayed pair uniquely. □

Theorem 62.3 (UB-VC3-03: Exact syzygy formula)**.** *For $1 \le \ell \le L-1$,*

$$\Omega M(i,\ell) \cong M(i+\ell, L-\ell),$$

and hence, for every $t \ge 0$,

$$\Omega^{2t} M(i,\ell) \cong M(i+tL, \ell),$$
$$\Omega^{2t+1} M(i,\ell) \cong M(i+tL+\ell, L-\ell).$$

All vertex indices are read modulo m.

Proof. The canonical projective cover

$$e_i A_{m,L} \twoheadrightarrow M(i,\ell)$$

has kernel $e_i J^\ell$. This kernel is uniserial of length $L-\ell$ and has top $S_{i+\ell}$, so it is $M(i+\ell, L-\ell)$. Applying the same formula twice gives

$$\Omega^2 M(i,\ell) \cong M(i+\ell+L-\ell, \ell) = M(i+L, \ell).$$

Induction yields the displayed even and odd formulas. □

Definition 62.4 (Syzygy type permutation)**.** On $\mathcal{I}_{m,L}$, define

$$T_{m,L}(i,\ell) := (i+\ell, L-\ell).$$

The preceding theorem says that syzygy acts on all nonprojective indecomposable types by the finite permutation $T_{m,L}$, with

$$T_{m,L}^2(i,\ell) = (i+L, \ell).$$

63 Finite syzygy-permutation theorem

Theorem 63.1 (UB-VC3-01: Finite syzygy-permutation Tachikawa theorem)**.** *Let A be a self-injective finite-dimensional algebra. Suppose the stable category contains only finitely many isomorphism classes of indecomposable nonzero objects, and syzygy permutes these classes. Let p be the order of that permutation, with $p = 1$ by convention when the stable category is zero. Then for every finitely generated A-module M,*

$$M \text{ is projective} \quad \Longleftrightarrow \quad \operatorname{Ext}_A^p(M, M) = 0.$$

In particular, Tachikawa's second conjecture holds for A, and the all-degree self-orthogonality premise is replaced by one explicitly computable degree.

Proof. Projective modules have vanishing positive Ext. Conversely, decompose the nonprojective part of M into indecomposables. Since Ω^p fixes each stable indecomposable isomorphism class, additivity gives a stable isomorphism

$$\Omega^p M \cong M.$$

If M is nonprojective, its stable identity is nonzero: if the identity factored through a projective, then M would be a direct summand of a projective. Transporting the stable identity across the displayed stable isomorphism gives

$$0 \neq \underline{\operatorname{Hom}}_A(\Omega^p M, M) \cong \operatorname{Ext}_A^p(M, M),$$

a contradiction. Thus vanishing in degree p forces projectivity. □

Corollary 63.2 (UB-VC3-02: Representation-finite self-injective closure)**.** *Tachikawa's second conjecture holds for every representation-finite self-injective finite-dimensional algebra. If r is the number of nonprojective indecomposable stable types and p is the order of the syzygy permutation on those types, then p is finite, computable from the stable Auslander–Reiten data, and*

$$\operatorname{Ext}_A^p(M, M) = 0 \Longrightarrow M \text{ is projective}.$$

Moreover p divides $\operatorname{lcm}(1, 2, \dots, r)$.

Remark 63.3 (Evolution beyond the finite-state theorem). The earlier UB finite-state theorem used a finite set of complete module states. That hypothesis is too restrictive for arbitrary direct sums with unbounded multiplicity. The present theorem needs only finitely many indecomposable stable types and a finite permutation of those types. Multiplicity vectors may be unbounded; the uniform permutation exponent still Returns every module to itself stably. VC3 therefore supplies a strictly more reusable Tachikawa bridge.

64 Uniform periodicity for self-injective Nakayama algebras

Definition 64.1 (Uniform Nakayama period bound)**.** Put

$$q_{m,L} := \frac{m}{\gcd(m, L)}, \qquad p_{m,L} := 2q_{m,L}.$$

Theorem 64.2 (UB-VC3-04: Uniform stable periodicity)**.** *For every finitely generated $A_{m,L}$-module M,*

$$\Omega^{p_{m,L}} M \cong M \qquad \textit{in } \underline{\mathrm{mod}} A_{m,L}.$$

Equivalently, the order of the syzygy type permutation divides $2m/\gcd(m, L)$.

Proof. For every nonprojective indecomposable,

$$\Omega^{2q_{m,L}} M(i, \ell) \cong M(i + q_{m,L}L, \ell).$$

By definition of $q_{m,L}$, the integer $q_{m,L}L$ is divisible by m, so the right-hand side is $M(i, \ell)$. Krull–Schmidt decomposition and additivity of syzygy in the stable category give the assertion for every finite module. Projective summands are zero in the stable category. □

Proposition 64.3 (Exact indecomposable period)**.** *For a nonprojective $M(i, \ell)$, its stable period is the least positive integer in the set*

$$\{2t : t \geq 1,\ tL \equiv 0 \pmod m\} \cup \{2t + 1 : t \geq 0,\ 2\ell = L \textit{ and } tL + \ell \equiv 0 \pmod m\}.$$

In particular it divides $p_{m,L}$.

Proof. The even and odd syzygy formulas show that an even return occurs exactly when the top index returns, namely $tL \equiv 0 \pmod m$. An odd return must preserve length, forcing $L - \ell = \ell$, and then must return the top index, forcing $tL + \ell \equiv 0 \pmod m$. Since top and length classify indecomposable modules, these conditions are also sufficient. □

65 Closed Tachikawa theorem for the truncated-cycle family

Theorem 65.1 (UB-VC3-05: VC3 finite-window Tachikawa closure)**.** *Let $A_{m,L} = kQ_m/J^L$, and let $p_{m,L} = 2m/\gcd(m, L)$. For every finitely generated right $A_{m,L}$-module M, the following are equivalent:*

1. *M is projective;*
2. *$\mathrm{Ext}^{p_{m,L}}_{A_{m,L}}(M, M) = 0$;*
3. *$\mathrm{Ext}^{i}_{A_{m,L}}(M, M) = 0$ for $1 \leq i \leq p_{m,L}$;*
4. *$\mathrm{Ext}^{i}_{A_{m,L}}(M, M) = 0$ for every $i \geq 1$.*

Consequently Tachikawa's second conjecture is true for every self-injective truncated cycle algebra $A_{m,L}$, with an explicit one-degree certificate.

Proof. Projectivity implies all positive Ext groups vanish, so $(1) \Rightarrow (4) \Rightarrow (3) \Rightarrow (2)$. Uniform stable periodicity gives $\Omega^{p_{m,L}} M \cong M$ stably. If M were nonprojective, the stable identity would produce

$$0 \neq \underline{\mathrm{Hom}}_{A_{m,L}}(\Omega^{p_{m,L}} M, M) \cong \mathrm{Ext}^{p_{m,L}}_{A_{m,L}}(M, M),$$

contradicting (2). Hence $(2) \Rightarrow (1)$. □

Corollary 65.2 (Module-specific witness degree)**.** *Let M be nonprojective, and let $\pi(M)$ be the order of its finite Krull–Schmidt multiplicity vector under the syzygy type permutation. Then*

$$1 \le \pi(M) \mid p_{m,L}, \qquad \mathrm{Ext}^{\pi(M)}_{A_{m,L}}(M,M) \neq 0.$$

Thus a verifier may use the smaller module-specific orbit period instead of the uniform family bound.

66 Sharper local subfamily and explicit cyclic execution

Theorem 66.1 (UB-VC3-06: Local truncated-polynomial sharpening)**.** *Let $A_{1,L} = k[x]/(x^L)$. For every finitely generated $A_{1,L}$-module M,*

$$M \text{ is projective} \iff \mathrm{Ext}^1_{A_{1,L}}(M,M) = 0.$$

For the indecomposable nonprojective module $M_r = A_{1,L}/(x^r)$, $1 \le r < L$,

$$\dim_k \mathrm{Ext}^1_{A_{1,L}}(M_r, M_r) = \min(r, L-r) > 0.$$

Proof. A minimal projective resolution of M_r begins with alternating multiplication maps

$$\cdots \xrightarrow{x^r} A_{1,L} \xrightarrow{x^{L-r}} A_{1,L} \xrightarrow{x^r} A_{1,L} \twoheadrightarrow M_r.$$

After applying $\mathrm{Hom}_{A_{1,L}}(-, M_r)$, multiplication by x^r is zero, so

$$\mathrm{Ext}^1(M_r, M_r) \cong \ker(x^{L-r} : M_r \to M_r).$$

The kernel has dimension r when $L-r \ge r$, and dimension $L-r$ when $L-r < r$, giving $\min(r, L-r)$. Every finite module is a direct sum of copies of $A_{1,L}$ and the M_r. Ext is additive on finite direct sums, so any nonprojective summand contributes a nonzero diagonal $\mathrm{Ext}^1(M_r, M_r)$ summand. □

Example 66.2 (The algebra $A_{3,4}$)**.** Here $q_{3,4} = 3$ and $p_{3,4} = 6$. The syzygy permutation is

$$T(i,1) = (i+1,3), \qquad T(i,2) = (i+2,2), \qquad T(i,3) = (i,1).$$

The length-two types form three-cycles, while the length-one and length-three types lie in six-cycles. Hence every module satisfies $\Omega^6 M \cong M$ stably, and

$$\mathrm{Ext}^6_{A_{3,4}}(M,M) = 0 \iff M \text{ is projective.}$$

This is a complete finite execution of the family theorem rather than an uninstantiated finite-state assumption.

67 VC3 verifier contract and AK–HT Return

Verifier contract 67.1 (UB-VC3-07: Finite Tachikawa certificate)**.** A VC3 certificate contains:

1. m, L, the oriented-cycle quiver, and the relation $J^L = 0$;

2. the uniserial summand decomposition of the input module, with matrices verifying the displayed isomorphism;
3. the finite type set $\mathcal{I}_{m,L}$ and the permutation matrix of $T_{m,L}$;
4. $q_{m,L} = m/\gcd(m, L)$ and the check $T_{m,L}^{2q_{m,L}} = I$;
5. projective-cover and kernel matrices verifying the syzygy formula;
6. either the uniform period $p_{m,L}$ or the smaller computed $\pi(M)$;
7. a stable isomorphism $\Omega^{\pi(M)}M \to M$;
8. a finite linear-algebra check that this map is not in the subspace of maps factoring through $\bigoplus_i P_i$;
9. the standard dimension-shift identification with the corresponding nonzero Ext class;
10. the requested projectivity or nonvanishing Return statement.

Every verification block is finite. No infinite Ext scan is required.

Theorem 67.2 (VC3 UB–HT route closure)**.** *The VC3 proof route is*

$$\begin{aligned}\textit{finite quiver realization} &\longrightarrow \textit{Krull–Schmidt type extraction}\\ &\longrightarrow \textit{finite syzygy permutation} \longrightarrow \textit{periodic stable Return}\\ &\longrightarrow \textit{nonzero Ext witness} \longrightarrow \textit{projectivity decision}.\end{aligned}$$

It is a complete UB external Return route. HC may compress the periodic resolution, and PE may serialize the stable-Hom/Ext witness, but neither a spectral sequence nor an inverse-limit theorem is required. Therefore SS and RL are correctly marked NOT-INVOKED rather than being inserted for cosmetic coverage.

Proof. Finite realization and module decomposition are verified over the finite algebra. The syzygy formula produces a finite permutation whose exponent gives stable return. Stable nonzero identity is exactly converted to a positive-degree Ext class by dimension shifting. Its vanishing is therefore equivalent to projectivity at the declared degree. All arrows and all non-invoked components are explicitly typed. □

Theorem 67.3 (UB-VC3-08: UB-VC3 success-case closure)**.** *For every field k, every $m \geq 1$, and every $L \geq 2$, Appendix UB completely proves and finitely verifies the following external statement:*

> *For the self-injective Nakayama algebra $A_{m,L} = kQ_m/J^L$, every finitely generated module M is projective if and only if $\operatorname{Ext}^{2m/\gcd(m,L)}_{A_{m,L}}(M, M) = 0$.*

The proof includes the algebra's self-injectivity, complete indecomposable classification, exact syzygy action, uniform periodicity, the nonzero stable identity witness, finite verifier, and Return to projectivity. No unresolved hypothesis remains inside the declared family theorem.

Non-claim 67.4 (VC3 novelty and promotion boundary). VC3 does not assert external-literature novelty for the classification of self-injective Nakayama algebras, uniserial modules, or periodic syzygies. Its validated contribution is the explicit AK–UB closure, the one-degree certificate, and the reusable finite syzygy-permutation theorem. It does not claim that:

1. $2m/\gcd(m, L)$ is the smallest possible uniform Ext degree;
2. every self-injective algebra has finite stable representation type;
3. finite generation of Ext alone forces a finite syzygy permutation;
4. the general Tachikawa conjecture is solved;
5. a periodic objectwise isomorphism automatically gives an unrecorded natural isomorphism of the entire syzygy functor;
6. compilation or finite enumeration replaces the stable-Hom nonfactorization proof.

Part XIV

Validated Success Case IV: Periodic and Twisted-Periodic Tachikawa Return

68 VC4 scope, roadmap gate, and external target

68.1 Challenge target

The VC4 Challenge Track asks whether the finite-window Tachikawa closure of VC3 can be extended beyond representation-finite self-injective algebras. The target is not a new conditional statement of the form "if every module is periodic, then Tachikawa holds." The target is a typed bridge from a standard algebra-level certificate—periodicity of the regular bimodule—to a uniform one-degree self-Ext certificate for every finitely generated module. The external statement to be Returned is

$$M \text{ projective} \quad \Longleftrightarrow \quad \mathrm{Ext}^q_A(M, M) = 0$$

for one degree q fixed by the algebra and valid even when A has infinitely many indecomposable modules.

68.2 Roadmap gate result

The mathematical Challenge target closes: bimodule periodicity gives a natural stable period for all modules and therefore a nonzero self-Ext witness for every non-projective module. This genuinely leaves the representation-finite regime. However, the implication is a direct homological consequence of standard bimodule-periodicity theory, and VC4 does not claim that this consequence is absent from the literature. Under the Challenge/Stable policy, the novelty gate therefore does not promote VC4 to Grade A. The same proof and source bindings are retained as a fully closed Stable Grade-B success case.

Definition 68.1 (VC4 right-module twist)**.** Let A be a finite-dimensional k-algebra and let $\sigma \in \mathrm{Aut}_k(A)$. For a right A-module M, write M_σ for the same vector space with action

$$m * a := m\sigma(a).$$

For an A–A-bimodule X, write ${}_1X_\sigma$ when the left action is unchanged and the right action is twisted by σ.

Definition 68.2 (Bimodule periodicity certificate)**.** A p-*periodicity certificate* for A is a finite projective A^e-resolution segment

$$0 \longrightarrow A \xrightarrow{u} P_{p-1} \longrightarrow \cdots \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

whose leftmost copy of A identifies with $\Omega^p_{A^e}(A)$ as an A–A-bimodule. A (p,σ)-*twisted periodicity certificate* replaces the leftmost A by ${}_1A_\sigma$.

69 Bimodule syzygy transport

Lemma 69.1 (Split-left tensor transport)**.** *Let*

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow A \longrightarrow 0$$

be a projective A^e-resolution of A. Viewed as a complex of left A-modules, every short exact syzygy sequence splits. Hence, for every right A-module M,

$$M \otimes_A P_\bullet \longrightarrow M$$

is an exact projective right A-resolution.

Proof. Every A^e-projective is projective as a left A-module. The augmentation quotient A is left projective, so the first syzygy sequence splits as left modules and its kernel is left projective. Induction gives left projectivity of every bimodule syzygy and split exactness of every short syzygy sequence. Tensoring a split exact sequence with an arbitrary right module preserves exactness. Finally, an A^e-projective is a direct summand of a sum of copies of $A \otimes_k A$, and

$$M \otimes_A (A \otimes_k A) \cong M \otimes_k A$$

is free as a right A-module. Thus all terms after tensoring are projective. □

Theorem 69.2 (UB-VC4-01: stable bimodule-to-module syzygy transport)**.** *Let A be finite-dimensional and let $X = \Omega^p_{A^e}(A)$ be represented by a projective bimodule resolution. For every finitely generated right A-module M there is a natural stable isomorphism*

$$\Omega^p_A(M) \simeq_{\mathrm{st}} M \otimes_A X.$$

Consequently:

1. *if $X \cong A$, then $\Omega^p_A \cong \mathrm{Id}$ on $\underline{\mathrm{mod}}\, A$;*
2. *if $X \cong {}_1A_\sigma$, then $\Omega^p_A(M) \simeq_{\mathrm{st}} M_\sigma$ naturally in M.*

Proof. By the preceding lemma, tensoring the chosen bimodule resolution with M gives a projective resolution of M whose pth kernel is $M \otimes_A X$. Comparison with a minimal projective resolution and Schanuel's lemma gives the natural stable isomorphism. If $X \cong A$, then $M \otimes_A X \cong M$. If $X \cong {}_1A_\sigma$, the map

$$M \otimes_A {}_1A_\sigma \longrightarrow M_\sigma, \qquad m \otimes a \longmapsto ma,$$

is a right A-module isomorphism. □

Lemma 69.3 (Stable identity witness)**.** *Let A be self-injective and let M be finitely generated. Then*

$$\underline{\mathrm{End}}_A(M) \neq 0 \quad \Longleftrightarrow \quad M \textit{ is nonprojective.}$$

Proof. If M is projective, its identity factors through a projective and the stable endomorphism space vanishes. Conversely, if the stable identity vanishes, then id_M factors through a projective module. Hence M is a direct summand of a projective and is projective. □

70 Periodic one-degree Tachikawa closure

Theorem 70.1 (UB-VC4-02: periodic one-degree Tachikawa theorem)**.** *Let A be a finite-dimensional self-injective k-algebra and suppose*

$$\Omega^p_{A^e}(A) \cong A$$

as bimodules for some $p \geq 1$. Then, for every finitely generated right A-module M,

$$\boxed{M \textit{ is projective} \quad \Longleftrightarrow \quad \mathrm{Ext}^p_A(M, M) = 0.}$$

In particular, Tachikawa's second conjecture holds for A, and it is enough to inspect one self-Ext degree.

Proof. Projective modules have vanishing positive-degree Ext. Suppose M is non-projective. By UB-VC4-01,

$$\Omega^p_A(M) \simeq_{\mathrm{st}} M.$$

Since A is self-injective, dimension shifting gives

$$\mathrm{Ext}^p_A(M, M) \cong \underline{\mathrm{Hom}}_A(\Omega^p_A M, M) \cong \underline{\mathrm{End}}_A(M).$$

The last space contains the nonzero stable identity by the preceding lemma. Thus $\mathrm{Ext}^p_A(M, M) \neq 0$ for every nonprojective M. □

Corollary 70.2 (Uniform multiple-degree certificate)**.** *Under the hypotheses of UB-VC4-02, for every $r \geq 1$,*

$$M \textit{ projective} \quad \Longleftrightarrow \quad \mathrm{Ext}^{rp}_A(M, M) = 0.$$

Proof. Iterating the natural stable period gives $\Omega^{rp}_A M \simeq_{\mathrm{st}} M$. The same stable-identity argument applies. □

Remark 70.3 (Exact strength)*.* The theorem does not require finite representation type, a classification of modules, a finite list of syzygy states, or a Noetherian generation bound for the total Ext algebra. The finite certificate is the regular-bimodule period itself. This is strictly broader than the VC3 finite-syzygy-permutation route.

71 Twisted periodicity and finite twist orbits

Theorem 71.1 (UB-VC4-03: orbit-sensitive twisted Tachikawa theorem)**.** *Let A be a finite-dimensional self-injective k-algebra and suppose*

$$\Omega^p_{A^e}(A) \cong {}_1A_\sigma$$

for some $p \geq 1$ and $\sigma \in \mathrm{Aut}_k(A)$. Let M be finitely generated. If there is an $r \geq 1$ such that

$$M_{\sigma^r} \simeq_{\mathrm{st}} M,$$

then

$$M \text{ projective} \iff \mathrm{Ext}^{pr}_A(M, M) = 0.$$

Proof. Twisting is exact, preserves projectives, and commutes with stable syzygy. Iterating UB-VC4-01 gives

$$\Omega^{pr}_A(M) \simeq_{\mathrm{st}} M_{\sigma^r} \simeq_{\mathrm{st}} M.$$

The stable-identity proof of UB-VC4-02 now applies in degree pr. □

Corollary 71.2 (Finite outer-order uniformization)**.** *Under the hypotheses of UB-VC4-03, suppose the class of σ has finite order r in $\mathrm{Out}_k(A)$. Then every finitely generated module satisfies*

$$M \text{ projective} \iff \mathrm{Ext}^{pr}_A(M, M) = 0.$$

Proof. The automorphism σ^r is inner, so twisting by σ^r is naturally isomorphic to the identity functor on mod A. Apply UB-VC4-03. □

Corollary 71.3 (Finite twist-orbit subcategory)**.** *Even when $[\sigma]$ has infinite order in $\mathrm{Out}(A)$, the full stable subcategory of modules with finite σ-orbit satisfies Tachikawa's conclusion with a module-specific finite degree $pr(M)$.*

Remark 71.4 (Challenge extension beyond algebra periodicity). The orbit-sensitive theorem is useful even when a global untwisted period of A is not available. It isolates the exact additional datum needed by a module: finite return under the twisting automorphism. VC4 does not assert that every twisted-periodic algebra has finite outer twist order; that question is not silently solved here.

72 Representation-infinite external family closures

72.1 Weighted surface algebras

Theorem 72.1 (UB-VC4-04: weighted-surface degree-four closure)**.** *Let k be algebraically closed and let Λ be a weighted surface algebra which is not a singular tetrahedral algebra, in the standard sense of Erdmann–Skowroński. The classical bimodule-periodicity theorem gives*

$$\Omega^4_{\Lambda^e}(\Lambda) \cong \Lambda.$$

Therefore, for every finitely generated Λ-module M,

$$\boxed{M \text{ projective} \iff \mathrm{Ext}^4_\Lambda(M, M) = 0.}$$

This includes representation-infinite tame symmetric families.

Proof. The source theorem supplies symmetry, hence self-injectivity, and bimodule period four. UB-VC4-02 applies with $p = 4$. □

72.2 Algebras of generalized quaternion type

Theorem 72.2 (UB-VC4-05: generalized-quaternion degree-four closure)**.** *Let k be algebraically closed and let A be a basic, indecomposable, 2-regular algebra of generalized quaternion type in the standard Erdmann–Skowroński class. Then A is a representation-infinite tame symmetric periodic algebra of period four, and every finitely generated module satisfies*

$$\boxed{M \text{ projective} \iff \operatorname{Ext}^4_A(M, M) = 0.}$$

The same conclusion applies to the wild period-four algebras supplied by the corresponding source construction.

Proof. The classification theorem identifies the displayed 2-regular generalized-quaternion class with symmetric periodic algebras of period four. Its explicit corollary also supplies wild symmetric periodic algebras of period four. Apply UB-VC4-02 in both cases. □

72.3 Dynkin preprojective algebras

Theorem 72.3 (UB-VC4-06: uniform degree-six preprojective closure)**.** *Let $\Pi(\Delta)$ be a finite-dimensional preprojective algebra of Dynkin type over a field in the standard periodicity regime. Then $\Pi(\Delta)$ is self-injective and its bimodule period divides six. Consequently, for every finitely generated module M,*

$$\boxed{M \text{ projective} \iff \operatorname{Ext}^6_{\Pi(\Delta)}(M, M) = 0.}$$

Except for small representation-finite cases, this is a representation-infinite or wild family closure.

Proof. Choose the actual bimodule period d with $d \mid 6$. UB-VC4-02 gives stable period d for all modules, and the uniform multiple-degree corollary applies in degree six. □

Remark 72.4 (Why VC4 is a genuine structural evolution). VC3 required a complete finite list of stable indecomposable types and computed a syzygy permutation on that list. VC4 replaces that finite-list mechanism by one algebra-level bimodule certificate. It therefore covers tame and wild algebras with infinitely many indecomposable modules. The proof still ends in the same rigid external Return: non-projective ⇒ nonzero stable identity ⇒ nonzero self-Ext.

73 Finite verifier and AK–HT route

Verifier contract 73.1 (VC4 periodicity-to-Ext certificate)**.** A replayable VC4 certificate contains:

1. a finite presentation of A and a declaration of right-module variance;
2. a self-injectivity certificate or a source theorem locator proving self-injectivity;
3. projective A^e-bimodules $P_0, \dots, P_{p-1}$ and differentials;
4. exactness checks for the finite bimodule resolution segment;
5. an explicit bimodule isomorphism $\Omega^p_{A^e}(A) \cong A$ or $\Omega^p_{A^e}(A) \cong {}_1A_\sigma$;
6. in the twisted case, either an outer-order certificate for σ or a module-specific stable isomorphism $M_{\sigma^r} \simeq_{\mathrm{st}} M$;
7. the tensor-transport matrices for the requested module presentation;
8. the stable identity witness and the dimension-shifting identification with $\mathrm{Ext}^q_A(M, M)$;
9. the final projective/nonprojective Return and its declared degree q.

Theorem 73.2 (UB-VC4-07: finite replay soundness)**.** *Every accepted VC4 certificate proves the declared one-degree projectivity criterion. The verifier terminates after finite linear-algebra, module-map, exactness, and bimodule-map checks.*

Proof. All presented modules and maps are finite-dimensional. Exactness is checked by ranks and kernel-image equality. Bimodule compatibility is checked on algebra generators. The syzygy transport is then certified by UB-VC4-01, and the final Ext witness by UB-VC4-02 or UB-VC4-03. No infinite module classification or all-degree Ext computation is required. □

Theorem 73.3 (UB-VC4-08: typed AK route)**.** *The validated VC4 route is*

$$\begin{aligned}
\textit{finite algebra presentation} &\longrightarrow \textit{bimodule periodicity certificate},\\
\textit{bimodule periodicity certificate} &\longrightarrow \textit{stable functorial period},\\
\textit{stable functorial period} &\longrightarrow \textit{stable identity witness},\\
\textit{stable identity witness} &\longrightarrow \textit{one-degree self-Ext witness},\\
\textit{one-degree self-Ext witness} &\longrightarrow \textit{projectivity Return}.
\end{aligned}$$

The route uses UB finite realization, UB rigidity, and the HT Return grammar. It does not invoke spectral-sequence or inverse-limit layers because no filtration or infinite tower is present.

74 VC4 closure, status, and non-claims

Classification 74.1 (Challenge/Stable status)**.** *VC4 receives the following status:*

Gate	*Result*	*Registration*
representation-infinite extension	*closed*	*mathematical Challenge success*
external-literature novelty	*not established*	*no Grade-A promotion*
full periodic-family theorem	*closed*	*Stable Grade B*
finite verifier	*closed*	*validated*
Return to Tachikawa conclusion	*closed*	*exact family strength*

Thus the roadmap fallback has been executed without discarding the Challenge work.

Non-claim 74.2 (VC4 boundary). VC4 does not claim that:

1. every self-injective algebra is periodic or twisted periodic;
2. every twisted-periodic algebra has a finite-order twist in $\mathrm{Out}(A)$;
3. a periodicity certificate of degree p is always minimal;
4. $\mathrm{Ext}^n_A(M, M) = 0$ for one arbitrary degree n implies projectivity;
5. Dynkin preprojective algebras all have minimal period exactly six;
6. the general Tachikawa second conjecture is solved;
7. the periodicity theorems for weighted surface, quaternion-type, or preprojective algebras are newly proved by AK;
8. the integrated one-degree corollaries are asserted to be literature-first results.

Theorem 74.3 (UB-VC4-09: UB-VC4 success-case closure)**.** *Appendix UB completely closes the following external family statement:*

> *For every finite-dimensional self-injective bimodule-periodic algebra A of period p, a finitely generated module M is projective if and only if $\mathrm{Ext}^p_A(M, M) = 0$. The statement extends to twisted-periodic modules with finite twist orbit in the corresponding multiplied degree. In particular, degree four detects projectivity on the displayed weighted-surface and generalized-quaternion families, and degree six is a uniform detector on Dynkin preprojective algebras.*

No representation-finiteness assumption or hidden all-degree Ext hypothesis remains.

Proof. UB-VC4-01 transports the regular-bimodule syzygy to every module. UB-VC4-02 and UB-VC4-03 convert stable return into a nonzero self-Ext witness for every nonprojective module. The external source bindings supply the declared periods and self-injectivity of the named families. UB-VC4-07 supplies finite replay, and the non-claim records the exact promotion ceiling. □

VC4 source theorem bindings

The source regime used by VC4 is explicitly limited to the following classical-strength inputs.

- Periodic algebras are self-injective, and bimodule periodicity induces module periodicity; see the periodic-algebra literature and the synthesis of Chan–Darpö–Iyama–Marczinzik, *Periodic dimensions and some homological properties of eventually periodic algebras*.
- Preprojective algebras of Dynkin type are periodic with periods dividing six; see the periodic-resolution literature summarized by Erdmann–Skowroński and Dugas.

- Weighted surface algebras other than the singular tetrahedral cases, and their declared socle deformations, are symmetric periodic algebras of period four; see Erdmann–Skowroński, *Weighted surface algebras*.
- Basic indecomposable 2-regular algebras of generalized quaternion type are representation-infinite tame symmetric periodic algebras of period four, and the same source produces wild symmetric period-four examples; see Erdmann–Skowroński, *Algebras of generalized quaternion type*.

The UB contribution in VC4 is the typed transport, exact one-degree Return, finite certificate contract, cross-family unification, and roadmap status discipline. No stronger reading of the source theorems is imported.

Part XV

Validated Success Case V: Finite Galois–Leopoldt Defect Return

75 VC5 scope, Challenge gate, and arithmetic target

75.1 Challenge target

The VC5 Challenge Track asks for a non-abelian arithmetic bridge in which the external object is not merely represented but is reconstructed from finitely many proper-subfield probes. Let L/K be a finite Galois extension with group G and let p be a prime. The target is to recover the Leopoldt kernel of L as a $\mathbb{Q}_p[G]$-module from Leopoldt kernels of selected intermediate fields, to derive an exact defect formula, and to prove a minimal zero-certificate in an explicit non-abelian group.

The first declared group is S_3. If

$$G = \mathrm{Gal}(L/K) \cong S_3,$$

write

$$F_2 = L^{A_3}, \qquad F_3 = L^{C_2},$$

where C_2 is any transposition subgroup. The requested external Return is stronger than an implication between conjectures: it is the exact reconstruction of the $\mathbb{Q}_p[S_3]$-module $\mathcal{L}(L,p)$ from the three integers

$$\delta(K,p), \qquad \delta(F_2,p), \qquad \delta(F_3,p).$$

75.2 Roadmap gate result

The mathematical Challenge target closes. The subgroup-fixed Leopoldt kernels are exactly the Leopoldt kernels of the fixed fields, and the three irreducible $\mathbb{Q}_p[S_3]$-types are separated by the fixed-point probes G, A_3, C_2. The resulting integral fixed-point matrix has determinant one. Consequently the full Leopoldt kernel, not only its dimension, is reconstructed by a finite formula. The two proper-subfield probes

A_3 and C_2 are jointly sufficient and atomically minimal for universal zero detection among subgroup-fixed probes.

The Frobenius-group defect relation and the S_3 propagation theorem are classical-strength results explicitly present in the source literature. VC5 therefore does not claim a literature-first solution of a new Leopoldt case. Under the Challenge/Stable policy, the new arithmetic bridge execution, exact module reconstruction, minimal certificate, and finite verifier are registered as a fully closed Stable Grade-B success case.

76 Leopoldt kernels as finite Galois modules

Definition 76.1 (Leopoldt map, kernel, and defect)**.** Let F be a number field and p a prime. Let $S_p(F)$ be the set of places of F above p and let $U^1_{F_v}$ be the principal local unit group. The rationalized Leopoldt map is

$$\Lambda_{F,p} : \mathbb{Q}_p \otimes_{\mathbb{Z}} \mathcal{O}_F^\times \longrightarrow \mathbb{Q}_p \otimes_{\mathbb{Z}_p} \prod_{v \in S_p(F)} U^1_{F_v}.$$

Define

$$\mathcal{L}(F,p) := \ker \Lambda_{F,p}, \qquad \delta(F,p) := \dim_{\mathbb{Q}_p} \mathcal{L}(F,p).$$

Leopoldt's conjecture for (F,p) is the assertion $\mathcal{L}(F,p) = 0$, equivalently $\delta(F,p) = 0$.

Lemma 76.2 (Galois equivariance)**.** *If L/K is finite Galois with group G, then $\Lambda_{L,p}$ is a map of $\mathbb{Q}_p[G]$-modules and $\mathcal{L}(L,p)$ is a finite-dimensional $\mathbb{Q}_p[G]$-module.*

Proof. The diagonal embedding of global units into the product of local units is equivariant under permutation of the places above p and the natural local Galois actions. Tensoring with $\mathbb{Q}_p$ preserves the action and the kernel is G-stable. Finite dimensionality follows from finite generation of $\mathcal{O}_L^\times$. □

Theorem 76.3 (UB-VC5-01: exact fixed-field descent for Leopoldt kernels)**.** *Let L/K be finite Galois with group G and let $H \le G$. Under the natural embeddings there is a canonical equality*

$$\boxed{\mathcal{L}(L,p)^H = \mathcal{L}(L^H,p).}$$

Consequently,

$$\delta(L^H,p) = \dim_{\mathbb{Q}_p} \mathcal{L}(L,p)^H.$$

Proof. The H-fixed subspace of $\mathbb{Q}_p \otimes_{\mathbb{Z}} \mathcal{O}_L^\times$ is canonically $\mathbb{Q}_p \otimes_{\mathbb{Z}} \mathcal{O}_{L^H}^\times$: the quotient between fixed global units and units of the fixed field is finite and is killed by tensoring with $\mathbb{Q}_p$. The restriction of the local diagonal map to this fixed subspace is the Leopoldt map of L^H. Since fixed vectors commute with taking the kernel of an equivariant map,

$$(\ker \Lambda_{L,p})^H = \ker(\Lambda_{L,p}|_{(\mathbb{Q}_p \otimes \mathcal{O}_L^\times)^H}) = \ker \Lambda_{L^H,p}.$$

□

Remark 76.4 (Source binding). UB-VC5-01 is the exact strength of Ferri–Johnston, Lemma 2.3, in *Applications of representation theory and of explicit units to Leopoldt's conjecture*. It is reproduced here because it is the arithmetic descent arrow used by every subsequent VC5 Return.

77 Finite fixed-point probe matrices

Definition 77.1 (Fixed-point probe matrix)**.** Let E be a characteristic-zero splitting field for a finite group G, let $W_1, \dots, W_r$ be representatives of the irreducible $E[G]$-modules, and let $H_1, \dots, H_s \le G$. Define

$$A_{ij} := \dim_E W_j^{H_i}.$$

For a semisimple module $V \cong \bigoplus_{j=1}^r W_j^{\oplus m_j}$, define the probe vector

$$d(V) := (\dim_E V^{H_i})_{i=1}^s.$$

Then $d(V) = Am$, where $m = (m_j)_j$ is the multiplicity vector.

Theorem 77.2 (UB-VC5-02: finite Galois defect reconstruction criterion)**.** *In the preceding setting:*

1. *if A has full column rank, the multiplicity vector of every semisimple $E[G]$-module is uniquely determined by the fixed-point probe vector;*
2. *if A has an integral left inverse, integral multiplicities are reconstructed by a finite exact integer calculation;*
3. *the selected probes detect zero modules if and only if every irreducible column of A contains a positive entry;*
4. *if a column is identically zero, the corresponding nonzero irreducible module is an explicit counterexample to zero reflection.*

Applied to $V = E \otimes_{\mathbb{Q}_p} \mathcal{L}(L, p)$ and combined with UB-VC5-01, this gives a finite intermediate-field reconstruction criterion for the Leopoldt kernel.

Proof. Semisimplicity follows from Maschke's theorem. Additivity of fixed-point dimensions gives $d = Am$. Full column rank gives uniqueness over the coefficient field, and an integral left inverse gives exact integer recovery. Since all entries and all multiplicities are nonnegative, $d = 0$ forces $m_j = 0$ for every j exactly when every column is seen by at least one row. A zero column gives the stated irreducible counterexample. □

Remark 77.3 (UB content)*.* The theorem separates three strengths which must not be conflated: zero detection needs only column coverage; numerical defect reconstruction needs enough linear rank; full semisimple module reconstruction needs a splitting field and the complete multiplicity vector. VC5 records the exact strength used at each Return.

78 Exact S_3 Leopoldt-kernel reconstruction

78.1 The fixed-point matrix

Over every $\mathbb{Q}_p$, the group S_3 has three absolutely irreducible modules:

$$\mathbf{1}, \qquad \varepsilon, \qquad \rho,$$

the trivial, sign, and two-dimensional standard modules. Their fixed-space dimensions are

H	$\mathbf{1}$	ε	ρ
S_3	1	0	0
A_3	1	1	0
C_2	1	0	1
1	1	1	2

where all transposition subgroups are conjugate.

Theorem 78.1 (UB-VC5-03: exact S_3 Leopoldt-module Return)**.** *Let L/K be an S_3-Galois extension, let $F_2 = L^{A_3}$ and let $F_3 = L^{C_2}$ for any transposition subgroup. Set*

$$d_0 = \delta(K, p), \qquad d_2 = \delta(F_2, p), \qquad d_3 = \delta(F_3, p).$$

Then the Leopoldt kernel is determined, as a $\mathbb{Q}_p[S_3]$-module, by

$$\boxed{\mathcal{L}(L, p) \cong \mathbf{1}^{\oplus d_0} \oplus \varepsilon^{\oplus(d_2-d_0)} \oplus \rho^{\oplus(d_3-d_0)}.}$$

In particular,

$$\boxed{\delta(L, p) = d_2 + 2d_3 - 2d_0.}$$

The consistency inequalities

$$d_2 \geq d_0, \qquad d_3 \geq d_0$$

are necessary for every valid arithmetic certificate.

Proof. Write

$$\mathcal{L}(L, p) \cong \mathbf{1}^{\oplus a} \oplus \varepsilon^{\oplus b} \oplus \rho^{\oplus c}.$$

By UB-VC5-01 and the displayed fixed-point table,

$$d_0 = a, \qquad d_2 = a + b, \qquad d_3 = a + c.$$

The coefficient matrix

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

has determinant one and inverse

$$B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}.$$

Thus $a = d_0$, $b = d_2 - d_0$, and $c = d_3 - d_0$. Taking total dimension gives

$$\delta(L, p) = a + b + 2c = d_2 + 2d_3 - 2d_0.$$

Nonnegativity of b and c gives the consistency inequalities. □

Corollary 78.2 (Exact defect relation)**.** *For every S_3-Galois extension L/K,*

$$\boxed{\delta(L, p) + 2\delta(K, p) = \delta(F_2, p) + 2\delta(F_3, p).}$$

Remark 78.3 (Strength beyond a scalar relation). The scalar defect relation follows by taking dimensions, but UB-VC5-03 retains the full semisimple Galois-module type. In particular, it identifies separately the trivial, sign, and standard defect multiplicities. This is the VC5 enriched Return analogue of the object-versus-scalar distinction established in VC1 and VC2.

79 Minimal proper-subfield zero certificates

Theorem 79.1 (UB-VC5-04: minimal S_3 two-subfield Leopoldt certificate)**.** *With the notation of UB-VC5-03,*

$$\boxed{\mathcal{L}(L,p)=0 \quad \Longleftrightarrow \quad \mathcal{L}(F_2,p)=0 \text{ and } \mathcal{L}(F_3,p)=0.}$$

Equivalently, Leopoldt's conjecture holds for L if and only if it holds for one quadratic and one cubic intermediate field.

Among universal zero tests formed only from fixed spaces of proper nontrivial subgroup types of S_3, two probes are necessary and sufficient. Up to conjugacy, the unique minimal pair is $\{A_3, C_2\}$.

Proof. If the top kernel vanishes, all fixed subspaces vanish. Conversely, $d_2 = d_3 = 0$ gives

$$a+b=0, \qquad a+c=0.$$

Since a, b, c are nonnegative, all vanish.

For minimality, the proper nontrivial subgroup types are A_3 and C_2; the base-field probe S_3 may also be considered but does not change the conclusion. The standard module ρ has no A_3-fixed vector, so the A_3 probe alone is not zero-reflecting. The sign module ε has no C_2-fixed vector, so the C_2 probe alone is not zero-reflecting. The S_3 probe sees only the trivial module. The pair $\{A_3, C_2\}$ covers all three irreducible columns and is therefore sufficient by UB-VC5-02. Conjugate transposition subgroups give the same probe type. □

Warning 79.2 (Meaning of minimality). The minimality statement is representation-theoretic: it is minimal among universal subgroup-fixed zero probes on the class of all semisimple $\mathbb{Q}_p[S_3]$-modules. It does not assert that every abstract multiplicity vector occurs as an actual Leopoldt kernel, nor that no different arithmetic invariant could certify Leopoldt with fewer data.

Corollary 79.3 (One-cubic-field Return over a known base)**.** *Suppose K is $\mathbb{Q}$ or an imaginary quadratic field. Then Leopoldt's conjecture is known for K and for the quadratic extension F_2/K. Consequently,*

$$\boxed{\mathrm{Leo}(L,p) \Longleftrightarrow \mathrm{Leo}(F_3,p)}$$

and

$$\boxed{\mathcal{L}(L,p) \cong \rho^{\oplus \delta(F_3,p)}, \qquad \delta(L,p) = 2\delta(F_3,p).}$$

For $K=\mathbb{Q}$, this applies to every S_3-extension and every prime p.

Proof. The known abelian cases give $d_0 = d_2 = 0$. Substitute these values in UB-VC5-03 and UB-VC5-04. □

80 Frobenius-group defect propagation

Lemma 80.1 (Frobenius permutation-module relation)**.** *Let $G = N \rtimes H$ be a finite Frobenius group with kernel N and complement H. Over every characteristic-zero field there is an equality in the representation ring*

$$[E[G/1]] + |H|[E[G/G]] = [E[G/N]] + |H|[E[G/H]].$$

Equivalently, the corresponding permutation characters agree.

Proof. Evaluate the characters on three types of elements. At the identity both sides have value $|G|+|H|$. A nonidentity element of N fixes every coset in G/N and no coset in G/H, giving $|H|$ on both sides. Every element outside N lies in a unique conjugate of a complement; it fixes no coset in G/N and exactly one coset in G/H, again giving $|H|$ on both sides. Equality of characteristic-zero characters gives the representation-ring identity. □

Theorem 80.2 (UB-VC5-05: exact Frobenius Leopoldt defect relation)**.** *Let L/K be finite Galois with $G = \mathrm{Gal}(L/K) = N \rtimes H$ a Frobenius group. Then, for every prime p,*

$$\boxed{\delta(L,p) + |H|\delta(K,p) = \delta(L^N,p) + |H|\delta(L^H,p).}$$

Moreover,

$$\boxed{\mathrm{Leo}(L,p) \iff \mathrm{Leo}(L^N,p) \text{ and } \mathrm{Leo}(L^H,p).}$$

Proof. Apply $\mathrm{Hom}_{E[G]}(-, E \otimes_{\mathbb{Q}_p} \mathcal{L}(L,p))$ to the permutation-module relation and take dimensions. Frobenius reciprocity identifies

$$\dim \mathrm{Hom}_{E[G]}(E[G/J], V) = \dim V^J.$$

UB-VC5-01 then identifies the four fixed-space dimensions with the four displayed Leopoldt defects. If the two proper-subfield kernels vanish, the dimension relation has nonnegative left-hand terms and forces $\delta(L,p) = \delta(K,p) = 0$; in particular the top kernel vanishes. The converse follows from fixed-field descent. □

Remark 80.3 (Classical-strength binding)*.* The defect formula is the Frobenius specialisation of the Brauer-relation theorem of Ferri–Johnston. VC5 supplies a self-contained permutation-character proof, types it as a finite UB certificate, and composes it with exact fixed-field descent and minimality data.

Corollary 80.4 (Odd-dihedral and S_3 one-subfield certificate)**.** *Let q be an odd prime and let L/K be Galois with*

$$\mathrm{Gal}(L/K) \cong D_{2q} = C_q \rtimes C_2.$$

Put $F_2 = L^{C_q}$ and let $F_q = L^{C_2}$ be any degree-q intermediate field. Then

$$\delta(L,p) + 2\delta(K,p) = \delta(F_2,p) + 2\delta(F_q,p),$$

and Leopoldt for L is equivalent to Leopoldt for F_2 and F_q. If $K = \mathbb{Q}$ or an imaginary quadratic field, then

$$\boxed{\mathrm{Leo}(L,p) \iff \mathrm{Leo}(F_q,p), \qquad \delta(L,p) = 2\delta(F_q,p).}$$

The case $q = 3$ is the S_3 certificate.

81 Finite verifier and AK–HT route

Verifier contract 81.1 (VC5 finite Galois–Leopoldt certificate)**.** A replayable VC5 certificate contains:

1. a finite group multiplication table or a certified presentation of G;
2. explicit subgroup generators for every probe subgroup and conjugacy-class data;
3. the fixed fields or exact source locators identifying them;
4. the rational irreducible character table, or permutation characters sufficient for the declared relation;
5. the fixed-point matrix $A_{ij} = \dim W_j^{H_i}$;
6. a rank, integral-inverse, or column-coverage calculation matching the requested Return strength;
7. subfield Leopoldt certificates, each supplied either by an accepted external theorem locator or by an independent arithmetic nonvanishing certificate;
8. the UB-VC5-01 descent identifiers;
9. reconstructed multiplicities, consistency inequalities, and the final top-field defect or zero Return;
10. a status declaration distinguishing propagation from new regulator nonvanishing.

Theorem 81.2 (UB-VC5-06: finite replay soundness)**.** *Every accepted VC5 certificate proves the declared Galois-module reconstruction, defect identity, or Leopoldt propagation statement. The representation-theoretic verifier terminates after finite group, character, integer-matrix, and nonnegativity checks.*

Proof. A finite group table permits exact subgroup, conjugacy, coset-action, and character calculations. Fixed-point dimensions are finite character averages. Matrix inversion and column coverage are finite rational or integer operations. UB-VC5-01 supplies the exact arithmetic interpretation of every fixed-space probe. The only arithmetic leaf inputs are the declared subfield Leopoldt certificates; the verifier does not replace them by numerical hope or finite-precision evidence. □

Theorem 81.3 (UB-VC5-07: typed arithmetic route)**.** *The validated VC5 route is*

$$\begin{aligned}
\textit{finite Galois extension} &\longrightarrow \mathbb{Q}_p[G]\textit{-Leopoldt kernel},\\
\mathbb{Q}_p[G]\textit{-Leopoldt kernel} &\longrightarrow \textit{subgroup-fixed probes},\\
\textit{subgroup-fixed probes} &\longrightarrow \textit{intermediate-field kernels},\\
\textit{finite character/Brauer matrix} &\longrightarrow \textit{module or defect reconstruction},\\
\textit{subfield zero certificates} &\longrightarrow \textit{top-field Leopoldt Return}.
\end{aligned}$$

The route invokes UB finite realization, coefficient discipline, descent, finite-generator reconstruction, and rigidity, together with the HT/IW Return grammar. It does not invoke HC, PE, spectral-sequence, or inverse-limit layers because no compression, filtration, or infinite tower is required in this finite Galois step.

82 VC5 status, closure, and non-claims

Classification 82.1 (Challenge/Stable status)**.** *VC5 receives the following status:*

Gate	*Result*	*Registration*
non-abelian arithmetic bridge	*closed*	*mathematical Challenge success*
full S_3 kernel reconstruction	*closed*	*exact enriched Return*
minimal proper-subfield zero probe	*closed*	*representation-theoretic Min-Cert*
new regulator nonvanishing	*not produced*	*general Leopoldt remains open*
external-literature novelty	*not established*	*no Grade-A promotion*
finite propagation verifier	*closed*	*Stable Grade B*

The Challenge mathematics is retained, while the novelty and general-Leopoldt gates enforce the Stable classification.

Non-claim 82.2 (VC5 boundary). VC5 does not claim that:

1. Leopoldt's conjecture is proved for arbitrary number fields;
2. subgroup representation theory proves the nonvanishing of an unknown p-adic regulator in a primitive field;
3. a finite amount of p-adic numerical precision certifies a determinant without a rigorous valuation or exact-nonvanishing bound;
4. every abstract semisimple $\mathbb{Q}_p[G]$-module occurs as a Leopoldt kernel;
5. the S_3 minimality theorem is minimal among all possible arithmetic invariants;
6. the Frobenius defect relation or the S_3 cubic-field equivalence is newly discovered in this appendix;
7. the choice of one cubic subfield suffices over an arbitrary base field without the quadratic and base-field Leopoldt inputs;
8. the finite Galois step settles any infinite Iwasawa tower or its $\varprojlim^1$ defect.

Theorem 82.3 (UB-VC5-08: UB-VC5 success-case closure)**.** *Appendix UB completely closes the following external arithmetic statements:*

1. *for every S_3-Galois extension L/K, the three subfield defects $\delta(K,p)$, $\delta(L^{A_3},p)$, and $\delta(L^{C_2},p)$ reconstruct the full $\mathbb{Q}_p[S_3]$-isomorphism class of $\mathcal{L}(L,p)$ by an explicit unimodular formula;*
2. *one quadratic and one cubic intermediate-field kernel form a necessary-and-sufficient universal proper-subfield zero certificate, minimal among subgroup-fixed probe types;*
3. *every finite Frobenius extension satisfies the exact displayed defect relation and a two-subfield Leopoldt propagation theorem;*
4. *over $\mathbb{Q}$ or an imaginary quadratic base, an odd-dihedral Galois closure is Leopoldt at p exactly when one degree-q complement-fixed subfield is Leopoldt, with top defect twice the subfield defect.*

All representation-theoretic and Return arrows are finite and explicit. The remaining primitive arithmetic obligation is exactly the Leopoldt certificate for the selected non-Galois subfield; it is not hidden inside the bridge.

Proof. UB-VC5-01 identifies subfield kernels with fixed spaces. UB-VC5-02 converts fixed-space data into a finite semisimple reconstruction problem. UB-VC5-03 computes the unimodular S_3 inverse and exact defect formula. UB-VC5-04 proves zero reflection and atomic minimality. UB-VC5-05 proves the Frobenius relation and propagation theorem, while the odd-dihedral corollary supplies the declared family. UB-VC5-06 gives finite replay and the non-claim fixes the arithmetic promotion ceiling. □

VC5 source theorem bindings

The source regime used by VC5 is explicitly limited to the following classical-strength inputs.

- Fabio Ferri and Henri Johnston, *Applications of representation theory and of explicit units to Leopoldt's conjecture*, especially Lemma 2.3 for fixed-field kernels, Theorem 1.10 for Brauer defect relations, Corollary 1.11 for Frobenius groups, and Corollary 4.27 for the $S_3/\mathbb{Q}$ cubic-field equivalence; preprint `https://arxiv.org/abs/2301.05700`.
- Alex Bartel and Tim Dokchitser, *Brauer relations in finite groups*, for the representation-ring framework behind permutation and Brauer relations; `https://arxiv.org/abs/1103.2047`.
- The classical Ax–Brumer theorem for Leopoldt's conjecture in finite abelian extensions of $\mathbb{Q}$ and of imaginary quadratic fields, used only in the one-subfield corollaries and imported at exactly that strength.

The UB contribution in VC5 is the typed fixed-point probe matrix, the exact $\mathbb{Q}_p[S_3]$-module reconstruction, the subgroup-probe MinCert, the finite verifier, and the explicit separation between propagation and primitive regulator nonvanishing. No stronger reading of the source theorems is imported.

Part XVI

Validated Success Case VI: Finite Object and Derived Descent Return

83 VC6 scope, Challenge gate, and descent target

83.1 Challenge target

The sixth validation cycle tests whether the abstract descent component of UB can be executed at object, morphism, exact, Yoneda, and derived strength rather than only at the level of global sections. The declared source class is a finite T_0 Alexandrov space, equivalently a finite poset, equipped with a finite open cover. Local objects are finite- dimensional sheaves of k-vector spaces or bounded complexes thereof, overlap maps are actual isomorphisms or chain isomorphisms, and triple-overlap coherence is part of the certificate.

The requested external conclusion is

$$\begin{aligned}&\text{finite local objects} + \text{ overlap isomorphisms} + \text{ triple cocycle}\\ &\quad\Longrightarrow \text{global object, morphisms, exact sequences, and Yoneda classes,}\\ &\qquad\qquad\qquad \text{together with bounded-derived and hypercohomology Return}.\end{aligned}$$

The cycle is not allowed to replace actual cocycle data by pairwise numerical invariants, and it is not allowed to identify naive descent data in derived categories with effective strict descent without chosen coherent representatives.

83.2 Roadmap gate result

The Challenge target closes on finite Alexandrov spaces at strict object and bounded- complex level. The proof is completely finite: a sheaf is a representation of the specialization poset, the global object is reconstructed pointwise from local representations and overlap matrices, and all exact and derived structure is transported by an exact equivalence. For principal-good covers, every nonempty finite intersection has a least point, so the Čech totalization computes derived global sections for every bounded complex without an additional acyclicity oracle.

The broader problem of gluing perfect complexes from isomorphisms supplied only inside an arbitrary derived category, with all higher homotopy coherence implicit, is not claimed. Because the constituent sheaf-gluing, Alexandrov-representation, and Čech arguments are classical-strength, VC6 is registered as a *Stable Grade-B* closure. Its AK contribution is the exact finite certificate, the strict-to-derived Return chain, the coherence-depth theorem, the enriched-probe composition, and the replay verifier.

Definition 83.1 (Finite Alexandrov descent instance)**.** A finite Alexandrov descent instance is a tuple

$$\mathfrak{D} = (P, k, \mathcal{U}, \mathbf{M}, \theta)$$

where:

1. P is a finite poset and X_P is the associated T_0 Alexandrov space whose opens are upward-closed subsets;
2. k is a field;
3. $\mathcal{U} = (P_i)_{i \in I}$ is a finite cover of P by upward-closed subposets;
4. M_i is a finite-dimensional representation of P_i;
5. for $P_{ij} = P_i \cap P_j$, the map
$$\theta_{ij} : M_i|_{P_{ij}} \xrightarrow{\sim} M_j|_{P_{ij}}$$
is a representation isomorphism, with $\theta_{ii} = \mathrm{Id}$ and $\theta_{ji} = \theta_{ij}^{-1}$;
6. on every triple intersection P_{ijk},
$$\theta_{jk} \circ \theta_{ij} = \theta_{ik}.$$

The category of such data and compatible local morphisms is denoted $\mathrm{Desc}(P, \mathcal{U}; k)$.

84 Finite Alexandrov realization

For $x \in P$, write
$$U_x = \{y \in P : x \le y\}.$$
This is the smallest open neighbourhood of x in X_P.

Theorem 84.1 (UB-VC6-01: exact Alexandrov sheaf–representation equivalence)**.** *There is an exact equivalence of abelian categories*

$$\mathrm{Sh}_k(X_P) \simeq \mathrm{Rep}_k(P),$$

where $\mathrm{Sh}_k(X_P)$ *denotes sheaves of finite-dimensional* k*-vector spaces and* $\mathrm{Rep}_k(P) = \mathrm{Fun}(P, \mathrm{Vect}_k^{\mathrm{fd}})$*. Under this equivalence, the value of the representation at* x *is the section space on the minimal open* U_x*.*

Proof. For a sheaf $\mathcal{F}$, define
$$\Phi(\mathcal{F})_x = \mathcal{F}(U_x).$$
If $x \le y$, then $U_y \subseteq U_x$, so restriction gives $\Phi(\mathcal{F})_x \to \Phi(\mathcal{F})_y$. This defines a representation.

Conversely, for a representation M, define

$$\Psi(M)(U) = \left\{ (m_x)_{x\in U} \in \prod_{x\in U} M_x : M(x \le y)m_x = m_y \text{ for every } x \le y \text{ in } U \right\}.$$

Restriction deletes components. Compatible families glue uniquely, so $\Psi(M)$ is a sheaf. On U_x, every compatible family is uniquely determined by its x-component; hence evaluation gives $\Psi(M)(U_x) \cong M_x$. The two constructions are mutually inverse on objects and morphisms. Kernels and cokernels of sheaves on a finite Alexandrov space, and of poset representations, are computed pointwise, so the equivalence is exact. □

Corollary 84.2 (Finite matrix realization)**.** *Every object, morphism, kernel, cokernel, short exact sequence, and bounded complex in* $\mathrm{Sh}_k(X_P)$ *has a finite incidence-matrix realization. The enriched projective-probe Return of VC2 therefore applies after Alexandrov realization.*

85 Effective object and morphism descent

Construction 85.1 (Explicit gluing operator)**.** Fix once and for all a choice function
$$c : P \longrightarrow I, \qquad x \in P_{c(x)}.$$
For a descent datum (M_i, θ_{ij}), define a representation $\mathrm{Gl}(M_i, \theta_{ij})$ by

$$M_x = (M_{c(x)})_x.$$

For $x \le y$, the openness of $P_{c(x)}$ implies $y \in P_{c(x)}$. Put

$$M(x \le y) = (\theta_{c(x),c(y)})_y \circ M_{c(x)}(x \le y).$$

The formula is a finite matrix operation.

Theorem 85.2 (UB-VC6-02: effective finite object descent)**.** *Restriction and explicit gluing define inverse exact equivalences*

$$\mathrm{Rep}_k(P) \xrightarrow{\sim} \mathrm{Desc}(P, \mathcal{U}; k).$$

Consequently:

1. *every certified finite descent datum has a global representation;*
2. *the global representation is unique up to a unique isomorphism compatible with the local identifications;*
3. *every compatible family of local morphisms Returns a unique global morphism;*
4. *monomorphisms, epimorphisms, kernels, cokernels, isomorphisms, and short exact sequences are preserved and reflected.*

Proof. Let $x \le y \le z$, and abbreviate $\alpha = c(x)$, $\beta = c(y)$, and $\gamma = c(z)$. Naturality of $\theta_{\alpha\beta}$ on the overlap and the triple cocycle give

$$\begin{aligned} M(y \le z)M(x \le y) &= (\theta_{\beta\gamma})_z M_\beta(y \le z)(\theta_{\alpha\beta})_y M_\alpha(x \le y) \\ &= (\theta_{\beta\gamma})_z(\theta_{\alpha\beta})_z M_\alpha(y \le z)M_\alpha(x \le y) \\ &= (\theta_{\alpha\gamma})_z M_\alpha(x \le z) = M(x \le z). \end{aligned}$$

Thus Gl is a representation. For $x \in P_i$, the maps $(\theta_{c(x),i})_x$ define a natural isomorphism $\mathrm{Gl}(M)|_{P_i} \cong M_i$; the cocycle gives compatibility on overlaps. Conversely, gluing the restrictions of a global representation returns it canonically. The same formulas glue compatible local morphisms. Pointwise exactness then shows that the equivalence is exact and preserves and reflects all listed structures. □

Theorem 85.3 (UB-VC6-03: coherence depth two is sufficient and sharp)**.** *For ordinary sheaves or strict complexes on a finite cover, pairwise overlap isomorphisms plus the triple-overlap cocycle are sufficient for all higher compatibility. No independent quadruple- or higher-overlap axiom is required.*

If a cover has a nonempty triple intersection and the local object has a nontrivial automorphism, the triple cocycle cannot in general be omitted. Hence coherence depth two is sharp in the declared one-categorical source class.

Proof. Sufficiency is contained in the construction and proof of UB-VC6-02: any composite of transition maps is reduced by repeated use of the triple cocycle, and the result is independent of the chosen chain of cover indices.

For sharpness, let P consist of one point and cover it three times by $P_1 = P_2 = P_3 = P$. Let every local object be a two-dimensional vector space V. Choose a nonidentity $g \in \mathrm{Aut}_k(V)$ and set

$$\theta_{12} = \mathrm{Id}, \qquad \theta_{23} = \mathrm{Id}, \qquad \theta_{13} = g.$$

All pairwise maps are isomorphisms, but $\theta_{23}\theta_{12} = \mathrm{Id} \ne g = \theta_{13}$. Therefore the data do not descend. □

86 Exact, Yoneda, and bounded-derived descent

Definition 86.1 (Strict bounded-complex descent category)**.** Let $\mathrm{Desc}_{\mathrm{Ch}^b}(P, \mathcal{U}; k)$ be the category whose objects are bounded complexes $K_i^\bullet$ in $\mathrm{Rep}_k(P_i)$ together with chain isomorphisms

$$\theta_{ij} : K_i^\bullet|_{P_{ij}} \xrightarrow{\sim} K_j^\bullet|_{P_{ij}}$$

satisfying the strict triple cocycle, and whose morphisms are compatible families of chain maps.

Theorem 86.2 (UB-VC6-04: exact and bounded-derived descent)**.** *Degreewise restriction and gluing give an exact equivalence*

$$\mathrm{Ch}^b(\mathrm{Rep}_k(P)) \xrightarrow{\sim} \mathrm{Desc}_{\mathrm{Ch}^b}(P, \mathcal{U}; k).$$

Chain homotopies, cones, and quasi-isomorphisms are preserved and reflected. After localization one obtains an equivalence

$$\mathrm{D^b}(\mathrm{Rep}_k(P)) \xrightarrow{\sim} \mathrm{D}^{\mathrm{b}}_{\mathrm{strDesc}}(P, \mathcal{U}; k),$$

where the right side is the localization of the strict descent-complex category at componentwise quasi-isomorphisms.

Proof. Apply UB-VC6-02 in every cohomological degree. Compatibility of the local differentials with the overlap maps makes the glued differential well-defined, and $d^2 = 0$ is checked locally. The inverse equivalence is again restriction. Every construction involved in a chain homotopy or mapping cone is additive and degreewise, so it is preserved and reflected. Cohomology is computed by pointwise kernels and cokernels, hence a chain map is a quasi-isomorphism exactly when all its local restrictions are. Localization of an exact equivalence at corresponding classes of morphisms yields the displayed derived equivalence. □

Corollary 86.3 (UB-VC6-05: all-degree Yoneda-class descent)**.** *For global representations M, N and every $n \geq 0$, restriction induces an isomorphism*

$$\mathrm{Ext}^n_{\mathrm{Rep}_k(P)}(M, N) \xrightarrow{\sim} \mathrm{Ext}^n_{\mathrm{Desc}(P,\mathcal{U};k)}(M|_{\mathcal{U}}, N|_{\mathcal{U}}),$$

and the isomorphism Returns individual Yoneda extension classes, Baer sums, connecting maps, and bounded derived triangles.

Proof. UB-VC6-02 is an exact equivalence of abelian categories. Exact equivalences preserve projective resolutions and Yoneda n-extensions up to the defining equivalence relation. Alternatively, use UB-VC6-04 and the standard identification $\mathrm{Ext}^n(M, N) \cong \mathrm{Hom}_{\mathrm{D^b}}(M, N[n])$. Both constructions preserve the represented class, not merely its dimension. □

Theorem 86.4 (Local enriched probes compose with descent)**.** *For each i, let*

$$G_i = \bigoplus_{x \in P_i} k[\mathrm{Hom}_{P_i}(x, -)]$$

be the VC2 enriched projective probe. The finite data consisting of

$$\mathrm{RHom}(G_i, K_i^\bullet), \quad \mathrm{RHom}(G_{ij}, \theta_{ij}), \quad \textit{and the triple cocycle matrices}$$

Returns the unique global bounded complex, every global chain map, every exact triangle, and every Yoneda class. No additional object-level oracle is required.

Proof. VC2 reconstructs every local complex and every overlap chain map from the enriched probe data. UB-VC6-04 then glues the reconstructed strict descent datum and Returns its global derived object. UB-VC6-05 Returns the individual extension classes. Each arrow is exact or fully faithful at the strength used, so the composite does not inflate Return strength. □

87 Principal-good covers and finite hypercohomology Return

Definition 87.1 (Principal-good finite cover)**.** A finite open cover $\mathcal{U} = (P_i)_{i\in I}$ of P is principal-good if every nonempty finite intersection

$$P_J = \bigcap_{j\in J} P_j$$

has a least element x_J.

Lemma 87.2 (Acyclicity of a least-point open)**.** *If an upward-closed subposet $Q \subseteq P$ has a least element q_0, then for every sheaf $\mathcal{F}$ of k-vector spaces on X_Q,*

$$\Gamma(Q, \mathcal{F}) \cong \mathcal{F}(U_{q_0})$$

and $\Gamma(Q, -)$ is exact. Hence

$$H^r(Q, \mathcal{F}) = 0 \qquad (r > 0).$$

Proof. A section over $Q = U_{q_0}$ is uniquely determined by its component at q_0, with every other component obtained by the representation map $q_0 \le y$. Under UB-VC6-01 this is evaluation at one vertex, which is exact in a functor category. Therefore all higher right-derived functors vanish. □

Theorem 87.3 (UB-VC6-06: finite Čech hypercohomology Return)**.** *Let $\mathcal{U}$ be a principal-good finite cover and let $K^\bullet$ be a bounded complex of sheaves of finite-dimensional k-vector spaces. Form the finite Čech double complex*

$$C^{p,q}(\mathcal{U}, K) = \prod_{i_0<\cdots<i_p} \Gamma(P_{i_0\cdots i_p}, K^q),$$

with the alternating Čech differential and the internal differential of $K^\bullet$. The canonical augmentation is a quasi-isomorphism

$$R\Gamma(P, K^\bullet) \xrightarrow{\sim} \operatorname{Tot} C^{\bullet,\bullet}(\mathcal{U}, K).$$

Thus every hypercohomology group, connecting map, and induced filtration is computable from finite local matrices.

Proof. Every nonempty intersection is acyclic for every sheaf by the preceding lemma. Apply a functorial injective or flabby resolution to $K^\bullet$ and form the augmented Čech double complex. The spectral sequence obtained by first taking sheaf cohomology on intersections has no positive intersection-cohomology rows, so the augmentation is a quasi-isomorphism. All index sets, complexes, and products are finite, so totalization uses finite direct sums and is a finite matrix complex. The standard filtration of the total complex Returns the induced hypercohomology filtration and its connecting maps. □

Corollary 87.4 (Finite nerve bound)**.** *If no point of P belongs to more than μ cover members, then the Čech degree is at most $\mu - 1$. Therefore the derived global-section certificate has a finite, explicitly known horizontal amplitude.*

88 Finite verifier, complexity bound, and explicit execution

Verifier contract 88.1 (UB-VC6 finite descent certificate)**.** A valid VC6 certificate serializes:

1. the finite poset relation and the upward-closed cover subsets P_i;
2. every local object or bounded complex as finite incidence matrices;
3. every overlap isomorphism as invertible matrices at overlap vertices;
4. local naturality or chain-map equations on overlap relations;
5. every triple-cocycle matrix equation;
6. the choice function $c(x)$ and the reconstructed global matrices;
7. for hypercohomology Return, a least-point witness for every nonempty cover intersection and the finite Čech totalization;
8. the requested object, morphism, exact, Yoneda, cone, or hypercohomology Return strength.

Theorem 88.2 (UB-VC6-07: finite replay soundness)**.** *The VC6 verifier accepts exactly strict finite Alexandrov descent certificates satisfying the displayed hypotheses. On acceptance, its reconstructed global representation or bounded complex is independent, up to the canonical compatible isomorphism, of the choice function c. Every accepted exact, Yoneda, cone, and hypercohomology output has the claimed external meaning.*

Proof. Upward-closedness, local functoriality, invertibility, naturality, and triple cocycles are finite matrix equalities. UB-VC6-02 proves that their acceptance is sufficient for unique global object and morphism Return. UB-VC6-04 and UB-VC6-05 prove the exact, cone, and Yoneda outputs. Least-point witnesses invoke UB-VC6-06. Two choice functions produce restrictions isomorphic to the same local descent datum, so uniqueness in UB-VC6-02 gives the canonical comparison. □

Proposition 88.3 (Crude polynomial replay bound)**.** *Assume every local vector space has dimension at most d, every bounded complex has at most a nonzero degrees, and let R, O_2, and O_3 be respectively the total numbers of local order relations, overlap vertices, and triple-overlap vertices represented in the certificate. Dense-matrix replay can be performed using*

$$O(ad^3(R + O_2 + O_3))$$

field operations, apart from the final cohomology row reductions, which are polynomial in the total Čech-complex dimension.

Proof. Each functoriality, chain-map, invertibility, and cocycle check uses a bounded number of matrix products of size at most d. The gluing formulas use the same scale. Čech assembly is linear in the number of represented intersections, and finite-dimensional cohomology is computed by Gaussian elimination. □

Example 88.4 (Two-open V-poset execution)**.** Let $P = \{a, b, c\}$ with $a < c$ and $b < c$, and cover it by

$$P_1 = \{a, c\}, \qquad P_2 = \{b, c\}.$$

The intersection is $\{c\}$. A local representation is

$$M_1 = (A \xrightarrow{f} C_1), \qquad M_2 = (B \xrightarrow{g} C_2),$$

together with an isomorphism $\theta : C_1 \xrightarrow{\sim} C_2$. VC6 glues it to

$$A \xrightarrow{\theta f} C_2 \xleftarrow{g} B.$$

A compatible pair of local morphisms glues exactly when its two maps at c satisfy the single overlap square. Short exact sequences and bounded complexes glue degreewise, and the cover is principal-good because P_1, P_2, and $P_1 \cap P_2$ have least elements a, b, and c respectively. Hence the two-column Čech totalization computes all hypercohomology.

89 VC6 status, closure, and non-claims

Classification 89.1 (VC6 roadmap status)**.** *VC6 is classified as follows:*

finite object and morphism descent	*PASS*
exact and all-degree Yoneda descent	*PASS*
strict bounded-derived descent	*PASS*
principal-good Čech hypercohomology Return	*PASS*
finite enriched-probe composition	*PASS*
sharp triple-coherence requirement	*PASS*
arbitrary higher-stack or infinity-categorical descent	*NOT-INVOKED*
external literature-first novelty	*not asserted*
roadmap registration	*Stable Grade B*

Non-claim 89.2 (VC6 promotion ceiling)*.* VC6 does not assert:

1. that pairwise overlap dimensions or isomorphism classes descend without actual transition maps and a triple cocycle;
2. that isomorphisms supplied only in local derived categories admit strict coherent representatives;
3. that the naive equalizer of local derived categories is the derived category of the global space;
4. that arbitrary infinite covers have a finite certificate;
5. that a non-principal-good cover computes hypercohomology without a separate acyclicity or convergence theorem;
6. that object-level descent reconstructs multiplicative, monoidal, or higher A_∞ structure unless those coherence cells are included;
7. that the classical sheaf and Čech constituents are literature-first results of VC6.

Theorem 89.3 (UB-VC6-08: UB-VC6 success-case closure)**.** *Appendix UB completely closes the following finite descent statement. For every finite poset P, every finite open cover $\mathcal{U}$, and every finite-dimensional strict sheaf or bounded-complex descent datum on $\mathcal{U}$, local objects, actual overlap isomorphisms, and the triple cocycle reconstruct a unique global object and every compatible morphism. The reconstruction preserves and reflects kernels, cokernels, short exact sequences, individual Yoneda classes in all degrees, mapping cones, quasi-isomorphisms, and bounded derived triangles. If the cover is principal-good, one finite Čech totalization also Returns the complete derived global-section object and hypercohomology. The entire route is replayable from finite incidence and overlap matrices.*

Proof. UB-VC6-01 gives finite Alexandrov realization. UB-VC6-02 proves effective exact descent, and UB-VC6-03 proves the exact coherence depth and its necessity. UB-VC6-04 and UB-VC6-05 Return complexes, triangles, and all Yoneda classes. UB-VC6-06 closes finite hypercohomology on principal-good covers. UB-VC6-07 supplies deterministic replay and choice independence. The non-claim fixes the boundary against higher-coherence and non-acyclic promotion. □

VC6 source theorem bindings

The classical-strength source regime used by VC6 is limited to:

- the equivalence between sheaves on a finite Alexandrov space and diagrams on its specialization poset;
- ordinary effective gluing of sheaves from overlap isomorphisms satisfying the cocycle condition;
- exact-equivalence invariance of Yoneda extensions and bounded derived categories;
- the Čech-to-derived comparison under acyclicity of all finite intersections.

Standard reference families include texts on finite topological spaces and Alexandrov spaces, Mac Lane–Moerdijk style sheaf theory, Weibel-style homological algebra, and the Stacks Project treatments of sheaf and perfect-complex descent. The UB contribution is the typed finite certificate, explicit choice-function gluing formula, coherence-depth boundary, exact VC2 enriched-probe composition, finite verifier, and single-route object-to-derived Return. No stronger source theorem is imported.

Part XVII

Validated Success Case VII: Finite-State Inverse Limits and Derived Apex Return

90 VC7 objective and Challenge-to-Stable gate

90.1 Roadmap target

VC7 addresses the first genuinely infinite transfer step in the validated UB sequence. The input is a countable inverse system

$$V_0 \xleftarrow{d_0} V_1 \xleftarrow{d_1} V_2 \xleftarrow{d_2} \cdots$$

or a tower of bounded complexes, while the output is an exact finite certificate for the ordinary inverse limit, the derived defect $\varprojlim^1$, and an external apex comparison. The required distinction is:

$$\text{a long finite prefix} \quad \not\Rightarrow \quad \text{an inverse-limit theorem},$$

unless a certified tail mechanism is supplied.

Classification 90.1 (VC7 Challenge gate)**.** *The Challenge lane attempted to apply the VC5 finite Galois–Leopoldt data directly to the natural towers of finite p-power quotients. That lane triggers* C-STOP-2 *and* C-STOP-3*: the quotient sizes grow, the typed transition data do not repeat in a finite state space, and VC5 supplies no valuation bound or arithmetic control theorem forcing a finite tail. Consequently VC7 does not promote finite-layer Leopoldt computations to primitive regulator nonvanishing. The same data are instead routed to the Stable lane below, where the tail hypothesis is explicit, finite, and independently replayable.*

90.2 Stable target

The Stable target is stronger than eventual constancy and weaker than an arbitrary tower theorem. A tower is encoded by a finite deterministic state machine whose states carry finite-dimensional vector spaces and whose transitions carry exact bonding matrices. Every such tower is eventually periodic, but the bonding maps need not be isomorphisms, injective, or surjective. VC7 proves:

1. a finite normal form for its inverse limit;
2. automatic vanishing of $\varprojlim^1$;
3. an explicit stable-image formula through one period monodromy;
4. a finite criterion for acyclicity and quasi-isomorphism of derived inverse limits;
5. a finite apex-agreement certificate;
6. deterministic replay and a sharp non-claim boundary.

91 Countable towers and the Roos model

Definition 91.1 (Linear inverse tower)**.** Let k be a field. A linear inverse tower is a sequence of finite-dimensional k-vector spaces and linear bonding maps

$$\mathbb{V} = (V_n, d_n)_{n\geq 0}, \qquad d_n : V_{n+1} \longrightarrow V_n.$$

For $m > n$, write

$$d_{n,m} = d_n d_{n+1} \cdots d_{m-1} : V_m \longrightarrow V_n.$$

Definition 91.2 (Roos operator)**.** The Roos operator of $\mathbb{V}$ is

$$\Delta_{\mathbb{V}} : \prod_{n\geq 0} V_n \longrightarrow \prod_{n\geq 0} V_n, \qquad \Delta_{\mathbb{V}}((x_n)_n) = (x_n - d_n x_{n+1})_n.$$

Define

$$\varprojlim V_n := \ker \Delta_{\mathbb{V}}, \qquad \varprojlim{}^1 V_n := \operatorname{coker} \Delta_{\mathbb{V}}.$$

For countable towers of vector spaces this agrees with the ordinary inverse limit and its first right-derived functor.

Definition 91.3 (Mittag–Leffler condition)**.** The tower $\mathbb{V}$ is Mittag–Leffler if, for every n, the descending chain

$$\operatorname{im}(d_{n,n+1}) \supseteq \operatorname{im}(d_{n,n+2}) \supseteq \operatorname{im}(d_{n,n+3}) \supseteq \cdots$$

stabilizes.

Lemma 91.4 (UB-VC7-01: finite-dimensional Mittag–Leffler)**.** *Every linear inverse tower is Mittag–Leffler. Consequently,*

$$\varprojlim{}^1 V_n = 0.$$

Proof. For fixed n, the displayed images form a descending chain of subspaces of the finite-dimensional space V_n. Its dimension can strictly decrease at most $\dim_k V_n$ times, so the chain stabilizes. The standard countable Roos theorem implies that the Mittag–Leffler condition makes $\Delta_{\mathbb{V}}$ surjective, hence its cokernel vanishes. Equivalently, one may solve the equations $y_n = x_n - d_n x_{n+1}$ recursively after choosing compatible lifts inside the stabilized image chains. □

Warning 91.5 (No uniform finite cutoff without a tail certificate). The preceding proof is pointwise in n. It does not provide one finite integer that certifies stabilization simultaneously at all levels of an arbitrary tower. A finite verifier therefore requires additional finite-state or surjective-tail data.

92 Finite-state towers and periodic tail normal form

Definition 92.1 (Typed finite-state linear tower)**.** A typed finite-state linear tower certificate consists of:

1. a finite state set S;
2. for every $s \in S$, a finite-dimensional based vector space V_s;
3. a deterministic transition $\tau : S \to S$;

4. for every $s \in S$, a matrix
$$D_s : V_{\tau(s)} \longrightarrow V_s;$$
5. an initial state $s_0 \in S$.

It generates the tower

$$s_{n+1} = \tau(s_n), \qquad V_n = V_{s_n}, \qquad d_n = D_{s_n}.$$

State equality includes equality of dimensions, chosen bases, and transition matrices; it is not merely equality of numerical invariants.

Lemma 92.2 (Finite entry and period)**.** *For every typed finite-state tower there exist integers N, r satisfying*

$$0 \le N < |S|, \qquad 1 \le r \le |S|, \qquad s_{N+r} = s_N,$$

and

$$s_{n+r} = s_n \quad (n \ge N).$$

The least such pair is found by the first repeated-state algorithm.

Proof. Among $s_0, \dots, s_{|S|}$, two states coincide. Determinism then forces equality of all subsequent state sequences with the corresponding shift. □

Definition 92.3 (Period monodromy and stable image)**.** Fix a certified entry-period pair (N, r). The period monodromy at the entry phase is

$$T = d_N d_{N+1} \cdots d_{N+r-1} : V_N \longrightarrow V_N.$$

Let

$$W = \mathrm{im}(T^e), \qquad e = \dim_k V_N.$$

The space W is called the stable image of the periodic tail.

Lemma 92.4 (Stable-image automorphism)**.** *The chain*

$$V_N \supseteq \mathrm{im}\, T \supseteq \mathrm{im}\, T^2 \supseteq \cdots$$

stabilizes by exponent e, and the restriction

$$T|_W : W \xrightarrow{\sim} W$$

is an automorphism.

Proof. Dimension can drop at most e times. Once $\mathrm{im}\, T^e = \mathrm{im}\, T^{e+1}$, the restriction of T to W is surjective, hence bijective because W is finite-dimensional. □

Theorem 92.5 (UB-VC7-02: exact finite-state inverse-limit Return)**.** *For a typed finite-state tower with entry N, period r, monodromy T, and stable image $W = \mathrm{im}\, T^e$, evaluation at level N induces a natural isomorphism*

$$\mathrm{ev}_N : \varprojlim V_n \xrightarrow{\sim} W.$$

Moreover,

$$\varprojlim{}^1 V_n = 0.$$

Both conclusions are determined by the finite state table and the single matrix T.

Proof. The second statement is UB-VC7-01. For the first, a compatible sequence $(x_n)_n$ satisfies

$$x_N = T^q x_{N+qr}$$

for every q, hence $x_N \in \bigcap_q \operatorname{im} T^q = W$. Thus evaluation lands in W.

Conversely, let $w \in W$. Since $T|_W$ is invertible, define

$$x_{N+qr} = T^{-q} w \qquad (q \geq 0).$$

For $0 < j < r$, define

$$x_{N+qr+j} = d_{N+qr+j} \cdots d_{N+(q+1)r-1} x_{N+(q+1)r},$$

and for $n < N$, define

$$x_n = d_{n,N} w.$$

These formulas satisfy $x_n = d_n x_{n+1}$ at every level. They are forced by compatibility and by invertibility of $T|_W$, so the construction is unique and inverse to evaluation. □

Corollary 92.6 (Finite dimension and zero criteria)**.** *Under the hypotheses of UB-VC7-02,*

$$\dim_k \varprojlim V_n = \operatorname{rank}(T^e).$$

Furthermore,

$$\varprojlim V_n = 0 \iff T^e = 0 \iff T \textit{ is nilpotent}.$$

Corollary 92.7 (Eventual-isomorphism case)**.** *If every bonding map on the periodic tail is an isomorphism, then $W = V_N$ and*

$$\varprojlim V_n \cong V_N.$$

VC7 therefore strictly extends eventual constancy and eventual isomorphism.

Example 92.8 (Non-isomorphic periodic tail)**.** Let $V_n = k^2$ and let every bonding map be

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The tower is constant as a typed state but not an isomorphism tower. VC7 gives

$$\varprojlim V_n \cong k \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \varprojlim{}^1 V_n = 0.$$

The disappearing coordinate is rejected by the stable-image calculation rather than by an unjustified eventual-constancy claim.

93 Finite-state apex agreement

Definition 93.1 (Apex comparison)**.** Let A be a finite-dimensional vector space. An apex comparison is a compatible family

$$\alpha_n : A \longrightarrow V_n, \qquad \alpha_n = d_n \alpha_{n+1}.$$

It induces

$$\alpha : A \longrightarrow \varprojlim V_n.$$

Theorem 93.2 (UB-VC7-03: finite apex-agreement criterion)**.** *In the setting of UB-VC7-02, the apex comparison α is an isomorphism if and only if*

$$\alpha_N(A) \subseteq W \quad \textit{and} \quad \alpha_N : A \xrightarrow{\sim} W.$$

Thus apex agreement is verified by one finite matrix at the periodic entry level.

Proof. Under the natural isomorphism $\mathrm{ev}_N : \varprojlim V_n \to W$, the composite $\mathrm{ev}_N \alpha$ is exactly α_N. The conclusion follows. □

Corollary 93.3 (Finite injectivity and surjectivity Return)**.** *The apex map is injective, respectively surjective, exactly when $\alpha_N : A \to W$ is injective, respectively surjective. No conclusion is drawn from agreement with a finite prefix outside the stable image.*

94 Bounded complexes and derived inverse limits

Definition 94.1 (Finite-state tower of bounded complexes)**.** A typed finite-state tower of bounded complexes assigns to every state s:

1. a bounded finite-dimensional cochain complex $C_s^\bullet$;
2. a chain map
$$D_s : C^\bullet_{\tau(s)} \longrightarrow C_s^\bullet;$$
3. chosen bases in every degree.

The generated tower is denoted $\mathbb{C} = (C_n^\bullet, D_n)$.

Definition 94.2 (Derived Roos object)**.** The derived inverse limit of $\mathbb{C}$ is represented by the shifted cone

$$\mathbf{R}\varprojlim C_n^\bullet := \mathrm{Cone}\left(\prod_n C_n^\bullet \xrightarrow{1-\mathrm{shift}(D)} \prod_n C_n^\bullet\right)[-1].$$

For every degree q, there is a Milnor exact sequence

$$0 \longrightarrow \varprojlim{}^1 H^{q-1}(C_n^\bullet) \longrightarrow H^q(\mathbf{R}\varprojlim C_n^\bullet) \longrightarrow \varprojlim H^q(C_n^\bullet) \longrightarrow 0.$$

Theorem 94.3 (UB-VC7-04: finite-state derived-limit Return)**.** *Let $\mathbb{C}$ be a typed finite-state tower of bounded finite-dimensional complexes. Fix an eventual period (N, r). For every q, let*

$$T_q : H^q(C_N^\bullet) \longrightarrow H^q(C_N^\bullet)$$

be the cohomological monodromy of one period, and set

$$W_q = \mathrm{im}(T_q^{e_q}), \qquad e_q = \dim_k H^q(C_N^\bullet).$$

Then

$$\varprojlim{}^1 H^q(C_n^\bullet) = 0$$

and the Milnor map is a natural isomorphism

$$H^q(\mathbf{R}\varprojlim C_n^\bullet) \xrightarrow{\sim} \varprojlim H^q(C_n^\bullet) \xrightarrow{\sim} W_q.$$

Hence the complete cohomology of the derived inverse limit is returned by finitely many stable-image matrices.

Proof. The state repetition at the chain level induces the same eventual period on every cohomology tower. Each cohomology group is finite-dimensional, so UB-VC7-01 gives vanishing of the $\varprojlim^1$ term. The Milnor sequence therefore collapses to an isomorphism. UB-VC7-02 applied to the cohomology tower identifies its inverse limit with W_q. □

Corollary 94.4 (Finite acyclicity criterion). *The derived inverse limit is acyclic if and only if*

$$T_q^{e_q} = 0$$

for every cohomological degree q. Equivalently, every cohomological period monodromy is nilpotent.

Corollary 94.5 (Finite quasi-isomorphism criterion). *Let $f_\bullet : \mathbb{C} \to \mathbb{D}$ be a morphism of typed finite-state towers sharing a common certified tail period. Let $K_n^\bullet = \mathrm{Cone}(f_n)$. Then*

$$\mathbf{R}\varprojlim f_n : \mathbf{R}\varprojlim C_n^\bullet \longrightarrow \mathbf{R}\varprojlim D_n^\bullet$$

is a quasi-isomorphism if and only if every stable image of the period monodromy on $H^q(K_N^\bullet)$ is zero.

Proof. The derived-limit map is a quasi-isomorphism exactly when the derived inverse limit of the cone tower is acyclic. Apply the preceding corollary. □

95 Periodic endomorphism towers

Theorem 95.1 (UB-VC7-05: one-endomorphism cohomology Return). *Let $C^\bullet$ be a bounded finite-dimensional complex and let*

$$f : C^\bullet \longrightarrow C^\bullet$$

be a chain endomorphism. For the constant-form tower

$$C^\bullet \xleftarrow{f} C^\bullet \xleftarrow{f} C^\bullet \xleftarrow{f} \cdots,$$

one has, naturally for every q,

$$H^q(\mathbf{R}\varprojlim_f C^\bullet) \cong \mathrm{im}((f^*)^{e_q} : H^q(C^\bullet) \to H^q(C^\bullet)),$$

where $e_q = \dim_k H^q(C^\bullet)$.

Proof. This is UB-VC7-04 with $N = 0$, $r = 1$, and monodromy $T_q = f^*$. □

Corollary 95.2 (Fitting decomposition interpretation). *Write*

$$H^q(C^\bullet) = H^q_{\mathrm{inv}} \oplus H^q_{\mathrm{nil}}$$

for the Fitting decomposition of f^*, where f^* is invertible on the first summand and nilpotent on the second. Then

$$H^q(\mathbf{R}\varprojlim_f C^\bullet) \cong H^q_{\mathrm{inv}}.$$

Thus the derived inverse limit keeps exactly the recurrent cohomological sector and rejects the transient nilpotent sector.

Example 95.3 (Finite dynamical cohomology certificate)**.** Let X be a finite CW complex, let $g : X \to X$ be cellular, and work over a field k. A finite cellular cochain matrix for g^* determines

$$H^q\Big(\mathbf{R}\varprojlim(C^\bullet(X;k), g^*)\Big)$$

by row reduction, cohomology computation, and one stable-image power. No infinite sequence of cochains is stored in the certificate.

96 Surjective towers and completion lane

Theorem 96.1 (UB-VC7-06: surjective-tail derived exactness)**.** *Let (M_n, d_n) be a countable inverse system of modules in which every bonding map is surjective from some level onward. Then*

$$\varprojlim{}^1 M_n = 0.$$

For a tower of bounded complexes whose cohomology bonding maps are eventually surjective in every degree,

$$H^q(\mathbf{R}\varprojlim C_n^\bullet) \cong \varprojlim H^q(C_n^\bullet).$$

Proof. An eventually surjective tower is Mittag–Leffler. Apply the Roos and Milnor exact sequences. □

Corollary 96.2 (Adic completion Return with external completeness)**.** *Let R be a ring, $I \subseteq R$ an ideal, and M an I-adically complete and separated module. Then the quotient tower is surjective and the canonical map is an isomorphism*

$$M \xrightarrow{\sim} \varprojlim_n M/I^nM, \qquad \varprojlim_n{}^1 M/I^nM = 0.$$

The completeness and separatedness hypotheses are external domain inputs; VC7 does not infer them from a finite quotient prefix.

Remark 96.3 (Relation to arithmetic towers)*.* This corollary is the correct UB–RL interface for completed unit, Selmer, and Iwasawa modules once a domain theorem proves completeness, separation, and control compatibility. VC7 supplies the derived-limit transport, not the missing arithmetic control theorem.

97 VC7 verifier and complexity

Verifier contract 97.1 (UB-VC7 finite inverse-limit certificate)**.** A valid finite-state certificate serializes:

1. the finite state set S, transition table τ, and initial state;
2. each based vector space or bounded complex attached to a state;
3. every bonding matrix and the equations asserting it is a chain map;
4. the first repeated-state pair $(N, N + r)$;
5. the period monodromy matrix T, or the induced cohomology matrices T_q;

6. stable-image bases for $\operatorname{im} T^e$ or $\operatorname{im} T_q^{e_q}$;
7. optional apex matrices α_N;
8. optional cone-tower matrices for a quasi-isomorphism claim;
9. the declared Return strength and all external completeness hypotheses.

Theorem 97.2 (UB-VC7-07: finite replay soundness)**.** *The VC7 verifier accepts exactly the declared finite-state inverse-limit, derived-limit, apex, zero, and quasi-isomorphism claims when the serialized matrix identities and stable-image conditions hold. Acceptance is sufficient for the corresponding mathematical conclusion.*

Proof. The repeated-state check certifies the entire infinite tail by determinism. Matrix multiplication constructs the period monodromy. Gaussian elimination verifies $W = \operatorname{im} T^e$, invariance, and invertibility of $T|_W$. UB-VC7-02 Returns the ordinary limit and kills $\varprojlim^1$. Chain-map verification and finite cohomology computation reduce the derived claims to UB-VC7-04. The apex and cone claims are respectively UB-VC7-03 and the finite quasi-isomorphism corollary. No verifier step extrapolates from an uncertified finite prefix. □

Remark 97.3 (Complexity bound). Let $s = |S|$, let a be the number of nonzero cochain degrees, and let d bound every vector-space dimension. Cycle detection costs $O(s)$ state comparisons. Monodromy and stable-image replay use at most $O(sd^3)$ field operations in the linear case and $O(asd^3)$ in the bounded-complex case, apart from coefficient bit complexity. The certificate size is finite and polynomial in the serialized state matrices.

98 AK–HT composition and exact status

Theorem 98.1 (UB-VC7-08: typed UB–RL–Return route)**.** *A VC7 certificate yields the typed route*

$$\begin{aligned} \textit{finite state presentation} &\xrightarrow{\text{UB-}\textit{STATE}} (N, r, T) \\ &\xrightarrow{\text{UB-}\textit{NOETH/Fitting}} W = \operatorname{im} T^e \\ &\xrightarrow{\text{HT-}\textit{RL}} (\varprojlim, \varprojlim^1 = 0) \\ &\xrightarrow{\text{UB-}\textit{RIG/apex}} \textit{external Return}. \end{aligned}$$

For bounded complexes the middle output is

$$H^q(\mathbf{R}\varprojlim C_n^\bullet) \cong \operatorname{im} T_q^{e_q}.$$

Every arrow is independently typed and replayable.

Proof. Finite-state determinism certifies the tail. Fitting stabilization computes the recurrent sector. UB-VC7-01 and UB-VC7-04 discharge the ordinary and derived inverse-limit defects. UB-VC7-03 or an externally supplied completeness comparison performs the final Return. The route never identifies finite prefix agreement with apex agreement. □

99 VC7 status, closure, and non-claims

Classification 99.1 (VC7 roadmap status)**.** *VC7 is classified as follows:*

Target	*Status*
Finite-state ordinary inverse limit	*PASS*
Finite-state $\varprojlim^1$ *vanishing*	*PASS*
Finite monodromy/stable-image formula	*PASS*
Finite apex-agreement criterion	*PASS*
Derived-limit cohomology Return	*PASS*
Acyclicity and quasi-isomorphism Return	*PASS*
Surjective-tail and completion interface	*PASS*
Primitive arithmetic tail theorem	*NOT-INVOKED*
General arbitrary-tower finite reduction	*REJECT*
Roadmap registration	*Stable Grade B*

Non-claim 99.2 (VC7 promotion ceiling)*.* VC7 does not assert:

1. that a long finite prefix determines an arbitrary inverse limit;
2. that bounded dimensions alone give one globally replayable cutoff without a finite-state or surjective-tail certificate;
3. that finite-state behavior of dimensions, ranks, or isomorphism classes replaces equality of the typed bonding data;
4. that $\varprojlim^1$ vanishes for arbitrary inverse systems outside a displayed Mittag–Leffler regime;
5. that cohomology commutes with inverse limit without the Milnor defect calculation;
6. that finite quotient data prove adic completeness, separatedness, Iwasawa control, or Leopoldt regulator nonvanishing;
7. that the Stable Grade-B theorem is a literature-first result rather than a finite AK–UB normal form of classical Roos, Mittag–Leffler, Fitting, and Milnor mechanisms.

Theorem 99.3 (UB-VC7-09: UB-VC7 success-case closure)**.** *For every typed finite-state tower of finite-dimensional vector spaces or bounded complexes, Appendix UB computes the ordinary and derived inverse limits from one finite periodic-tail monodromy, proves* $\varprojlim^1 = 0$*, gives exact zero, acyclicity, quasi-isomorphism, and apex-agreement criteria, and supplies a finite deterministic verifier. For eventually surjective towers it supplies the exact Milnor-collapse interface, including adic completion once completeness and separatedness are externally proved. No core infinite hypothesis is hidden in the declared finite-state success case.*

Proof. UB-VC7-01 proves the defect vanishing. UB-VC7-02 reconstructs the ordinary limit by the stable image. UB-VC7-03 closes apex Return. UB-VC7-04 computes derived inverse-limit cohomology and its cone tests. UB-VC7-05 provides a concrete periodic endomorphism specialization. UB-VC7-06 supplies the surjective and completion lane. UB-VC7-07 gives replay soundness, and UB-VC7-08 composes the result with HT-RL and external rigidity. The non-claim explicitly blocks the failed primitive arithmetic Challenge promotion. □

VC7 source theorem bindings

The classical-strength source regime used by VC7 is limited to the countable Roos exact sequence, the Mittag–Leffler vanishing theorem for $\varprojlim^1$, the Milnor exact sequence for derived inverse limits, finite-dimensional Fitting decomposition, and standard adic completion. VC7's contribution is the exact typed finite-state normal form, stable-image replay certificate, apex criterion, cone criterion, and integration with the UB–HT closure contract. No literature-first claim is attached to the classical constituents.

Part XVIII

Validated Success Case VII-L: Iwasawa Measures and p-adic L-Function Return

100 VC7L objective and Challenge-to-Stable gate

100.1 Roadmap insertion

VC7L is inserted between VC7 and the analytic compactness lane VC8. Its purpose is not to claim a new construction of a primitive arithmetic p-adic L-function. Its purpose is to close, at exact strength, the universal Return chain

$$
\begin{gathered}
\text{finite quotient distributions} \longleftrightarrow \text{compatible finite group-ring elements} \\
\text{compatible finite group-ring elements} \longleftrightarrow \text{an Iwasawa-algebra element} \\
\text{an Iwasawa-algebra element} \longleftrightarrow \text{a bounded measure} \longleftrightarrow \text{an Amice power series} \\
\text{an Amice power series} \longrightarrow \text{finite-order and weight specializations.}
\end{gathered}
$$

A pseudomeasure denominator, an arithmetic regulator map, or a characteristic-ideal comparison is admitted only as an explicitly typed external input.

Classification 100.1 (VC7L Challenge gate)**.** *The Challenge lane attempted to infer an arithmetic p-adic L-function from a finite list of special values or a finite prefix of group-ring layers. This triggers* C-STOP-2 *and* C-STOP-3*. A finite list of values does not determine an unrestricted power series, and a finite quotient prefix does not certify boundedness, Kummer-type congruences, distribution compatibility, or an arithmetic explicit reciprocity law. VC7L therefore does not promote finite numerical interpolation to a primitive p-adic L-function. The Stable lane instead takes the exact integrality and compatibility conditions as replayable certificate fields and proves that they are necessary and sufficient for the complete Return chain above.*

100.2 Coefficient and group conventions

Let $\mathcal{O}$ be the valuation ring of a finite extension $E/\mathbb{Q}_p$, let ϖ be a uniformizer, and normalize $v_\varpi(\varpi) = 1$. Put

$$e_{\mathcal{O}} := v_\varpi(p).$$

Let $\Gamma \cong \mathbb{Z}_p$ be a procyclic group with chosen topological generator γ, and write

$$\Gamma_n := \Gamma/\Gamma^{p^n}, \qquad \Lambda_{\mathcal{O}}(\Gamma) := \varprojlim_n \mathcal{O}[\Gamma_n].$$

The transition maps are the natural quotient maps and are surjective.

Definition 100.2 (Finite quotient distribution)**.** A level-n distribution is an element

$$\Theta_n = \sum_{a \in \mathbb{Z}/p^n\mathbb{Z}} c_{n,a}[\gamma^a] \in \mathcal{O}[\Gamma_n].$$

A family $(\Theta_n)_{n\geq 0}$ is distribution-compatible if

$$c_{n,a} = \sum_{b \equiv a \pmod{p^n}} c_{n+1,b}$$

for every n and every residue class a mod p^n.

101 Exact finite-layer, measure, and Iwasawa-algebra Return

Theorem 101.1 (UB-VC7L-01: compatible finite group-ring Return)**.** *There are natural $\mathcal{O}$-algebra isomorphisms*

$$\Lambda_{\mathcal{O}}(\Gamma) \cong \mathcal{M}(\Gamma, \mathcal{O}) \cong \{(\Theta_n)_n : \Theta_{n+1} \mapsto \Theta_n\},$$

where $\mathcal{M}(\Gamma, \mathcal{O})$ denotes bounded $\mathcal{O}$-valued measures on Γ, with convolution as multiplication. Explicitly,

$$\Theta_n = \sum_{a \bmod p^n} \mu(\gamma^a \Gamma^{p^n})[\gamma^a],$$

and conversely a compatible system defines the unique measure satisfying

$$\mu(\gamma^a \Gamma^{p^n}) = c_{n,a}.$$

Proof. A bounded measure is finitely additive on the compact-open cosets. Refining a coset at level n into its p level-$n+1$ subcosets gives precisely the displayed compatibility relation, so the associated group-ring elements form an inverse system. Conversely, compatibility makes the coefficient assignment finitely additive on the Boolean algebra of compact-open subsets. Since the coefficients lie in the complete valuation ring $\mathcal{O}$ and are uniformly bounded, the assignment extends uniquely by continuity to an $\mathcal{O}$-valued bounded measure. Convolution of measures agrees with multiplication in every finite group ring, hence with multiplication in the inverse limit. □

Corollary 101.2 (Roos-defect closure for the Iwasawa tower). *For the tower $(\mathcal{O}[\Gamma_n])_n$,*

$$\varprojlim_n{}^1 \mathcal{O}[\Gamma_n] = 0.$$

Consequently the Iwasawa algebra is an exact inverse-limit apex rather than a finite-prefix extrapolation.

Proof. Every transition map is surjective. Apply UB-VC7-06. □

102 Amice–Mahler Return

Identify Γ with $\mathbb{Z}_p$ by $x \mapsto \gamma^x$.

Definition 102.1 (Amice transform). For $\mu \in \mathcal{M}(\Gamma, \mathcal{O})$, define

$$\mathcal{A}_\mu(T) := \int_{\mathbb{Z}_p} (1+T)^x \, d\mu(x) = \sum_{m \geq 0} \left(\int_{\mathbb{Z}_p} \binom{x}{m} d\mu(x) \right) T^m.$$

Theorem 102.2 (UB-VC7L-02: Amice power-series equivalence). *The Amice transform is an $\mathcal{O}$-algebra isomorphism*

$$\mathcal{A} : \mathcal{M}(\Gamma, \mathcal{O}) \xrightarrow{\sim} \mathcal{O}[[T]].$$

Under the chosen generator, the composite

$$\Lambda_{\mathcal{O}}(\Gamma) \xrightarrow{\sim} \mathcal{M}(\Gamma, \mathcal{O}) \xrightarrow{\mathcal{A}} \mathcal{O}[[T]]$$

sends $[\gamma]$ to $1+T$.

Proof. Mahler's theorem gives a unique expansion

$$f(x) = \sum_{m \geq 0} b_m(f) \binom{x}{m}, \qquad b_m(f) \longrightarrow 0,$$

for every continuous $f : \mathbb{Z}_p \to E$. A bounded measure is therefore determined by the bounded coefficient sequence

$$a_m = \int \binom{x}{m} d\mu(x) \in \mathcal{O}.$$

Conversely, for any $F(T) = \sum_{m \geq 0} a_m T^m \in \mathcal{O}[[T]]$, the rule

$$\mu_F(f) := \sum_{m \geq 0} b_m(f) a_m$$

converges and defines a bounded measure. These constructions are inverse. The convolution identity follows from

$$(1+T)^{x+y} = (1+T)^x (1+T)^y.$$

For the Dirac measure at 1, the transform is $1+T$, proving the generator statement. □

Corollary 102.3 (Exact coefficient Return). *If $F_\Theta(T) = \sum_{m \geq 0} a_m T^m$ is the Iwasawa series attached to Θ, then*

$$a_m = \int_\Gamma \binom{x}{m} d\mu_\Theta(x).$$

Thus the formal coefficients, the Mahler moments, the measure, and the compatible finite group-ring system determine each other without loss.

103 Finite-order specialization and Fourier Return

Let E_n/E be a finite extension containing the values of every character of Γ_n.

Theorem 103.1 (UB-VC7L-03: specialization triangle)**.** *Let $\Theta \in \Lambda_{\mathcal{O}}(\Gamma)$, let μ_Θ be its measure, and let $F_\Theta(T)$ be its Amice series. If $\chi : \Gamma \to E_n^\times$ has conductor dividing p^n, then*

$$\chi(\Theta) = \chi(\Theta_n) = \int_\Gamma \chi(x)\, d\mu_\Theta(x) = F_\Theta(\chi(\gamma) - 1).$$

Proof. Write $\chi(\gamma) = \zeta$. Since $\chi(\gamma^x) = \zeta^x$, the Amice identity at $T = \zeta - 1$ gives the last equality. The first two equalities follow because χ factors through Γ_n, and the finite sum defining the integral is exactly character evaluation on Θ_n. □

Theorem 103.2 (UB-VC7L-04: finite Fourier reconstruction and scalar minimality)**.** *For fixed n, the full character-specialization vector*

$$(\chi(\Theta_n))_{\chi \in \widehat{\Gamma_n}}$$

uniquely reconstructs $\Theta_n \otimes_{\mathcal{O}} E_n$ by

$$c_{n,a} = \frac{1}{p^n} \sum_{\chi \in \widehat{\Gamma_n}} \chi(\Theta_n)\chi(\gamma^a)^{-1}.$$

Among scalar E_n-linear probes on $E_n[\Gamma_n]$, at least p^n independent scalar values are necessary to reconstruct an arbitrary level-n element, and the full character family has exactly this size.

Proof. This is the Fourier inversion formula on the finite cyclic group Γ_n. The group algebra has E_n-dimension p^n, so an injective linear scalar-probe map must have target dimension at least p^n. The character table is invertible and has p^n rows. □

Remark 103.3 (Precision budget). If the character values are known modulo $\varpi^{N+ne_{\mathcal{O}}}$, then Fourier inversion reconstructs every coefficient $c_{n,a}$ modulo ϖ^N. The loss is exactly the division by p^n; additional ramification in a chosen splitting field must be accounted for in its own normalized valuation.

104 One-layer finite-precision Return

Lemma 104.1 (UB-VC7L-05: binomial valuation)**.** *For $0 < j < p^n$,*

$$v_p\binom{p^n}{j} = n - v_p(j).$$

Proof. Use

$$\binom{p^n}{j} = \frac{p^n}{j}\binom{p^n - 1}{j - 1}.$$

The base-p digits of $p^n - 1$ are all $p - 1$, so Lucas' theorem shows that $\binom{p^n-1}{j-1}$ is a p-adic unit. □

For $M \geq 2$, put
$$b_p(M) := \max_{1 \leq j < M} v_p(j) = \lfloor \log_p(M-1) \rfloor,$$
and put $b_p(1) = 0$.

Theorem 104.2 (UB-VC7L-06: one-layer truncated Iwasawa Return)**.** *Fix $N, M \geq 1$. Assume*
$$p^n \geq M, \qquad e_{\mathcal{O}}(n - b_p(M)) \geq N.$$
Then the image $\Theta_n \bmod \varpi^N$ of an Iwasawa element Θ at the single level n uniquely determines
$$F_\Theta(T) \pmod{(\varpi^N, T^M)}.$$
Equivalently, the relation
$$(1+T)^{p^n} - 1$$
vanishes in $\mathcal{O}[[T]]/(\varpi^N, T^M)$.

Proof. For $1 \leq j < M$, UB-VC7L-05 gives
$$v_\varpi \binom{p^n}{j} = e_{\mathcal{O}}(n - v_p(j)) \geq e_{\mathcal{O}}(n - b_p(M)) \geq N.$$
All terms of degree at least M vanish modulo T^M. Hence
$$(1+T)^{p^n} - 1 = 0 \quad \text{in} \quad \mathcal{O}[[T]]/(\varpi^N, T^M).$$
Two Iwasawa lifts of the same level-n element differ by a multiple of this relation, so their truncations agree. $\square$

Corollary 104.3 (Finite special-value to power-series certificate)**.** *Under the hypotheses of UB-VC7L-06, all conductor-dividing-p^n character values known to the Fourier precision budget of UB-VC7L-04 determine*
$$F_\Theta(T) \pmod{(\varpi^N, T^M)}.$$
Thus a finite character table yields a finite certified approximation of the full p-adic analytic object.

105 Weight branches on $\mathbb{Z}_p^\times$ and Mellin Return

Assume p is odd. Write
$$\mathbb{Z}_p^\times = \Delta \times \Gamma, \qquad \Delta \cong \mu_{p-1}, \qquad \Gamma = 1 + p\mathbb{Z}_p,$$
and choose a topological generator γ of Γ. Enlarge $\mathcal{O}$ so that it contains the values of every character of Δ.

Definition 105.1 (Teichmuller branch)**.** For $\eta \in \widehat{\Delta}$, let
$$e_\eta := \frac{1}{|\Delta|} \sum_{\delta \in \Delta} \eta(\delta)^{-1}[\delta] \in \mathcal{O}[\Delta].$$
For $\Theta \in \Lambda_{\mathcal{O}}(\mathbb{Z}_p^\times)$, write
$$e_\eta \Theta \longleftrightarrow F_{\Theta,\eta}(T) \in \mathcal{O}[[T]].$$

Theorem 105.2 (UB-VC7L-07: branch and Mellin Return)**.** *There is an exact decomposition*

$$\Lambda_{\mathcal{O}}(\mathbb{Z}_p^{\times}) \cong \bigoplus_{\eta\in\widehat{\Delta}} \mathcal{O}[[T]]e_{\eta}.$$

For $s \in \mathbb{Z}_p$, define

$$L_{\Theta,\eta}(s) := F_{\Theta,\eta}(\gamma^s - 1).$$

Then $L_{\Theta,\eta}$ is p-adic analytic on $\mathbb{Z}_p$. For every finite-order character ψ of Γ,

$$(\eta\psi)(\Theta) = F_{\Theta,\eta}(\psi(\gamma) - 1).$$

Hence the branch power series, the analytic weight function, and all finite-order branch specializations Return each other at their declared strengths.

Proof. Since $|\Delta| = p - 1$ is a unit in $\mathcal{O}$, the idempotents e_η are integral, orthogonal, and sum to one. The completed group algebra therefore decomposes into the displayed branches. The map $s \mapsto \gamma^s - 1$ is analytic from $\mathbb{Z}_p$ to the open unit disc, so composition with $F_{\Theta,\eta}$ is analytic. The specialization formula is UB-VC7L-03 applied on the η-idempotent component. □

106 Special-value interpolation as an exact Return criterion

Theorem 106.1 (UB-VC7L-08: special-value distribution criterion)**.** *For every n, suppose values*

$$y_{n,\chi} \in E_n \qquad (\chi \in \widehat{\Gamma_n})$$

are given. Define their Fourier coefficients by

$$c_{n,a} := \frac{1}{p^n} \sum_{\chi\in\widehat{\Gamma_n}} y_{n,\chi}\chi(\gamma^a)^{-1}.$$

The following are equivalent.

1. *There exists a unique $\Theta \in \Lambda_{\mathcal{O}}(\Gamma)$ such that*

$$\chi(\Theta) = y_{n,\chi}$$

   for all n and all $\chi \in \widehat{\Gamma_n}$.

2. *The coefficients satisfy:*
   - *(a) $c_{n,a} \in \mathcal{O}$ for every n, a;*
   - *(b) the Fourier coefficient vector is Galois invariant over E;*
   - *(c) the distribution relation*

$$c_{n,a} = \sum_{b\equiv a \pmod{p^n}} c_{n+1,b}$$

     holds for every n, a.

When these conditions hold, UB-VC7L-01 and UB-VC7L-02 construct the unique bounded measure and Iwasawa power series interpolating the declared values.

Proof. If Θ exists, UB-VC7L-04 recovers the coefficients of Θ_n, which are integral and compatible by definition. Conversely, the three conditions define a compatible inverse system $\Theta_n = \sum_a c_{n,a}[\gamma^a] \in \mathcal{O}[\Gamma_n]$. UB-VC7L-01 gives a unique Θ, and finite Fourier inversion gives the prescribed specializations. Uniqueness follows at every level from invertibility of the character table and globally from the inverse-limit universal property. □

Remark 106.2 (Kummer congruences as source obligations). In concrete constructions, Kummer-type congruences are one mechanism that proves the integrality and compatibility conditions in UB-VC7L-08. VC7L does not infer those arithmetic congruences from interpolation values; it records them as the exact source-side certificate needed to Return a bounded measure.

107 Pseudomeasures and controlled poles

Definition 107.1 (One-denominator pseudomeasure certificate)**.** Let $D \in \Lambda_{\mathcal{O}}(\Gamma)$ be a non-zero-divisor. A D-normalized pseudomeasure is an element

$$\Xi = D^{-1}\Theta \in \mathrm{Frac}(\Lambda_{\mathcal{O}}(\Gamma))$$

with $\Theta \in \Lambda_{\mathcal{O}}(\Gamma)$. The certificate stores (D, Θ), not only the quotient symbol.

Theorem 107.2 (UB-VC7L-09: pseudomeasure specialization Return)**.** *Let $\Xi = D^{-1}\Theta$. For every finite-order character χ with $\chi(D) \neq 0$,*

$$\chi(\Xi) = \frac{\chi(\Theta)}{\chi(D)}.$$

Under the coordinate $\Lambda \cong \mathcal{O}[[T]]$, the branch Mellin function is meromorphic with poles contained in the zero locus of $D(T)$. Replacing (D, Θ) by $(uD, u\Theta)$ for a unit $u \in \Lambda^\times$ leaves the pseudomeasure and every defined specialization unchanged.

Proof. All assertions follow from localization at the multiplicative set generated by D, together with UB-VC7L-03. The unit-rescaling statement is immediate. □

Warning 107.3 (Pole versus divergence). A pole in a branch Mellin function is not an inverse-limit failure and is not a Type-IV Return obstruction by itself. It is represented by the declared denominator D. Return is undefined only at specializations where D vanishes, unless an independent cancellation theorem supplies a regularized value.

108 Kubota–Leopoldt exact-strength recovery

Theorem 108.1 (UB-VC7L-10: Kubota–Leopoldt Return binding)**.** *Fix the standard Teichmuller normalization for a Dirichlet character θ. The classical Kubota–Leopoldt construction supplies a bounded measure or, on the zeta branch, a controlled pseudomeasure whose Mellin transform satisfies the standard interpolation formula*

$$L_p(1-k, \theta) = \big(1 - \theta\omega^{-k}(p)p^{k-1}\big)L(1-k, \theta\omega^{-k})$$

for the admissible positive integers k and the corresponding parity branch. Once the classical Kummer-congruence and boundedness theorem is bound as *the source input, UB-VC7L-01–UB-VC7L-09 Return, without further arithmetic assumptions:*

1. *the compatible finite group-ring distributions;*
2. *the unique Iwasawa-algebra element or normalized pseudomeasure;*
3. *the branch Amice power series;*
4. *every finite-order character specialization;*
5. *every finite-precision truncation covered by UB-VC7L-06.*

Proof. The classical source theorem supplies exactly the integrality, compatibility, and interpolation fields required by UB-VC7L-08, with a denominator on the zeta branch when necessary. The asserted Returns are then UB-VC7L-01, UB-VC7L-02, UB-VC7L-03, UB-VC7L-07, UB-VC7L-09, and UB-VC7L-06. No part of this proof replaces the classical construction of the measure or its Kummer congruences. □

Classification 108.2 (Recovery strength)**.** *UB-VC7L-10 is an exact-strength Known-Theorem Recovery and a validated Return case. It is not a new construction of the Kubota–Leopoldt function and does not claim a new nonvanishing theorem.*

109 Arithmetic, algebraic, and Leopoldt interfaces

109.1 Arithmetic regulator interface

Definition 109.1 (Arithmetic p-adic L-adapter)**.** An arithmetic adapter consists of:

1. a norm-compatible class z in a declared Iwasawa cohomology or unit tower;
2. a continuous Λ-linear regulator map
$$\mathcal{R} : H^1_{\mathrm{Iw}} \longrightarrow S^{-1}\Lambda;$$
3. a proved finite-level compatibility square;
4. an explicit reciprocity theorem identifying $\mathcal{R}(z)$ with the analytic Iwasawa element.

Theorem 109.2 (UB-VC7L-11: arithmetic-to-analytic Return)**.** *Given an arithmetic adapter, the norm-compatible finite layers of z, the regulator compatibility, and the explicit reciprocity equality Return the same pseudomeasure, branch power series, and finite-order specializations as the analytic lane. Without the explicit reciprocity equality, VC7L Returns two typed Iwasawa elements but does not identify them.*

Proof. VC7 and the surjective inverse-limit lane reconstruct the Iwasawa cohomology class from its compatible layers when the source control theorem applies. Applying $\mathcal{R}$ gives an element of $S^{-1}\Lambda$. The explicit reciprocity theorem identifies this element with the analytic one, after which UB-VC7L-03, UB-VC7L-07, and UB-VC7L-09 transport all declared specializations. In the absence of that equality there is no valid comparison arrow. □

109.2 Algebraic characteristic-element interface

Theorem 109.3 (UB-VC7L-12: determinant and characteristic-element Return)**.** *Let X be a torsion $\Lambda_{\mathcal{O}}(\Gamma)$-module with an exact square presentation*

$$0 \longrightarrow \Lambda^r \xrightarrow{A} \Lambda^r \longrightarrow X \longrightarrow 0,$$

and assume $\det A \neq 0$*. Then*

$$\mathrm{Fitt}_{\Lambda,0}(X) = (\det A).$$

If a proved comparison gives

$$\det A = u\Theta \qquad (u \in \Lambda^\times),$$

for an analytic Iwasawa element Θ, then the algebraic and analytic ideals agree and every specialization away from the zero locus is related by the same unit factor. VC7L does not manufacture the unit comparison.

Proof. The zeroth Fitting ideal of a module with the displayed square presentation is generated by the maximal minor $\det A$. The comparison equality identifies its principal ideal with (Θ). Character specialization is UB-VC7L-03 after applying the specialization homomorphism to the equality. □

Corollary 109.4 (Main-conjecture-strength interface)**.** *An Iwasawa main conjecture enters UB as the external comparison*

$$\mathrm{char}_\Lambda(X) = (\Theta_{\mathrm{an}})$$

or as a determinant-line equality. Once supplied, UB Returns its finite-layer, measure, power-series, and character-specialization consequences. The equality itself remains an external theorem unless proved independently in the declared domain.

109.3 Leopoldt interface

Theorem 109.5 (UB-VC7L-13: p-adic L-value to Leopoldt Return interface)**.** *Suppose an external p-adic class-number or regulator theorem for a number field F gives an equality*

$$\mathrm{Reg}_p(F) = u_F \prod_{\eta \in \mathcal{X}} L_{p,\eta}(s_\eta), \qquad u_F \neq 0,$$

where all omitted local, class-number, and period factors are proved nonzero. If VC7L certificates show that every displayed specialization is nonzero, then $\mathrm{Reg}_p(F) \neq 0$*, hence the corresponding Leopoldt kernel vanishes. VC5 may then propagate that zero certificate through its finite Galois fixed-field Return.*

Proof. The product equality and nonzero auxiliary factor imply regulator nonvanishing. The declared external regulator theorem identifies this with Leopoldt kernel zero. VC5 then transports the kernel-zero statement along its proved fixed-field and Galois representation formulas. □

Non-claim 109.6 (Leopoldt boundary). VC7L-13 does not prove the class-number formula, the nonvanishing of the specializations, or general Leopoldt. It closes the Return chain once these exact arithmetic source obligations are discharged.

110 VC7L verifier and finite replay

Verifier contract 110.1 (UB-VC7L certificate)**.** A VC7L certificate contains:

1. $p, \mathcal{O}, \varpi, e_{\mathcal{O}}$, a generator γ, and the declared quotient levels;
2. group-ring coefficient vectors $c_{n,a}$ and projection-compatibility identities;
3. optional character tables and specialization vectors;
4. Fourier-integrality and Galois-invariance checks;
5. a requested precision pair (N, M) and a level n satisfying UB-VC7L-06;
6. optional branch idempotents e_η;
7. optional denominator data (D, Θ);
8. optional arithmetic regulator or determinant comparison equalities;
9. the exact Return strength requested.

Theorem 110.2 (UB-VC7L-14: finite replay soundness)**.** *For every finite requested precision and conductor bound, the VC7L verifier decides and reconstructs the claimed finite quotient distributions, character values, branch components, and power-series truncations by finite linear algebra and valuation checks. If an infinite compatible system is declared by an external symbolic rule, the verifier checks the rule's compatibility proof rather than extrapolating from a finite prefix.*

Proof. Finite quotient compatibility is a finite coefficient-summing calculation. Fourier inversion is finite matrix multiplication. UB-VC7L-05 and the displayed valuation inequality certify the one-level truncation bound. Branch decomposition uses a finite idempotent table. Denominator and comparison claims reduce to multiplication in the requested finite quotient or truncation. Infinite existence is accepted only through a proof object establishing the universal compatibility relation. □

Remark 110.3 (Complexity)*.* At level n, direct Fourier replay costs $O(p^{2n})$ scalar operations, or quasi-linear complexity with a suitable finite Fourier transform implementation. Compatibility from level $n+1$ to n costs $O(p^{n+1})$. Truncated power-series Return to degree $M-1$ costs polynomial time in M, N, and the coefficient-ring precision.

111 VC7L status, closure, and non-claims

Classification 111.1 (VC7L roadmap status).

Finite group-ring to measure Return	*PASS*
Measure to Iwasawa power series	*PASS*
Finite-order character Return	*PASS*
Special-value distribution criterion	*PASS*
One-layer finite-precision theorem	*PASS*
Teichmuller branch and Mellin Return	*PASS*
Controlled pseudomeasure pole lane	*PASS*
Kubota–Leopoldt exact recovery	*PASS*
Arithmetic regulator interface	*PARTIAL-RETURN*
Characteristic-ideal interface	*PARTIAL-RETURN*
Primitive explicit reciprocity or main conjecture	*NOT-INVOKED*
General nonvanishing	*REJECT*
Roadmap registration	*Stable Grade B*

Non-claim 111.2 (VC7L promotion ceiling). VC7L does not assert:

1. that finitely many special values determine an unrestricted p-adic L-function without a precision bound or distribution certificate;
2. that arbitrary interpolation data satisfy Kummer congruences or boundedness;
3. a new construction or nonvanishing theorem for Kubota–Leopoldt, Katz, modular-form, Rankin–Selberg, or other p-adic L-functions;
4. an explicit reciprocity law, Euler-system theorem, Iwasawa main conjecture, or p-adic BSD formula;
5. that a pole is a failed inverse limit or that a divergent specialization has a canonical regularized Return;
6. general Leopoldt from the existence of a p-adic L-function alone.

Theorem 111.3 (UB-VC7L-15: success-case closure). *VC7L closes the universal p-adic L-Return chain at the strongest level justified by bounded distribution data:*

$$\boxed{\begin{aligned}&\textit{compatible finite distributions} \longleftrightarrow \Lambda_{\mathcal{O}}(\Gamma) \longleftrightarrow \mathcal{M}(\Gamma,\mathcal{O}) \longleftrightarrow \mathcal{O}[[T]]\\ &\qquad\qquad\qquad\qquad \longrightarrow \textit{Mellin branches and finite-order values}.\end{aligned}}$$

It also supplies a sharp finite Fourier certificate, an explicit one-level precision bound, a controlled pseudomeasure lane, a Kubota–Leopoldt exact recovery, and typed interfaces to arithmetic regulators, characteristic elements, VC5 Leopoldt propagation, VC7 inverse limits, and HT-IW/Return.

Proof. UB-VC7L-01 closes finite distributions, measures, and the Iwasawa inverse limit. UB-VC7L-02 closes the Amice equivalence. UB-VC7L-03 and UB-VC7L-04 close character specialization and finite reconstruction. UB-VC7L-05 and UB-VC7L-06 close finite precision. UB-VC7L-07 closes weight branches. UB-VC7L-08 gives necessary and sufficient interpolation-certificate conditions. UB-VC7L-09 handles denominators. UB-VC7L-10 binds the classical Kubota–Leopoldt source theorem. UB-VC7L-11–UB-VC7L-13 type the arithmetic, algebraic, and Leopoldt Returns without importing their missing source theorems. UB-VC7L-14 supplies finite replay. □

VC7L source theorem bindings

The classical-strength source regime used by VC7L consists of: Mahler expansion for continuous functions on $\mathbb{Z}_p$; the equivalence between bounded measures, completed group rings, and Amice power series; finite Fourier inversion; the Kubota–Leopoldt interpolation theorem and its Kummer-congruence construction; standard Coleman or Perrin–Riou regulator maps when an arithmetic lane is invoked; and Fitting or characteristic ideals for torsion Iwasawa modules. The VC7L contribution is the typed bidirectional Return contract, the exact special-value criterion, the explicit one-layer finite-precision bound, the denominator discipline, the finite verifier, and the integration with VC5, VC7, HT-RL, and HT-IW/Return.

Part XIX

Validated Success Case VIII: Analytic Compactness, Defect Annihilation, and Continuation Return

112 VC8 objective and Challenge-to-Stable gate

VC8 is the analytic validation lane of Appendix UB. Its purpose is to determine exactly which parts of a nonlinear regularity proof can be standardized as reusable bridge components and which parts remain irreducibly analytic. The declared benchmark is the unforced three-dimensional incompressible Navier–Stokes system on $\mathbb{R}^3 \times (0,T)$,

$$\partial_t u + (u\cdot\nabla)u - \nu\Delta u + \nabla p = 0, \qquad \nabla\cdot u = 0, \qquad \nu > 0,$$

with divergence-free L^2 initial data and Leray–Hopf weak solutions.

The frozen source criterion is the strict-subcritical fixed-exponent implication

$$u \in L^6(0,T;L^6(\mathbb{R}^3)) \quad\Longrightarrow\quad u \text{ is regular on } \mathbb{R}^3\times(0,T].$$

The exponent identity

$$\frac{2}{6}+\frac{3}{6}=\frac{5}{6}<1$$

keeps VC8 away from endpoint issues. The classical regularity conclusion is quarantined until the final Return arrow.

Classification 112.1 (VC8 Challenge gate)**.** *The Challenge lane attempted to derive a Serrin-class bound, an epsilon-regularity estimate, or a continuation modulus from the AK projection profile, finite energy data, or persistence information alone. This triggers* C-STOP-1 *and* C-STOP-3. *The energy inequality does not imply an $L^6_tL^6_x$ bound, and a finite topological or persistence profile does not by itself control concentration, pressure, or the nonlinear convection term. VC8 therefore does not promote AK-internal collapse or finite observed regularity into a new Navier–Stokes regularity theorem. The Stable lane instead accepts exact analytic witnesses—energy bounds, time-compactness bounds, residual convergence, and* *a Serrin norm certificate—and proves the complete finite-window, compactness, defect-annihilation, uniqueness, continuation, and Return chain.*

Non-claim 112.2 (Challenge non-promotion). VC8 does not infer a Serrin bound from the Leray energy inequality, does not prove an endpoint criterion, does not prove epsilon-regularity from AK persistence data, and does not prove unconditional global regularity. Any such implication would require a new analytic estimate external to the present UB component library.

113 Analytic objects and the VC8 defect stack

Let

$$H = L^2_\sigma(\mathbb{R}^3), \qquad V = H^1_\sigma(\mathbb{R}^3), \qquad V' = H^{-1}_\sigma(\mathbb{R}^3),$$

where the subscript σ denotes divergence-free closure. For a compact ball $B_R \subset \mathbb{R}^3$, the corresponding local spaces are denoted $H(B_R)$, $V(B_R)$, and $V'(B_R)$.

Definition 113.1 (VC8 analytic certificate)**.** A VC8 analytic certificate for a time interval $(0,T)$ contains:

1. a typed weak solution or approximation sequence and a weak-form residual;
2. a uniform energy bound in $L^\infty(0,T;H) \cap L^2(0,T;V)$;
3. local time-derivative bounds in $L^{4/3}(0,T;V'(B_R))$ for every declared radius R;
4. a finite time atlas and rational upper-bound witnesses for the declared local Serrin norms;
5. initial-data compatibility and the energy-inequality proof object;
6. source locators for compactness, weak–strong uniqueness, and the fixed-exponent Serrin Return;
7. the requested regularity, uniqueness, continuation, or approximation Return strength.

A numerical sample without a certified norm upper bound is not an analytic certificate.

Definition 113.2 (VC8 defect stack)**.** For an analytic route define

$$\mathfrak{D}_{\mathrm{VC8}} = (D_{\text{energy}}, D_{\text{time}}, D_{\text{comp}}, D_{\text{nl}}, D_{\text{Serrin}}, D_{\text{glue}}, D_{\text{uniq}}, D_{\text{restart}}, D_{\text{return}}),$$

where:

- D_{energy} records failure of uniform energy control;
- D_{time} records failure of a compactness-compatible time derivative bound;
- D_{comp} records failure of strong local compactness;
- D_{nl} records a surviving nonlinear product or Reynolds defect;
- D_{Serrin} records absence of a verified Serrin-class witness;
- D_{glue} records an uncovered time point or incompatible local solution;
- D_{uniq} records failure of weak–strong uniqueness;

- D_{restart} records failure of continuation beyond a maximal time;
- D_{return} records any mismatch between the AK analytic packet and the native PDE conclusion.

114 Finite time-atlas reduction

Lemma 114.1 (UB-VC8-01: finite norm-window partition)**.** *Let $f \in L^1(0,T)$ be non-negative, let*

$$M = \int_0^T f(t)\,dt < \infty,$$

and let $\delta > 0$. There is a partition

$$0 = t_0 < t_1 < \cdots < t_N = T$$

such that

$$\int_{t_{j-1}}^{t_j} f(t)\,dt \le \delta \qquad (1 \le j \le N)$$

and

$$N \le \left\lceil \frac{M}{\delta} \right\rceil + 1.$$

Proof. Set $F(t) = \int_0^t f(s)\,ds$. The map F is continuous and nondecreasing. Choose the partition points by successive first hitting times of the levels $j\delta$, stopping at T. Every complete window carries mass exactly δ, the last carries at most δ, and the displayed bound follows. □

Corollary 114.2 (UB-VC8-02: finite $L^6_t L^6_x$ atlas)**.** *If*

$$\|u\|_{L^6(0,T;L^6)} \le M,$$

then for every $\varepsilon > 0$ there is a finite partition with

$$\|u\|_{L^6(t_{j-1},t_j;L^6)} \le \varepsilon$$

and

$$N \le \left\lceil \frac{M^6}{\varepsilon^6} \right\rceil + 1.$$

Proof. Apply UB-VC8-01 to $f(t) = \|u(t)\|_{L^6}^6$ and $\delta = \varepsilon^6$. □

Theorem 114.3 (UB-VC8-03: finite Serrin-atlas Return)**.** *Let u be a Leray–Hopf weak solution on $\mathbb{R}^3 \times (0,T)$. Suppose there is a finite family of open time intervals $I_1, \dots, I_N$ covering $(0,T]$ such that*

$$u|_{I_j} \in L^6(I_j; L^6(\mathbb{R}^3)) \qquad (1 \le j \le N).$$

Then u is regular on $\mathbb{R}^3 \times (0,T]$. If another Leray–Hopf solution has the same initial data, the two solutions coincide on $[0,T]$.

Proof. Apply the frozen Serrin source theorem on each interval after time translation. The intervals cover every positive time, so local regularity Returns globally. On each interval the regular solution belongs to a Prodi–Serrin uniqueness class; weak–strong uniqueness identifies any second Leray–Hopf solution with u. Equality on overlapping intervals and continuity in the weak energy topology glue the local equalities. □

Remark 114.4 (Why the atlas is a genuine UB component). The theorem does not replace the analytic Serrin estimate. Its contribution is to convert a potentially nonuniform local regularity proof into a finite descent object with explicit coverage, norm witnesses, and uniqueness cells. The time atlas is the analytic analogue of VC6 descent.

115 Compactness and nonlinear defect annihilation

Theorem 115.1 (UB-VC8-04: local Aubin–Lions Return)**.** *Let (u_n) be a sequence such that for every $R > 0$,*

$$\sup_n \|u_n\|_{L^2(0,T;H^1(B_R))} < \infty$$

and

$$\sup_n \|\partial_t u_n\|_{L^{4/3}(0,T;H^{-1}(B_R))} < \infty.$$

Then a subsequence converges strongly in

$$L^2(0,T;L^2(B_R))$$

for every $R > 0$, after a diagonal extraction.

Proof. For fixed R, the embedding

$$H^1(B_R) \Subset L^2(B_R) \hookrightarrow H^{-1}(B_R)$$

meets the Aubin–Lions–Simon compactness hypotheses. Apply compactness on B_R and diagonalize over $R \in \mathbb{N}$. □

Theorem 115.2 (UB-VC8-05: nonlinear product-defect annihilation)**.** *Under the hypotheses of UB-VC8-04, suppose $u_n \to u$ strongly in $L^2(0,T;L^2(B_R))$. Then*

$$u_n \otimes u_n \longrightarrow u \otimes u$$

strongly in $L^1(0,T;L^1(B_R))$. Consequently the nonlinear convection term has no local Reynolds defect in the distributional limit.

Proof. Use

$$\|u_n \otimes u_n - u \otimes u\|_{L^1} \le (\|u_n\|_{L^2} + \|u\|_{L^2})\|u_n - u\|_{L^2}$$

on $(0,T)\times B_R$. The first factor is uniformly bounded and the second tends to zero. □

Corollary 115.3 (UB-VC8-06: compactness-defect closure)**.** *For an approximation sequence satisfying the VC8 energy and local time-derivative fields, the implications*

$$D_{\text{energy}} = D_{\text{time}} = 0 \Longrightarrow D_{\text{comp}} = D_{\text{nl}} = 0$$

are valid after subsequence extraction.

Warning 115.4 (What compactness does not Return). Strong local L^2 convergence passes the nonlinear term, but it does not imply that the limit is regular. The singularity question remains in D_{Serrin} or another source regularity criterion. Compactness is not a substitute for a continuation estimate.

116 Approximation-to-regularity closure

Definition 116.1 (Admissible VC8 approximation)**.** A sequence (u_n) is an admissible approximation to the Navier–Stokes system if:

1. each u_n is divergence free;
2. (u_n) is uniformly bounded in $L^\infty(0,T;H) \cap L^2(0,T;V)$;
3. the local time-derivative bounds of UB-VC8-04 hold;
4. the weak-form residuals tend to zero against every compactly supported smooth divergence-free test field;
5. the initial data converge to u_0 in the declared topology;
6. the energy inequality passes by lower semicontinuity;
7. there is a uniform bound

$$\sup_n \|u_n\|_{L^6(0,T;L^6(\mathbb{R}^3))} \leq M < \infty.$$

Theorem 116.2 (UB-VC8-07: approximation-to-regularity Return)**.** *Every admissible VC8 approximation has a subsequence converging to a Leray–Hopf weak solution u with*

$$\|u\|_{L^6(0,T;L^6)} \leq M.$$

The limit is regular on $\mathbb{R}^3 \times (0,T]$ and is the unique Leray–Hopf solution with initial data u_0. Consequently every convergent subsequence has the same limit, and the full sequence converges to u in the local L^2 topology whenever the sequence is relatively compact there.

Proof. UB-VC8-04 gives strong local L^2 compactness. Weak compactness in the energy spaces and UB-VC8-05 pass the linear and nonlinear terms to the limit; the residual and initial-data fields identify the Navier–Stokes weak formulation. Lower semicontinuity Returns the energy inequality. Weak compactness in $L^6_t L^6_x$ gives the displayed norm bound for the limit. UB-VC8-03 gives regularity and weak–strong uniqueness. Relative compactness plus uniqueness of every cluster point yields convergence of the full sequence. □

Corollary 116.3 (UB-VC8-08: finite-window approximation certificate)**.** *Under the hypotheses of UB-VC8-07, a requested local threshold $\varepsilon > 0$ admits at most*

$$\left\lceil \frac{M^6}{\varepsilon^6} \right\rceil + 1$$

regularity windows. On each window the limit is in the fixed Serrin class, and the windows glue uniquely.

117 Continuation and blow-up Return

Theorem 117.1 (UB-VC8-09: fixed-exponent continuation Return)**.** *Let u be a maximal strong Navier–Stokes solution on $[0, T_*)$ with $T_* < \infty$. If*

$$\|u\|_{L^6(0,T_*;L^6)} < \infty,$$

then u extends as a strong solution beyond T_*. Equivalently,

$$T_* < \infty \quad \Longrightarrow \quad \|u\|_{L^6(0,T_*;L^6)} = \infty.$$

Proof. The finite norm gives a finite Serrin atlas by UB-VC8-02. The source criterion gives regularity up to the terminal time on the last window. The terminal state therefore lies in the local strong-solution restart class, and the local existence theorem extends the solution beyond T_*. The second statement is the contrapositive. □

Theorem 117.2 (UB-VC8-10: defect-free continuation package)**.** *For the fixed-exponent Navier–Stokes benchmark, suppose an analytic certificate verifies*

$$D_{\text{energy}} = D_{\text{time}} = D_{\text{Serrin}} = D_{\text{glue}} = 0.$$

Then, after the compactness extraction when an approximation lane is used,

$$D_{\text{comp}} = D_{\text{nl}} = D_{\text{uniq}} = D_{\text{restart}} = D_{\text{return}} = 0.$$

The resulting external Return is regularity and uniqueness on the declared interval, together with continuation whenever the interval is maximal and the Serrin norm remains finite.

Proof. Compactness and nonlinear-defect vanishing are UB-VC8-04–UB-VC8-06. Regularity and uniqueness are UB-VC8-03. Continuation is UB-VC8-09. These conclusions are native PDE statements, so the final comparison is identity at the declared source strength. □

118 AK–UB–HT route and finite replay

Theorem 118.1 (UB-VC8-11: typed analytic bridge route)**.** *The validated VC8 route is*

$$\begin{aligned}
\textit{approximation or weak solution} &\xrightarrow{\textit{energy/time certificate}} \textit{compact analytic packet}\\
&\xrightarrow{\textit{Aubin–Lions}} \textit{strong local limit}\\
&\xrightarrow{\textit{product Return}} \textit{defect-free weak solution}\\
&\xrightarrow{L_t^6 L_x^6\ \textit{witness}} \textit{finite Serrin atlas}\\
&\xrightarrow{\textit{source theorem}} \textit{regular unique solution}\\
&\xrightarrow{\textit{restart}} \textit{continuation Return}.
\end{aligned}$$

Every arrow has an explicit source theorem, certificate field, and promotion ceiling. HC, PE, SS, and RL are not invoked merely for naming uniformity; VC6 is used only for the finite local-to-global atlas logic, and the final IW/Return layer records the native analytic conclusion.

Verifier contract 118.2 (UB-VC8 certificate)**.** A finite replay certificate contains:

1. rational time endpoints and cover incidence data;
2. rational norm upper bounds and proof objects for the energy and Serrin fields;
3. finite-dimensional approximation matrices or symbolic residual identities when an approximation lane is used;
4. local compactness-bound locators for each declared spatial ball;
5. weak-form residual convergence and initial-data compatibility proofs;
6. source theorem identifiers for Aubin–Lions, Prodi–Serrin uniqueness, the fixed $L_t^6 L_x^6$ criterion, and local restart;
7. the defect-stack status and requested Return strength.

The verifier checks algebraic consistency, cover completeness, the finite-window count, and the dependency graph. Infinite-dimensional norm inequalities are accepted only through a rigorous proof object or certified interval enclosure; sample values alone do not pass.

Theorem 118.3 (UB-VC8-12: finite replay soundness)**.** *If the VC8 verifier accepts a certificate, then every finite combinatorial claim in the time atlas, defect graph, approximation residual, and Return route is correct, and the external analytic conclusion follows from the bound source theorems at exactly their registered strength.*

Proof. Time-cover and threshold claims are finite rational comparisons. Residual and matrix claims are finite symbolic or interval computations. The compactness, regularity, uniqueness, and restart arrows are invoked only when their exact hypothesis fields are present. The status precedence rule rejects a missing analytic witness rather than extrapolating it from sampled data. □

119 VC8 status, closure, and non-claims

Classification 119.1 (VC8 roadmap status)**.** *The Challenge lane does not close a new analytic regularity estimate and is stopped under C-STOP-1 and C-STOP-3. The Stable lane closes:*

- *finite Serrin-window reduction with an explicit window-count bound;*
- *local compactness and nonlinear-defect annihilation;*
- *approximation-to-regularity and uniqueness Return;*
- *fixed-exponent blow-up alternative and continuation;*
- *a typed finite verifier and exact-strength source quarantine.*

VC8 is therefore registered as Stable Grade B*.*

Non-claim 119.2 (VC8 promotion ceiling)*.* VC8 does not assert:

- unconditional three-dimensional Navier–Stokes regularity;
- a new proof that every Leray–Hopf solution belongs to $L_t^6 L_x^6$;

- endpoint $L_t^\infty L_x^3$ regularity or any endpoint replacement;
- a new epsilon-regularity theorem, pressure estimate, or concentration-exclusion theorem;
- that finite numerical sampling certifies an infinite-dimensional norm bound;
- that topological persistence, Type IV data, or AK collapse alone rules out blow-up.

Theorem 119.3 (UB-VC8-13: analytic success-case closure)**.** *VC8 closes the analytic bridge at the strongest level justified by exact source witnesses:*

$$\boxed{\begin{aligned}&\text{energy/time compactness} + \text{residual convergence} + L_t^6 L_x^6 \text{ certificate}\\&\quad\Longrightarrow \text{defect-free Leray–Hopf limit}\\&\quad\Longrightarrow \text{regularity and uniqueness} \Longrightarrow \text{continuation}.\end{aligned}}$$

The route is finite-windowed, replayable, and compatible with the UB component contract. Its only unresolved path toward a Clay-level result is the primitive analytic production of a global regularity witness, not a missing Return or gluing arrow.

Proof. UB-VC8-01 and UB-VC8-02 give the finite atlas. UB-VC8-04–UB-VC8-06 close compactness and the nonlinear defect. UB-VC8-07 and UB-VC8-08 close the approximation route. UB-VC8-03 and UB-VC8-09 close regularity, uniqueness, and continuation. UB-VC8-10–UB-VC8-12 close the defect package, typed route, and replay. The non-claims isolate the absent primitive estimate. □

VC8 source theorem bindings

The exact frozen benchmark is the AK-HPDST source track `NS-KTR-SERRIN-L6`, bound to Serrin's 1962 interior-regularity theorem and the Prodi uniqueness lineage. The compactness lane uses the classical Aubin–Lions–Simon compactness theorem; the nonlinear product passage is proved directly in UB-VC8-05; weak–strong uniqueness and the continuation implication are used only at their Prodi–Serrin source strength. VC8 contributes the typed defect stack, the finite-window count, the local-to-global analytic atlas, the approximation-to-regularity closure, the finite replay contract, and the exact isolation of the missing primitive analytic estimate.

Part XX

Validated Success Case IX: Certified Nonvanishing, Valuation, and Regulator Return

120 VC9 objective and Challenge–panorama–Stable gate

The preceding success cases expose a common final obstruction. The object, comparison, limit, and Return arrows may all be closed, while the external theorem still requires a primitive assertion of the form

$$x \neq 0, \qquad \det A \neq 0, \qquad v_{\varpi}(x) = m, \qquad F(t) \neq 0, \qquad \|u\| < \infty.$$

VC9 develops a reusable certified-nonvanishing layer. Its certificates are exact residue classes or non-Archimedean balls, never floating-point plausibility.

Classification 120.1 (Challenge panorama)**.** *Four mathematically plausible lanes were tested.*

1. ***p-adic logarithmic independence.*** *Baker–Brumer type theorems can prove nonvanishing in large algebraic classes, but their general use does not by itself provide a small effective verifier for a requested concrete regulator.*
2. ***Explicit-unit and representation descent.*** *This is powerful for special Galois families and composes with VC5, but the construction of suitable units remains domain-specific.*
3. ***Newton polygon, Weierstrass, and Strassmann.*** *These give exact zero-count and zero-free criteria once coefficient valuations and analytic tails are certified.*
4. ***Complete-DVR ball arithmetic.*** *Valuation separation, determinant minors, and logarithm truncations admit finite replay and apply directly to regulators and VC7L power series.*

The family-level Challenge of producing a new infinite Leopoldt family from UB structure alone stops under C-STOP-1: an effective source of independent units or logarithmic lower bounds is still required. The panoramic review therefore selects lanes (3) and (4), and uses lane (2) through the already validated VC5 Return. Unlike a fallback containing only generic lemmas, VC9 also closes one explicit totally real S_3 arithmetic instance.

Definition 120.2 (VC9 defect stack)**.** The ordered defect stack is

$$D_{\text{prec}}, \quad D_{\text{val}}, \quad D_{\text{det}}, \quad D_{\text{tail}}, \quad D_{\text{log}}, \quad D_{\text{reg}}, \quad D_{\text{spec}}, \quad D_{\text{return}}.$$

They denote insufficient precision, unresolved valuation, determinant cancellation, uncertified analytic tail, invalid logarithm truncation, missing regulator comparison, singular specialization, and failure of the external Return theorem.

121 Complete-DVR balls and finite nonvanishing

Let K be a complete discretely valued field with valuation $v : K^\times \to \mathbb{Z}$, valuation ring $\mathcal{O}$, uniformizer ϖ, and residue field k.

Definition 121.1 (Certified ball)**.** A precision-N ball is

$$[a]_N := a + \varpi^N \mathcal{O}.$$

The certificate records a, N, the normalization of v, and the arithmetic expression whose exact value lies in the ball.

Theorem 121.2 (UB-VC9-01: ball-stable valuation and nonvanishing)**.** *If $x \in [a]_N$ and $v(a) < N$, then*

$$\boxed{v(x) = v(a) \quad \textit{and} \quad x \neq 0.}$$

If $v(a) \geq N$, the ball does not decide whether x vanishes.

Proof. Write $x = a + \varpi^N b$. Since $v(a) < N \leq v(\varpi^N b)$, the ultrametric equality gives $v(x) = v(a)$. When $v(a) \geq N$, the ball contains zero whenever $a \in \varpi^N \mathcal{O}$, so no stronger conclusion is valid. □

Theorem 121.3 (UB-VC9-02: determinant-ball certificate)**.** *Let $A \in M_r(\mathcal{O})$ and let*

$$A \in \widetilde{A} + \varpi^N M_r(\mathcal{O}).$$

Then

$$\det A \in [\det \widetilde{A}]_N.$$

Consequently, if $v(\det \widetilde{A}) < N$, then

$$\boxed{\det A \neq 0, \qquad v(\det A) = v(\det \widetilde{A}).}$$

Every nonzero certified minor gives a certified lower bound for the rank.

Proof. The determinant is a polynomial with integral coefficients in the matrix entries. In the difference of the two determinant expansions, every monomial contains at least one entry difference in $\varpi^N \mathcal{O}$ and all remaining factors are integral. Thus the determinant difference lies in $\varpi^N \mathcal{O}$, and UB-VC9-01 applies. □

Theorem 121.4 (UB-VC9-03: unique tropical-minimum determinant)**.** *For $A = (a_{ij}) \in M_r(K)$, define*

$$w(\sigma) = \sum_{i=1}^{r} v(a_{i,\sigma(i)}) \qquad (\sigma \in S_r),$$

with $v(0) = +\infty$. If one permutation σ_0 has strictly smaller weight than every other permutation, then

$$\boxed{v(\det A) = w(\sigma_0), \qquad \det A \neq 0.}$$

This is a finite sufficient certificate that avoids cancellation-sensitive determinant expansion.

Proof. The σ_0 determinant monomial has strictly smaller valuation than the sum of all other monomials. The ultrametric equality therefore identifies the valuation of the sum with the unique smallest valuation. □

Corollary 121.5 (Certified rank and cokernel lane)**.** *For a square integral matrix, UB-VC9-02 certifies full rank and the determinant valuation. For a rectangular matrix, a certified nonzero $s \times s$ minor proves rank at least s. When exact DVR row and column operations produce Smith diagonal entries $\varpi^{e_i} u_i$, the same certificate Returns the elementary divisors and the finite-length cokernel*

$$\mathrm{length}_{\mathcal{O}}(\mathrm{coker}\, A) = \sum_i e_i.$$

122 Certified p-adic logarithms

Let p be odd. The standard logarithm on $\mathbb{Z}_p^\times$ kills the Teichmuller part and is characterized by

$$\log_p(u) = \frac{1}{p-1} \log(u^{p-1}).$$

Set $z = u^{p-1} - 1 \in p\mathbb{Z}_p$.

Lemma 122.1 (UB-VC9-04: finite logarithm truncation)**.** *Let $s = v_p(z) \geq 1$. For a requested precision p^N, choose K such that*

$$ns - v_p(n) \geq N \qquad \textit{for all } n > K.$$

Then

$$\boxed{\log_p(u) \equiv \frac{1}{p-1} \sum_{n=1}^{K} (-1)^{n+1} \frac{z^n}{n} \pmod{p^N}.}$$

A sufficient computable choice is any K after which $ns - \lfloor \log_p n \rfloor \geq N$.

Proof. The nth logarithm term has valuation

$$v_p(z^n/n) = ns - v_p(n).$$

Every omitted term lies in $p^N\mathbb{Z}_p$. The series converges p-adically, and division by $p - 1$ does not change precision. □

Verifier contract 122.2 (Logarithm replay)**.** A certified logarithm record contains:

1. p, the target precision N, and a residue $u \bmod p^N$;
2. the check $p \nmid u$;
3. $z = u^{p-1} - 1 \bmod p^N$ and its valuation lower bound;
4. a truncation index K satisfying UB-VC9-04;
5. every rational term evaluated by removing the exact p-power from its denominator and inverting the remaining unit;
6. the final residue and an independent Teichmuller-decomposition replay when requested.

Acceptance proves the displayed residue class of $\log_p(u)$; it does not infer algebraic independence from numerical precision alone.

123 Power-series nonvanishing, zero-free discs, and Iwasawa order

Let $F(T) = \sum_{n\geq 0} a_n T^n$ converge on the closed unit disc of K.

Theorem 123.1 (UB-VC9-05: certified zero-free disc)**.** *Fix $\rho \in \mathbb{Q}_{\geq 0}$. If*

$$v(a_0) < v(a_n) + n\rho \qquad (n \geq 1),$$

with the infinitely many inequalities supplied by a finite coefficient list and a certified tail bound, then

$$\boxed{F(t) \neq 0 \quad \text{for every } t \text{ with } v(t) \geq \rho.}$$

Moreover $v(F(t)) = v(a_0)$ throughout that disc.

Proof. For $v(t) \geq \rho$, every nonconstant term has valuation strictly greater than $v(a_0)$. The ultrametric equality gives the claim. □

Theorem 123.2 (UB-VC9-06: finite Strassmann certificate)**.** *Suppose a finite index d and a certified tail bound prove that*

$$v(a_d) = \min_{n\geq 0} v(a_n),$$

and that d is the largest index attaining this minimum. Then F has at most d zeros in $\mathcal{O}$, counted without multiplicity in the usual Strassmann bound.

Proof. This is Strassmann's theorem after the finite tail certificate verifies that no omitted coefficient ties or improves the dominant absolute value. □

Corollary 123.3 (UB-VC9-07: certified Iwasawa μ–λ data)**.** *Let $F(T) \in \mathcal{O}[[T]]$ and suppose a finite coefficient record plus a tail lower bound certifies*

$$\mu = \min_{n\geq 0} v_\varpi(a_n), \qquad \lambda = \min\{n : v_\varpi(a_n) = \mu\}.$$

Then μ and λ are the exact Iwasawa invariants of the nonzero series at the declared branch. If a_0 is the unique coefficient of minimal valuation, UB-VC9-05 gives a zero-free closed unit disc.

Remark 123.4 (VC7L composition)*.* VC7L supplies coefficients of an Iwasawa series to a requested finite precision from one sufficiently deep finite layer. VC9 supplies the missing logical step: a coefficient or determinant is declared nonzero only when its valuation is separated from the precision ceiling, and a zero-count is declared only when the analytic tail is bounded. Thus finite-layer reconstruction and analytic nonvanishing remain distinct but composable certificates.

124 Regulator and Leopoldt Return

Let F be a number field of unit rank r, let p be a prime, and choose units $\varepsilon_1, \dots, \varepsilon_r \in \mathcal{O}_F^\times$. For p-adic embeddings $\iota_1, \dots, \iota_s$ define the logarithm matrix

$$R_p(\varepsilon; \iota) = \left(\log_p \iota_i(\varepsilon_j)\right)_{i,j}.$$

Theorem 124.1 (UB-VC9-08: certified regulator-minor Return)**.** *If an $r \times r$ minor of $R_p(\varepsilon;\iota)$ has a UB-VC9-02 nonvanishing certificate, then the p-adic logarithm map on global units has rank r. Therefore*

$$\boxed{\delta(F,p)=0,}$$

so Leopoldt's conjecture holds for (F,p).

Proof. The chosen r global units have linearly independent logarithm vectors over $\mathbb{Q}_p$. Dirichlet's unit theorem bounds the rational rank of the global unit group by r, so the completed logarithm map has full possible rank. Its kernel has zero $\mathbb{Q}_p$-dimension, which is precisely the vanishing of the Leopoldt defect. □

Corollary 124.2 (Special-value to regulator Return)**.** *Suppose an external, explicitly registered p-adic class-number or regulator formula gives*

$$R_p(F)=u\prod_{j=1}^{m}L_{p,j}(s_j),\qquad u\in\mathcal{O}^{\times}.$$

If VC7L reconstructs the relevant branches and VC9 certifies every factor as nonzero, then $R_p(F)\neq 0$ and UB-VC9-08 Returns Leopoldt. The formula itself, including exceptional-zero corrections, is an external source theorem and is not manufactured by UB.

125 Explicit totally real cubic certificate at $p=61$

Construction 125.1 (The arithmetic object)**.** Let

$$f(X)=X^3+2X^2-3X-1,\qquad F=\mathbb{Q}(\alpha),\qquad f(\alpha)=0.$$

The rational-root test proves irreducibility. The discriminant is

$$\mathrm{disc}(f)=257>0,$$

which is squarefree and not a square. Hence F is a totally real cubic field, $\mathcal{O}_F=\mathbb{Z}[\alpha]$, and the Galois closure $L/\mathbb{Q}$ has group S_3. The elements

$$\varepsilon_1=\alpha,\qquad \varepsilon_2=\alpha-1$$

are units because

$$N_{F/\mathbb{Q}}(\alpha)=1,\qquad N_{F/\mathbb{Q}}(\alpha-1)=-f(1)=1.$$

Lemma 125.2 (UB-VC9-09: complete splitting and Hensel certificate)**.** *Modulo 61,*

$$f(X)\equiv(X-26)(X-38)(X-56).$$

The roots are distinct. Their Hensel lifts modulo

$$61^4=13\,845\,841$$

are

$$\begin{aligned} r_1&=6\,676\,781,\\ r_2&=4\,998\,073,\qquad f(r_i)\equiv 0 \pmod{61^4}.\\ r_3&=2\,170\,985,\end{aligned}$$

Thus 61 splits completely in F, and each r_i defines an embedding $F\hookrightarrow\mathbb{Q}_{61}$ to precision 61^4.

Proof. Direct substitution gives the three roots modulo 61; their distinctness and $61 \nmid 257$ give simple Hensel lifting. Substitution of the displayed residues verifies the congruences modulo 61^4. □

Theorem 125.3 (UB-VC9-10: explicit 61-adic regulator certificate)**.** *Use the embeddings corresponding to r_1 and r_2. UB-VC9-04 gives, modulo 61^4,*

$$\begin{array}{c|cc} & \log_{61}(\alpha) & \log_{61}(\alpha-1) \\ \hline \iota_1 & 13\,731\,222 & 4\,425\,489 \\ \iota_2 & 4\,164\,226 & 1\,317\,417 \end{array} \qquad \pmod{61^4}.$$

Every entry is divisible by 61. Dividing by 61 gives the normalized matrix modulo $61^3 = 226\,981$,

$$\widetilde{R} = \begin{pmatrix} 225\,102 & 72\,549 \\ 68\,266 & 21\,597 \end{pmatrix} \pmod{61^3}.$$

Its determinant satisfies

$$\det \widetilde{R} \equiv 144\,222 \equiv 18 \pmod{61}.$$

Consequently

$$\boxed{v_{61} \det \begin{pmatrix} \log_{61} \iota_1(\alpha) & \log_{61} \iota_1(\alpha-1) \\ \log_{61} \iota_2(\alpha) & \log_{61} \iota_2(\alpha-1) \end{pmatrix} = 2,}$$

and the regulator minor is nonzero.

Proof. The Hensel residues from UB-VC9-09 are units and are not congruent to 1 modulo 61, so both global units are local units in the selected embeddings. Apply

$$\log_{61}(u) = \frac{1}{60} \sum_{n \geq 1} (-1)^{n+1} \frac{(u^{60}-1)^n}{n}$$

with UB-VC9-04 to precision 61^4. The displayed residues are obtained by finite modular arithmetic. Their quotients by 61 are known modulo 61^3. The determinant residue is a unit modulo 61, so UB-VC9-02 proves exact valuation zero for the normalized determinant and valuation two before normalization. □

Corollary 125.4 (UB-VC9-11: explicit Leopoldt closure)**.** *For the field*

$$F = \mathbb{Q}(\alpha), \qquad \alpha^3 + 2\alpha^2 - 3\alpha - 1 = 0,$$

Leopoldt's conjecture holds at $p = 61$:

$$\boxed{\delta(F, 61) = 0.}$$

If L is the S_3-Galois closure of F, then VC5 gives

$$\boxed{\delta(L, 61) = 0.}$$

Thus one finite regulator certificate closes a non-abelian Galois Return.

Proof. The field is totally real cubic, so its unit rank is two. UB-VC9-10 gives two units with independent 61-adic logarithm vectors; UB-VC9-08 proves the first assertion. Over the base $\mathbb{Q}$, VC5 identifies Leopoldt for an S_3-closure with Leopoldt for a cubic fixed field because the base and quadratic lanes are already zero. Apply that Return to $F \subset L$. □

Remark 125.5 (Meaning of the explicit success). This is a genuine finite proof of a named conjecture for one explicit non-abelian arithmetic object and prime. It is not presented as a literature-first instance: computational verification of Leopoldt for fixed fields and primes is classical in principle. The UB contribution is the complete certificate architecture, the exact valuation witness, and its composition with the validated S_3 Return.

126 VC9 verifier, AK–HT composition, and status

Verifier contract 126.1 (VC9 certificate)**.** A VC9 certificate serializes:

1. the complete-DVR identity, valuation normalization, precision ideal, and residue representatives;
2. every ball operation and precision-loss rule;
3. determinant or minor matrices and either a separated determinant residue or unique tropical-minimum witness;
4. logarithm units, Teichmuller or $(p-1)$-power normalization, truncation bound, and modular term list;
5. for a power series, the coefficient window, tail valuation bound, disc radius, and selected Strassmann or dominance criterion;
6. for a number field, defining polynomial, irreducibility and discriminant witnesses, units and norm checks, prime factorization, Hensel roots, and regulator minor;
7. the VC5 or VC7L locator used for the final external Return;
8. all non-claims and the precision ceiling.

Theorem 126.2 (UB-VC9-12: replay soundness)**.** *Acceptance of a VC9 certificate proves every declared valuation, nonvanishing, rank, zero-free, Strassmann, regulator, and external Return statement at the exact recorded precision and source strength.*

Proof. Ball and determinant steps are UB-VC9-01–UB-VC9-03. Logarithm replay is UB-VC9-04. Analytic claims are UB-VC9-05–UB-VC9-07 and require the recorded tail. Regulator and Leopoldt claims are UB-VC9-08 and the named external comparison. No accepted rule upgrades an unresolved ball or absent tail to nonvanishing. □

Classification 126.3 (VC9 roadmap status)**.** *The strongest family-level Challenge stops because no general effective logarithmic-independence or explicit-unit theorem is generated by UB alone. The panoramic alternative succeeds in two ways:*

- *a reusable certified nonvanishing and analytic-zero component library is closed;*
- *an explicit totally real cubic field and its S_3 closure are proved Leopoldt at $p = 61$ by a finite regulator certificate.*

VC9 is registered as Stable Grade B with a concrete external arithmetic closure. *No literature-first novelty claim is made.*

Non-claim 126.4 (VC9 promotion ceiling). VC9 does not assert:

- general Leopoldt, a new infinite Leopoldt family, or an effective general Baker–Brumer theorem;
- that a floating-point decimal or a residue with valuation at least the precision proves nonvanishing;
- that finitely many power-series coefficients determine zeros without a tail bound;
- a main conjecture, explicit reciprocity law, or regulator formula not separately bound as a source theorem;
- that the explicit $(F, 61)$ computation is a literature-first example;
- that GPT generation or successful compilation replaces independent mathematical checking.

Theorem 126.5 (UB-VC9-13: certified-nonvanishing success-case closure)**.** *VC9 closes the route*

$$\boxed{\begin{array}{c}\textit{finite DVR residues and certified tails}\\ \Downarrow \\ \textit{exact valuations, ranks, and zero bounds}\\ \Downarrow \\ \textit{regulator or special-value nonvanishing}\\ \Downarrow \\ \textit{VC5/VC7L arithmetic Return}\end{array}}$$

For $F = \mathbb{Q}(\alpha)$ with $\alpha^3 + 2\alpha^2 - 3\alpha - 1 = 0$, the route proves Leopoldt for $(F, 61)$ and for its S_3-Galois closure.

Proof. The first arrow is UB-VC9-01–UB-VC9-07. The regulator arrow is UB-VC9-08. The explicit witness is UB-VC9-09–UB-VC9-10. The named arithmetic Return is UB-VC9-11. Replay and strength control are UB-VC9-12 and the non-claim ceiling. □

VC9 source theorem bindings

The classical source regime consists of complete-DVR arithmetic, Hensel lifting, the p-adic logarithm, Strassmann's theorem, Newton polygon and Weierstrass preparation, Dirichlet's unit theorem, and the standard regulator formulation of Leopoldt. Baker–Brumer logarithmic independence and explicit-unit methods motivate the Challenge panorama but are not imported into the finite certificate unless a precise source theorem supplies their hypotheses and conclusion. VC5 supplies the previously validated S_3 fixed-field Return, and VC7L supplies the previously validated Iwasawa-series and specialization Return.

Part XXI

Validated Success Case X: Effective Iwasawa Layer Growth and Finite Coinvariant Return

127 VC10 objective and Challenge–panorama–Stable gate

Classification 127.1 (Challenge target)**.** *The strongest VC10 target is an exact finite-layer growth theorem for arbitrary finitely generated torsion modules over multivariable or noncommutative Iwasawa algebras, with the analytic characteristic element reconstructed by VC7L and its nonvanishing certified by VC9. A panoramic comparison was made among:*

1. *multivariable Cuoco–Monsky growth and higher-codimension support;*
2. *noncommutative Weierstrass and Akashi-series methods;*
3. *pseudo-isomorphism and elementary-module reduction;*
4. *one-variable square presentations and determinant resultants.*

In the first three lanes, a single determinant does not control pseudo-null error, higher-codimension support, all Tor terms, or an exact finite-layer constant without additional source theorems. The unrestricted Challenge therefore stops under C-STOP-1 and C-STOP-2. The Stable lane selects the strongest setting in which all arrows can be proved internally: one-variable Iwasawa algebras and square-presented torsion modules. Unlike a merely asymptotic fallback, the result below gives an exact formula from an explicit finite threshold and a replayable cyclotomic/logarithmic certificate.

Definition 127.2 (VC10 defect stack)**.** The VC10 defects are

$$D_{\mathrm{sq}}, \quad D_{\mathrm{cyc}}, \quad D_{\log}, \quad D_{\mathrm{Tor}}, \quad D_{\mathrm{growth}}, \quad D_{\mathrm{control}}.$$

They record, respectively, failure of a square presentation, a cyclotomic zero, an uncertified logarithmic norm, nonzero finite-layer Tor, failure of the exact length formula, and failure of the external arithmetic control map.

128 Square-presented Iwasawa modules

Let p be odd, let $\mathcal{O}$ be the ring of integers in a finite extension $K/\mathbb{Q}_p$, let ϖ be a uniformizer, and normalize

$$v_\varpi(\varpi) = 1, \qquad e = v_\varpi(p).$$

Set

$$\Lambda = \mathcal{O}[[T]], \qquad \omega_n(T) = (1+T)^{p^n} - 1, \qquad R_n = \Lambda/(\omega_n).$$

Then R_n is finite free of rank p^n over $\mathcal{O}$.

Definition 128.1 (Square-presented torsion object)**.** A VC10 object is an exact sequence

$$0 \longrightarrow \Lambda^r \xrightarrow{A(T)} \Lambda^r \longrightarrow X \longrightarrow 0$$

with $\det A(T) \neq 0$. Write its Weierstrass factorization as

$$\det A(T) = \varpi^\mu U(T) P(T),$$

where $\mu \geq 0$, $U(T) \in \Lambda^\times$, and $P(T) \in \mathcal{O}[T]$ is distinguished monic of degree λ.

Remark 128.2 (Why the square hypothesis is structural)*.* The square presentation converts every finite layer into one finite free endomorphism. Its $\mathcal{O}$-determinant is a norm/resultant, so length, Tor, and growth are controlled by one typed object. Without a square presentation or a separately certified perfect determinant complex, the characteristic element need not determine all finite-layer errors.

129 Finite cyclotomic-obstruction scan

For $m \geq 0$, let μ_{p^m} denote the p^m-power roots of unity, with $\mu_{p^0} = \{1\}$. Define

$$m_{\text{cyc}}(K, \lambda) = \max\{m \geq 0 : [K(\mu_{p^m}) : K] \leq \lambda\},$$

where the maximum is finite; for $\lambda = 0$ the set is empty and no scan is required.

Theorem 129.1 (UB-VC10-01: finite cyclotomic scan)**.** *Let P be distinguished of degree $\lambda > 0$. The following are equivalent:*

1. *$P(T)$ and $\omega_n(T)$ are coprime in $K[T]$ for every $n \geq 0$;*
2. *no root α of P satisfies $1 + \alpha \in \mu_{p^\infty}$;*
3. *for every $0 \leq m \leq m_{\text{cyc}}(K, \lambda)$, P has no root of the form $\zeta - 1$ with $\zeta \in \mu_{p^m}$.*

Thus the absence of all finite-layer cyclotomic zeros is decided by a finite factor/gcd certificate.

Proof. The roots of ω_n are exactly $\zeta - 1$ with $\zeta \in \mu_{p^n}$. Hence (1) and (2) are equivalent. If $P(\zeta - 1) = 0$, the minimal polynomial of $\zeta - 1$ over K divides P, so

$$[K(\zeta) : K] \leq \deg P = \lambda.$$

Therefore the order of ζ occurs within the finite range in (3). The converse is immediate. □

Corollary 129.2 (Cyclotomic failure witness)**.** *If the scan finds $P(\zeta - 1) = 0$ with $\zeta^{p^m} = 1$, then for every $n \geq m$ the finite layer $X/\omega_n X$ is not finite over $\mathcal{O}$ unless an independently displayed cancellation removes that factor. VC10 records this as D_{cyc} rather than assigning an Iwasawa growth formula.*

130 Logarithmic norm and effective exponential threshold

Assume the cyclotomic scan passes. Put

$$B = K[T]/(P), \qquad t = T \bmod P.$$

Because P is distinguished, t is topologically nilpotent and

$$\ell_P = \log(1+t) = \sum_{j\geq 1}(-1)^{j+1}\frac{t^j}{j}$$

converges in the finite K-algebra B. The cyclotomic scan implies that ℓ_P is a unit in B.

Definition 130.1 (Logarithmic growth data)**.** Let

$$C_P = v_\varpi\Big(N_{B/K}(\ell_P)\Big).$$

After extending v_ϖ to an algebraic closure and listing the roots $\alpha_1, \ldots, \alpha_\lambda$ of P with multiplicity, set

$$\rho_P = \min_i v_\varpi(\log(1+\alpha_i)).$$

Let $n_0(P)$ be the smallest nonnegative integer satisfying

$$n_0(P)e + \rho_P > \frac{e}{p-1}.$$

For $\lambda = 0$, set $C_P = 0$ and $n_0(P) = 0$.

Theorem 130.2 (UB-VC10-02: finite logarithmic-norm certificate)**.** *The values C_P, a certified lower bound for ρ_P, and hence a valid threshold $n_0(P)$ are finite VC9 certificates. Concretely, one may:*

1. *represent multiplication by a sufficiently long truncation of* $\log(1+t)$ *on the K-basis $1, t, \ldots, t^{\lambda-1}$;*
2. *use the VC9 logarithm-tail bound to enclose the exact multiplication matrix;*
3. *certify the determinant valuation C_P by UB-VC9-02;*
4. *use the Newton polygon of its characteristic polynomial to bound the smallest eigenvalue valuation ρ_P.*

No root approximation is logically required.

Proof. The logarithm truncation is controlled by UB-VC9-04 in the finite Banach algebra B. Multiplication and determinant are polynomial operations on the enclosed coefficients, so UB-VC9-01 and UB-VC9-02 certify the norm. The eigenvalues of multiplication by ℓ_P are $\log(1+\alpha_i)$ with multiplicity; Newton-polygon slopes therefore give their valuations. The cyclotomic scan makes every eigenvalue nonzero. □

Lemma 130.3 (UB-VC10-03: resultant valuation after the threshold)**.** *For every $n \geq n_0(P)$,*

$$v_\varpi \operatorname{Res}(P, \omega_n) = \lambda e n + C_P.$$

Proof. For each root α_i, put $L_i = \log(1+\alpha_i)$. The threshold gives

$$v_\varpi(p^n L_i) > \frac{e}{p-1}.$$

On this ball the p-adic exponential converges, is inverse to the logarithm, and satisfies

$$v_\varpi(\exp z - 1) = v_\varpi(z).$$

Hence
$$(1+\alpha_i)^{p^n} - 1 = \exp(p^n L_i) - 1$$
has valuation $ne + v_\varpi(L_i)$. Multiplying over the roots of the monic polynomial P gives
$$\mathrm{Res}(P, \omega_n) = \prod_i ((1+\alpha_i)^{p^n} - 1),$$
and summing valuations gives $\lambda en + C_P$. □

Corollary 130.4 (Small-root simplification)**.** *If every root α of P satisfies*
$$v_\varpi(\alpha) > \frac{e}{p-1},$$
then
$$C_P = v_\varpi(P(0)), \qquad n_0(P) = 0.$$

Proof. On the displayed ball, the first term of $\log(1+\alpha)$ has strictly smaller valuation than every later term, so $v_\varpi(\log(1+\alpha)) = v_\varpi(\alpha)$. The product of the roots is $P(0)$ up to sign, and the exponential threshold already holds at $n = 0$. □

131 Exact finite-layer growth theorem

Theorem 131.1 (UB-VC10-04: exact square-presentation growth)**.** *Let*
$$0 \to \Lambda^r \xrightarrow{A(T)} \Lambda^r \to X \to 0$$
be a VC10 square presentation, and suppose the cyclotomic scan for its distinguished factor P passes. Then for every $n \geq n_0(P)$:

1. *the specialized map*
$$A_n : R_n^r \longrightarrow R_n^r$$
is injective;
2. $$\mathrm{Tor}_1^\Lambda(X, R_n) = 0;$$
3. *$X/\omega_n X$ is a finite $\mathcal{O}$-module; and*
4. *its exact length is*
$$\boxed{\mathrm{length}_{\mathcal{O}}(X/\omega_n X) = \mu p^n + \lambda en + C_P.}$$

Thus the classical Iwasawa growth shape is not merely asymptotic in this lane: after the certified threshold it is an exact Return with a computable constant.

Proof. The $\mathcal{O}$-module R_n is free of rank p^n. Restriction of scalars gives
$$\det_{\mathcal{O}}(A_n) = N_{R_n/\mathcal{O}}(\det A \bmod \omega_n).$$
The unit U contributes an $\mathcal{O}$-unit, multiplication by ϖ^μ contributes valuation μp^n, and the distinguished factor contributes
$$v_\varpi \, \mathrm{Res}(P, \omega_n) = \lambda en + C_P$$
by UB-VC10-03. Hence $\det_{\mathcal{O}}(A_n) \neq 0$, so A_n is injective and its finite cokernel has length equal to the determinant valuation. Tensoring the square presentation with R_n identifies the kernel with $\mathrm{Tor}_1^\Lambda(X, R_n)$ and the cokernel with $X/\omega_n X$. □

Corollary 131.2 (UB-VC10-05: exact Fitting-to-layer Return)**.** *Suppose a perfect torsion complex or arithmetic Iwasawa module has a certified square presentation whose determinant generates its zeroth Fitting ideal. If an external control theorem identifies its finite arithmetic layer with $X/\omega_n X$, then for all $n \geq n_0(P)$ the arithmetic exponent is exactly*

$$\mu p^n + \lambda e n + C_P.$$

The square presentation, the determinant/Fitting comparison, and the arithmetic control map remain separately typed obligations.

132 Composition with VC7L and VC9

Theorem 132.1 (UB-VC10-06: finite-data Iwasawa growth closure)**.** *Assume the following certificates are present:*

1. *VC7L reconstructs an Iwasawa series or characteristic element from compatible finite distributions;*
2. *VC9 certifies its Weierstrass data (μ, P), the cyclotomic scan, the logarithmic norm C_P, and a valid threshold $n_0(P)$;*
3. *a square presentation identifies the element with $\det A$ up to a unit;*
4. *an external control theorem identifies the target arithmetic layer with $X/\omega_n X$.*

Then a finite UB certificate proves, uniformly for every $n \geq n_0(P)$,

$$\boxed{\mathrm{length}_{\mathcal{O}}(\textit{arithmetic layer}_n) = \mu p^n + \lambda e n + C_P.}$$

No unverified tower prefix, numerical interpolation, or omitted pseudo-null error is permitted in this Return.

Proof. VC7L supplies the exact Iwasawa element, VC9 supplies finite nonvanishing and valuation data, UB-VC10-04 supplies the algebraic finite-layer formula, and the named control theorem supplies the last external comparison. Every strength increase occurs at a recorded theorem boundary. □

Classification 132.2 (v21 promotion candidate)**.** *The composite*

$$\textit{finite distributions} \longrightarrow \textit{Iwasawa element} \longrightarrow (\mu, \lambda, C_P, n_0) \longrightarrow \textit{exact layer growth}$$

is the first UB component used across inverse-limit, p-adic analytic, nonvanishing, and arithmetic-layer lanes. It is therefore registered as the Finite Iwasawa Return Kernel*, a v21 Core-promotion candidate. Promotion is not automatic: it requires a separate Main/HT compatibility audit and does not import an arithmetic main conjecture or control theorem.*

133 Explicit finite replay example

Construction 133.1 (A square-presented module over the one-variable 3-adic Iwasawa algebra)**.** Let

$$\mathcal{O} = \mathbb{Z}_3, \qquad P(T) = T^2 + 9T + 9, \qquad A(T) = \begin{pmatrix} 3 & 0 \\ 0 & 3P(T) \end{pmatrix}.$$

Then
$$\det A(T) = 3^2 P(T), \qquad \mu = 2, \qquad \lambda = 2.$$

The Newton polygon of P has one segment of slope -1, so both roots have 3-adic valuation one. Since
$$1 > \frac{1}{3-1},$$
the small-root corollary gives
$$C_P = v_3(P(0)) = 2, \qquad n_0(P) = 0.$$

The only possible nontrivial 3-power cyclotomic factor of degree at most two is
$$\Phi_3(1+T) = T^2 + 3T + 3,$$
and it is coprime to P; also $T \nmid P$.

Theorem 133.2 (UB-VC10-07: explicit exact layer sequence)**.** *For*
$$X = \operatorname{coker} A(T)$$
and every $n \geq 0$*,*
$$\boxed{\operatorname{length}_{\mathbb{Z}_3}(X/\omega_n X) = 2 \cdot 3^n + 2n + 2.}$$
The first four values are

n	0	1	2	3
length	4	10	24	62.

Proof. All hypotheses of UB-VC10-04 hold with $(\mu, \lambda, e, C_P, n_0) = (2, 2, 1, 2, 0)$. Direct Sylvester-determinant replay gives
$$v_3 \operatorname{Res}(P, \omega_n) = 2n + 2$$
for $n = 0, 1, 2, 3$, and the theorem proves the same identity for every n. □

134 VC10 verifier, status, and non-claims

Verifier contract 134.1 (VC10 certificate)**.** A VC10 certificate serializes:

1. p, K, $\mathcal{O}$, ϖ, and $e = v_\varpi(p)$;
2. the square presentation matrix $A(T)$ and an exact determinant replay;
3. the Weierstrass tuple (μ, U, P, λ);
4. the finite cyclotomic scan and every gcd/factor witness;
5. a VC9 enclosure for multiplication by $\log(1+t)$ in $K[T]/P$;
6. C_P, a lower bound for ρ_P, and the threshold $n_0(P)$;
7. finite-layer resultant or norm determinants;
8. the external Fitting, characteristic-element, and control locators when arithmetic Return is requested;
9. the declared precision ceilings and all non-claims.

Theorem 134.2 (UB-VC10-08: VC10 replay soundness)**.** *Acceptance of a VC10 certificate proves the cyclotomic-free range, logarithmic norm, threshold, Tor vanishing, exact finite-layer length formula, and every declared arithmetic Return at the exact recorded source strength.*

Proof. The finite obstruction scan is UB-VC10-01. The analytic matrix and threshold are UB-VC10-02. The resultant valuation is UB-VC10-03. Tor and length are UB-VC10-04. External arithmetic statements are emitted only through UB-VC10-05 or UB-VC10-06 with named source locators. □

Classification 134.3 (VC10 roadmap status)**.** *The multivariable or noncommutative Challenge is not closed and is not converted into a conditional success claim. The one-variable square-presentation lane closes an exact theorem stronger than a bare asymptotic formula and executes a complete finite example. VC10 is registered as* Stable Grade B with a v21 Finite Iwasawa Return Kernel candidate.

Non-claim 134.4 (VC10 promotion ceiling). VC10 does not assert:

1. that every torsion Iwasawa module has a square presentation;
2. that characteristic ideals determine pseudo-null or higher-codimension errors;
3. an exact multivariable or noncommutative growth formula;
4. an arithmetic control theorem, main conjecture, or explicit reciprocity law;
5. finite coinvariant length when a cyclotomic factor survives;
6. that the displayed exact constant C_P equals $v_\varpi(P(0))$ outside the small-root regime;
7. literature-first novelty for the classical Weierstrass, resultant, or Iwasawa-growth constituents.

Theorem 134.5 (UB-VC10-09: effective Iwasawa-growth success-case closure)**.** *VC10 closes the route*

$$\boxed{\begin{array}{c}\textit{square presentation} \longrightarrow \textit{finite cyclotomic/log certificate}\\ \longrightarrow \textit{Tor-free finite layers} \longrightarrow \mu p^n + \lambda en + C_P\end{array}}$$

for every n beyond an explicit finite threshold, and composes the result with VC7L, VC9, Fitting/characteristic Return, and external arithmetic control without hiding any source obligation.

Proof. Combine UB-VC10-01 through UB-VC10-08. The explicit witness is UB-VC10-07, and the strength ceiling is the preceding non-claim. □

VC10 source theorem bindings

The classical source regime consists of one-variable Weierstrass preparation, finite free restriction of scalars, determinant norms and resultants, local p-adic logarithm/exponential theory, and the standard Iwasawa growth formalism. Multivariable Cuoco–Monsky, Akashi-series, pseudo-null, and noncommutative Weierstrass theories belong to the Challenge panorama only and are not imported into the Stable theorem. VC7L supplies finite-distribution and Iwasawa-series reconstruction;

VC9 supplies certified valuation and logarithmic computations; any arithmetic Fitting or control statement remains an external source theorem.

Part XXII

Validated Success Case XI: Cyclotomic-Singular Layers, Bockstein Return, and Reduced Euler Data

135 VC11 objective and Challenge–panorama–Stable gate

VC10 proves exact finite-layer length after a finite scan has excluded all cyclotomic zeros. VC11 addresses the branch that VC10 deliberately marked as D_{cyc}: a characteristic element may contain one or more factors

$$C_m(T) := \Phi_{p^m}(1+T), \qquad m \geq 0,$$

where $C_0(T) = T$. Such a factor divides every sufficiently deep

$$\omega_n(T) := (1+T)^{p^n} - 1 = \prod_{j=0}^{n} C_j(T),$$

so the ordinary layer is not finite over the coefficient DVR. The strongest Challenge target was an exact singular-layer theorem for every square presentation using only its determinant. That target is false.

Counterexample 135.1 (determinant data do not determine singular rank)**.** Over $\Lambda = \mathcal{O}[[T]]$, consider

$$A_1(T) = \begin{pmatrix} T & 0 \\ 0 & T \end{pmatrix}, \qquad A_2(T) = \begin{pmatrix} T & 1 \\ 0 & T \end{pmatrix}.$$

Both determinants equal T^2. Nevertheless,

$$\operatorname{coker} A_1 \cong \Lambda/(T) \oplus \Lambda/(T), \qquad \operatorname{coker} A_2 \cong \Lambda/(T^2).$$

At the trivial specialization, the first coinvariant has $\mathcal{O}$-rank two and the second has rank one. Therefore determinant order and leading coefficient alone cannot recover the complete singular layer of an arbitrary square presentation.

The panoramic alternatives are higher Bockstein spectral sequences, determinant-line and refined Euler-characteristic methods, local Smith theory at cyclotomic height-one primes, and exact cyclic elementary divisors. The first three give a natural research route, but an integral all-multiplicity theorem requires elementary-divisor or higher-Fitting data that are not contained in one determinant. VC11 therefore closes the strongest safe pair:

1. a square-presentation Bockstein theorem for transverse finite-character zeros;

2. an exact all-layer theorem for cyclic elementary divisors with arbitrary finite cyclotomic support, extended additively to exact finite direct sums.

Definition 135.2 (VC11 defect stack)**.** The VC11 defects are

$$D_{\mathrm{elem}}, \quad D_{\mathrm{trans}}, \quad D_{\mathrm{char}}, \quad D_{\mathrm{lead}}, \quad D_{\mathrm{ctrl}}.$$

They respectively record missing elementary-divisor data, failure of first-order transversality, unresolved character twisting, unresolved leading-coefficient precision, and a missing arithmetic control theorem.

136 Finite-character Bockstein and leading determinant coefficient

Let $\mathcal{O}$ be a complete DVR with uniformizer ϖ, fraction field K, and valuation v_ϖ. Put $\Lambda = \mathcal{O}[[T]]$.

Construction 136.1 (presentation Bockstein)**.** Let

$$A(T) = A_0 + TA_1 + T^2A_2 + \cdots \in M_d(\Lambda), \qquad \det A(T) \neq 0.$$

Set

$$H_0^{-1} := \ker(A_0 : \mathcal{O}^d \to \mathcal{O}^d), \qquad H_0^0 := \operatorname{coker}(A_0 : \mathcal{O}^d \to \mathcal{O}^d).$$

The first Bockstein map is

$$\beta_A : H_0^{-1} \longrightarrow H_0^0, \qquad v \longmapsto A_1 v \bmod A_0\mathcal{O}^d.$$

It is the connecting map attached to first-order motion in the T-direction.

Theorem 136.2 (UB-VC11-01: transverse Bockstein leading-term theorem)**.** *Let*

$$s = d - \operatorname{rank}_K(A_0 \otimes_{\mathcal{O}} K).$$

The following are equivalent:

1. $\operatorname{ord}_{T=0} \det A(T) = s$*;*
2. $\beta_A \otimes K$ *is an isomorphism between two* s*-dimensional* K*-spaces.*

When these conditions hold, β_A *is injective, its cokernel is finite, and*

$$\boxed{\operatorname{length}_{\mathcal{O}} \operatorname{coker}(\beta_A) = v_\varpi([T^s] \det A(T)).}$$

Thus a transverse cyclotomic zero Returns a finite Bockstein index rather than an invalid finite coinvariant length.

Proof. Choose $U, V \in \mathrm{GL}_d(\mathcal{O})$ putting UA_0V in Smith form

$$\begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix},$$

where D is a nonsingular diagonal matrix of size $d - s$. Unit row and column operations preserve the valuation of the leading determinant coefficient and the isomorphism class of the Bockstein cokernel. In the corresponding block decomposition write

$$UA_1V = \begin{pmatrix} B & C \\ E & F \end{pmatrix}.$$

A block determinant expansion gives

$$\det(UA(T)V) = T^s \det(D)\det(F) + O(T^{s+1}).$$

Hence the order is exactly s if and only if F is nonsingular over K, which is precisely the condition that $\beta_A \otimes K$ be an isomorphism. The cokernel of β_A has the square presentation

$$\begin{pmatrix} D & C \\ 0 & F \end{pmatrix}.$$

Its length is the valuation of the determinant $\det(D)\det(F)$, equal to the valuation of $[T^s]\det A(T)$. Since the source of β_A is free and the map is injective over K, its integral kernel is zero. $\square$

Corollary 136.3 (UB-VC11-02: finite-character twist Return)**.** *Let χ be a finite-order character of the procyclic group generated by $\gamma = 1+T$, and enlarge $\mathcal{O}$ to a complete DVR $\mathcal{O}_\chi$ containing $\chi(\gamma)$. Define*

$$A_\chi(S) := A\big(\chi(\gamma)(1+S) - 1\big).$$

If

$$s_\chi = d - \operatorname{rank}_{K_\chi} A_\chi(0) = \operatorname{ord}_{S=0} \det A_\chi(S),$$

then the twisted Bockstein $\beta_{A,\chi}$ has finite cokernel and

$$\operatorname{length}_{\mathcal{O}_\chi} \operatorname{coker}(\beta_{A,\chi}) = v_{\varpi_\chi}\left([S^{s_\chi}]\det A_\chi(S)\right).$$

Every simple finite-character zero is therefore reduced to the trivial-character theorem by a typed twist.

137 Exact cyclic cyclotomic-singular layer theorem

For $m \geq 1$ put

$$d_m = \deg C_m = p^{m-1}(p-1), \qquad d_0 = 1.$$

The prime-power cyclotomic resultant identity gives, for $m \neq j$,

$$v_\varpi \operatorname{Res}(C_m, C_j) = e\, d_{\min(m,j)}, \qquad e = v_\varpi(p).$$

Definition 137.1 (cyclotomic elementary-divisor certificate)**.** A VC11 cyclic certificate consists of

$$f(T) = \varpi^\mu P(T) \prod_{m\in S} C_m(T)^{a_m},$$

where $S \subset \mathbb{Z}_{\geq 0}$ is finite, every $a_m \geq 1$, and P is a distinguished polynomial of degree λ with no p-power cyclotomic factor. A unit factor is omitted because it does not change the principal ideal. Let

$$R_S(T) = \prod_{m\in S} C_m(T), \qquad D_S = \deg R_S = \sum_{m\in S} d_m.$$

For $n \geq \max S$, set

$$u_{n,S}(T) = \frac{\omega_n(T)}{R_S(T)}.$$

Finally define

$$\kappa_m(n,S) := e \sum_{\substack{0\leq j\leq n \\ j\notin S}} d_{\min(m,j)}.$$

All terms are finite and explicitly computable.

Lemma 137.2 (UB-VC11-03: exact singular-layer sequence)**.** *Let $X = \Lambda/(f)$ be a VC11 cyclic certificate and let $n \geq \max S$. Put*

$$h(T) = \varpi^{\mu} P(T) \prod_{m \in S} C_m(T)^{a_m - 1}.$$

Then multiplication by R_S gives a short exact sequence of $\mathcal{O}$-modules

$$\boxed{0 \longrightarrow \Lambda/(h, u_{n,S}) \xrightarrow{\cdot R_S} X/\omega_n X \longrightarrow \Lambda/(R_S) \longrightarrow 0.}$$

The left term has finite $\mathcal{O}$-length and the right term is free of rank D_S.

Proof. The two defining equations are $f = R_S h$ and $\omega_n = R_S u_{n,S}$. If $R_S x$ lies in $(R_S h, R_S u_{n,S})$, cancellation in the domain Λ gives $x \in (h, u_{n,S})$, proving injectivity. Quotienting the middle term by the image of multiplication by R_S gives $\Lambda/(R_S)$. The latter is free because R_S is monic. The polynomials h and $u_{n,S}$ are coprime over $K[T]$: every cyclotomic factor occurring in h has been removed from $u_{n,S}$, and P is cyclotomic-free. Hence the left term is finite. □

Theorem 137.3 (UB-VC11-04: exact cyclotomic rank and torsion formula)**.** *Assume the VC10 regular-factor certificate for P gives*

$$v_{\varpi} \operatorname{Res}(P, \omega_n) = \lambda e n + C_P \qquad (n \geq n_0(P)).$$

Then for every

$$n \geq N_0 := \max(\max S, n_0(P))$$

one has

$$\boxed{\operatorname{rank}_{\mathcal{O}}(X/\omega_n X) = D_S}$$

and

$$\boxed{\begin{aligned} \operatorname{length}_{\mathcal{O}}(X/\omega_n X)_{\text{tors}} = \mu(p^n - D_S) + \sum_{m \in S} (a_m - 1)\kappa_m(n, S) \\ + \lambda e n + C_P - \sum_{m \in S} v_{\varpi} \operatorname{Res}(P, C_m). \end{aligned}}$$

Thus every persistent cyclotomic zero is separated into a free rank contribution and an exact finite reduced sector.

Proof. By UB-VC11-03 the torsion submodule of the middle term is the finite left term, because the right term is $\mathcal{O}$-free. Since $u_{n,S}$ is monic of degree $p^n - D_S$, restriction of scalars identifies the left-term length with

$$v_{\varpi} \operatorname{Res}(h, u_{n,S}).$$

Multiplicativity of the resultant gives the contribution $\mu(p^n - D_S)$ from ϖ^{μ}, the contribution $(a_m - 1)\kappa_m(n, S)$ from each repeated cyclotomic factor, and

$$v_{\varpi} \operatorname{Res}(P, u_{n,S}) = v_{\varpi} \operatorname{Res}(P, \omega_n) - \sum_{m \in S} v_{\varpi} \operatorname{Res}(P, C_m).$$

Insert the VC10 formula. The rank statement follows from the exact sequence. □

Corollary 137.4 (UB-VC11-05: exact direct-sum singular profile)**.** *For an exact finite direct sum*

$$X = \bigoplus_{i=1}^{q} \Lambda/(f_i)$$

of VC11 cyclic certificates, the finite-layer rank and torsion length are the sums of the corresponding UB-VC11-04 formulas. No pseudo-isomorphism is substituted for the displayed exact decomposition.

138]

replay]Complete $\mathbb{Z}_3[[T]]$ replay

Take $p = 3$, $\mathcal{O} = \mathbb{Z}_3$, and

$$C_0(T) = T, \qquad C_1(T) = T^2 + 3T + 3, \qquad P(T) = T^2 + 9T + 9.$$

Let

$$X = \Lambda/(3T^2 C_1(T)P(T)).$$

Thus

$$\mu = 1, \quad S = \{0, 1\}, \quad a_0 = 2, \quad a_1 = 1, \quad \lambda = 2, \quad D_S = 3.$$

VC10 gives

$$v_3 \operatorname{Res}(P, \omega_n) = 2n + 2.$$

Direct calculation gives

$$v_3 \operatorname{Res}(P, C_0) = 2, \qquad \operatorname{Res}(P, C_1) = 36, \qquad v_3(36) = 2.$$

Moreover

$$\kappa_0(n, S) = n - 1 \qquad (n \geq 1).$$

Theorem 138.1 (UB-VC11-06: explicit singular layer sequence)**.** *For every* $n \geq 1$*,*

$$\boxed{\operatorname{rank}_{\mathbb{Z}_3}(X/\omega_n X) = 3}$$

and

$$\boxed{\operatorname{length}_{\mathbb{Z}_3}(X/\omega_n X)_{\mathrm{tors}} = 3^n + 3n - 6.}$$

The first values are

n	1	2	3	4
rank	3	3	3	3
length(tors)	0	9	30	87.

At $n = 1$ *the layer is purely free; deeper layers acquire an exactly controlled torsion sector.*

Proof. Apply UB-VC11-04:

$$\begin{aligned} \operatorname{length}(\mathrm{tors}) &= (3^n - 3) + (n - 1) + (2n + 2) - 2 - 2 \\ &= 3^n + 3n - 6. \end{aligned}$$

For independent replay, put

$$h(T) = 3TP(T), \qquad u_{n,S}(T) = \frac{\omega_n(T)}{TC_1(T)}.$$

The Sylvester resultants satisfy

$$v_3 \operatorname{Res}(h, u_{n,S}) = 0, 9, 30 \quad \text{for } n = 1, 2, 3,$$

matching the formula exactly. ∎

139 VC11 verifier, status, and non-claims

Verifier contract 139.1 (VC11 certificate)**.** A VC11 certificate serializes:

1. the complete DVR, p, e, and precision ceiling;
2. the presentation matrix and its first jet for every Bockstein claim;
3. the rank drop, determinant order, leading coefficient, and Bockstein matrix;
4. the finite cyclotomic support S, exponents a_m, and factors C_m;
5. the regular factor P and its VC10 tuple (λ, C_P, n_0);
6. every Res(P, C_m) and $\kappa_m(n, S)$ calculation;
7. the exact short sequence and resultant determinant for each replayed layer;
8. any external arithmetic control theorem as a separately named source binding.

Theorem 139.2 (UB-VC11-07: VC11 replay soundness)**.** *Acceptance of a VC11 certificate proves every declared transverse Bockstein index, finite-layer rank, and reduced torsion length at the exact displayed strength.*

Proof. UB-VC11-01 and UB-VC11-02 verify the square-presentation transverse lane. UB-VC11-03 verifies the exact singular decomposition. UB-VC11-04 computes the rank and torsion length. UB-VC11-06 provides the complete numerical replay. No arithmetic object is identified with the module without a named external control theorem. □

Classification 139.3 (VC11 roadmap status)**.** *The determinant-only arbitrary-square-presentation Challenge is false and is rejected by an explicit counterexample. The panorama identifies higher Bockstein and refined determinant methods as the correct future theory. The transverse square-presentation lane and the arbitrary-finite-cyclotomic-support cyclic lane are completely closed. VC11 is registered as* Stable Grade B and closes the VC10 cyclotomic defect on the declared exact elementary-divisor class.

Non-claim 139.4 (VC11 promotion ceiling). VC11 does not assert:

1. that a determinant determines the singular layer of an arbitrary presentation;
2. an integral higher-Bockstein theorem for arbitrary repeated elementary divisors in a nonsplit square presentation;
3. that a pseudo-isomorphism preserves the exact finite-layer rank or torsion constant;
4. an arithmetic control theorem, main conjecture, or exceptional-zero formula;
5. a finite ordinary length when the layer has positive $\mathcal{O}$-rank.

Theorem 139.5 (UB-VC11-08: cyclotomic-singular success-case closure)**.** *VC11 closes the route*

$$\begin{aligned} \textit{cyclotomic characteristic factors} &\longrightarrow \textit{typed free sector} \\ &\longrightarrow \textit{Bockstein or exact cyclic reduction} \\ &\longrightarrow \textit{leading coefficient/resultant} \\ &\longrightarrow \textit{rank and reduced torsion Return}. \end{aligned}$$

The branch rejected by VC10 is therefore no longer unresolved on the declared exact elementary-divisor class.

Proof. Combine UB-VC11-01 through UB-VC11-07. The determinant-only ceiling and the exact class boundary are enforced by the counterexample and the preceding non-claim. ☐

VC11 source theorem bindings

The classical source regime consists of prime-power cyclotomic resultant identities, Smith normal form over a DVR, determinant and Bockstein descriptions of leading terms, and the truncated Euler-characteristic relation between a characteristic-series leading coefficient and positive-rank Iwasawa cohomology. Higher Bockstein spectral sequences, refined Euler characteristics, and noncommutative leading-term conjectures remain panorama inputs only. The exact cyclic short sequence and the closed rank–torsion formula are proved above; any arithmetic interpretation remains a named external control obligation.

Part XXIII

Governance, Registry, Non-Claims, and Final Closure

140 Minimal UB verifier certificate

Verifier contract 140.1 (Canonical UB certificate)**.** A canonical certificate serializes at least:

1. source and target identities, versions, categories, and coefficient rings;
2. requested Return strength and equivalence relation;
3. realization functors or procedures and variance;
4. finite generator family, amplitude, and probe data;
5. coefficient fibers, flatness, Jacobson, perfection, and completion data;
6. generation bounds, finite-state transition tables, or other tail theorem;
7. local objects, overlap maps, and cocycle cells;
8. natural transformations and generator-level coherence checks;
9. AK Core and HT local theorem locators and interface certificates;
10. native target comparison, rigidity theorem, and promotion ceiling;
11. analytic energy, compactness, residual, norm-atlas, and continuation fields when VC8 is invoked;
12. complete-DVR precision, valuation, determinant, logarithm, power-series tail, and regulator fields when VC9 is invoked;

13. square-presentation, Weierstrass, cyclotomic-scan, logarithmic-norm, threshold, resultant, Tor, and control fields when VC10 is invoked;
14. Bockstein-jet, character-twist, cyclotomic-support, free-rank, reduced-resultant, and exact elementary-divisor fields when VC11 is invoked;
15. the complete defect stack and deterministic primary status;
16. explicit non-claims and excluded strengths.

Protocol 140.2 (GPT-assisted construction discipline)**.** *A GPT-assisted UB construction must proceed in the following order:*

1. *type the source and target before proposing a bridge;*
2. *choose the weakest sufficient Return strength;*
3. *select only components whose positive regimes match the source;*
4. *prove finite coverage rather than infer it from finiteness of observed data;*
5. *construct actual maps, functors, cocycles, and naturality cells;*
6. *expose every remaining domain theorem as a named non-scalar obligation;*
7. *run the status precedence rule without optimistic filling of missing data;*
8. *emit the composite theorem only after component closure.*

141 Canonical theorem locator registry

Every locator in this registry is a stable textual identifier attached to an actual theorem, lemma, corollary, verifier contract, or closure environment in this source. Layout-derived theorem numbers may change under optimization; the UB locators do not.

UB locator	Statement	Strength
UB-CORE-01	Conservative UB–HT compatibility	conditional composition
UB-GEN-01	Finite-amplitude generator detection	zero predicate
UB-GEN-02	Finite naturality-cell extension	global naturality
UB-MOR-01	Finite-category projective-generator realization	object, morphism, exact, derived
UB-FIN-01	UB finite realization and isomorphism closure	finite conservative realization
UB-FF-01	Faithfully flat derived zero-reflection	derived zero and isomorphism
UB-NAK-01	Perfect derived Nakayama	finite-fiber zero and isomorphism
UB-NAK-02	Finite semilocal fiber atlas	finite coefficient atlas
UB-COMP-01	Perfect completion zero-reflection	source, completion, special fiber
UB-COEFF-01	Finite coefficient-completion Return	composite coefficient Return
UB-NOETH-01	Noetherian finite-window zero theorem	all-degree vanishing
UB-NOETH-02	Finite-window morphism theorem	all-degree isomorphism

UB locator	Statement	Strength
UB-STATE-01	Finite-state terminality theorem	exact finite tail decision
UB-RL-01	Eventually constant inverse-system Return	limit and $\varprojlim^1$
UB-RL-02	Finite-state derived-limit closure	finite-to-infinite Return
UB-DESC-01	Effective faithfully flat module descent	object/morphism Return
UB-DESC-02	Perfect-complex descent	derived object Return
UB-DESC-03	Finite Čech-to-derived Return	global derived sections
UB-DESC-04	UB finite descent closure	finite local-to-global Return
UB-KUN-01	Field Kunneth Return	tensor cohomology
UB-DUAL-01	Perfect duality zero-reflection	dual derived Return
UB-RIG-01	Generator criterion for rigidity	fully faithful and conservative
UB-RIG-02	Rigidity promotion theorem	stronger external Return
UB-COH-01	Finite coherence-cell certification	global coherence
UB-HT-01	UB–HT component composition	composite Return
UB-CLOSE-01	Universal Finite Bridge Closure	finite external Return
UB-CLOSE-02	Certificate complexity bound	explicit finite block bound
UB-SC-01	Integral box-rank zeta transform	exact finite reconstruction input
UB-SC-02	Explicit $2d$-difference inversion	exact object Return
UB-SC-03	Minimal additive scalar probe theorem	sharp MinCert lower bound
UB-SC-04	Box-rank realizability and verifier theorem	finite replay certificate
UB-SC-05	2×2 nine-probe computation	complete numerical success case
UB-SC-06	UB exact rank-probe success-case closure	fully executed external theorem
UB-VC2-01	Exact enriched projective-probe equivalence	object and morphism Return
UB-VC2-02	Exact structural and cone Return	kernels, cokernels, exact, derived
UB-VC2-03	Yoneda-class Return	all-degree extension-class Return
UB-VC2-04	Minimal basic projective-generator theorem	sharp atomic probe minimum
UB-VC2-05	Two-by-two enriched execution	complete finite matrix success case
UB-VC2-06	Scalar-to-enriched UB composition	strength-separated VC1–VC2 pipeline
UB-VC2-07	Enriched projective-probe closure	fully executed structural theorem
UB-VC3-01	Finite syzygy-permutation Tachikawa theorem	one-degree projectivity detection
UB-VC3-02	Representation-finite self-injective closure	explicit finite Tachikawa window
UB-VC3-03	Exact Nakayama syzygy formula	finite type permutation

UB locator	Statement	Strength
UB-VC3-04	Uniform truncated-cycle periodicity	period dividing $2m/\gcd(m, L)$
UB-VC3-05	Truncated-cycle Tachikawa closure	complete external family theorem
UB-VC3-06	Local truncated-polynomial sharpening	degree-one projectivity detection
UB-VC3-07	Finite Tachikawa verifier and route	replayable stable-Ext certificate
UB-VC3-08	UB-VC3 success-case closure	fully executed conjecture-family theorem
UB-VC4-01	Stable bimodule-to-module syzygy transport	natural stable Return
UB-VC4-02	Periodic one-degree Tachikawa theorem	exact projectivity detector
UB-VC4-03	Orbit-sensitive twisted Tachikawa theorem	module-specific finite Return
UB-VC4-04	Weighted-surface degree-four closure	representation-infinite family theorem
UB-VC4-05	Generalized-quaternion degree-four closure	tame symmetric family theorem
UB-VC4-06	Uniform degree-six preprojective closure	Dynkin preprojective theorem
UB-VC4-07	Finite replay soundness	verifier theorem
UB-VC4-08	Typed AK route	exact bridge composition
UB-VC4-09	UB-VC4 success-case closure	periodic-family Tachikawa theorem
UB-VC5-01	Exact fixed-field Leopoldt descent	arithmetic fixed-point Return
UB-VC5-02	Finite Galois defect reconstruction criterion	fixed-point probe theorem
UB-VC5-03	Exact S_3 Leopoldt-module Return	unimodular module reconstruction
UB-VC5-04	Minimal S_3 two-subfield certificate	subgroup-probe MinCert
UB-VC5-05	Exact Frobenius Leopoldt defect relation	non-abelian propagation theorem
UB-VC5-06	Finite replay soundness	arithmetic verifier theorem
UB-VC5-07	Typed arithmetic route	exact bridge composition
UB-VC5-08	UB-VC5 success-case closure	Galois–Leopoldt Return theorem
UB-VC6-01	Exact Alexandrov sheaf–representation equivalence	finite conservative realization
UB-VC6-02	Effective finite object descent	object, morphism, and exact Return
UB-VC6-03	Coherence-depth-two theorem	sharp descent coherence
UB-VC6-04	Exact bounded-derived descent	complexes, cones, and triangles
UB-VC6-05	All-degree Yoneda-class descent	individual extension-class Return

UB locator	Statement	Strength
UB-VC6-06	Principal-good Čech hypercohomology Return	finite derived global sections
UB-VC6-07	Finite replay soundness	descent verifier theorem
UB-VC6-08	UB-VC6 success-case closure	object-to-derived descent theorem
UB-VC7-01	Finite-dimensional Mittag–Leffler theorem	exact $\varprojlim^1$ vanishing
UB-VC7-02	Finite-state inverse-limit Return	stable-image reconstruction
UB-VC7-03	Finite apex-agreement criterion	one-level rigidity Return
UB-VC7-04	Finite-state derived-limit Return	Milnor-defect closure
UB-VC7-05	One-endomorphism cohomology Return	recurrent-sector theorem
UB-VC7-06	Surjective-tail derived exactness	completion interface
UB-VC7-07	Finite replay soundness	inverse-limit verifier theorem
UB-VC7-08	Typed UB–RL–Return route	exact bridge composition
UB-VC7-09	UB-VC7 success-case closure	finite-state inverse-limit theorem
UB-VC7L-01	Compatible finite group-ring Return	measure/Iwasawa equivalence
UB-VC7L-02	Amice power-series equivalence	exact Mahler Return
UB-VC7L-03	Specialization triangle	character/Mellin Return
UB-VC7L-04	Finite Fourier reconstruction	sharp scalar probe count
UB-VC7L-05	Binomial valuation theorem	precision bound input
UB-VC7L-06	One-layer truncated Iwasawa Return	finite precision closure
UB-VC7L-07	Branch and Mellin Return	weight-space specialization
UB-VC7L-08	Special-value distribution criterion	exact interpolation certificate
UB-VC7L-09	Pseudomeasure specialization Return	controlled pole lane
UB-VC7L-10	Kubota–Leopoldt Return binding	exact known-theorem recovery
UB-VC7L-11	Arithmetic-to-analytic Return	explicit reciprocity interface
UB-VC7L-12	Determinant and characteristic Return	algebraic Iwasawa interface
UB-VC7L-13	p-adic L-value to Leopoldt Return	VC5 composition
UB-VC7L-14	Finite replay soundness	Iwasawa verifier theorem
UB-VC7L-15	VC7L success-case closure	complete p-adic L Return chain
UB-VC8-01	Finite norm-window partition	explicit finite atlas bound
UB-VC8-02	Finite $L^6_t L^6_x$ atlas	local norm certificate
UB-VC8-03	Finite Serrin-atlas Return	regularity and uniqueness
UB-VC8-04	Local Aubin–Lions Return	strong compactness

UB locator	Statement	Strength
UB-VC8-05	Nonlinear product-defect annihilation	convection-term Return
UB-VC8-06	Compactness-defect closure	defect-free weak limit
UB-VC8-07	Approximation-to-regularity Return	regular solution closure
UB-VC8-08	Finite-window approximation certificate	replayable analytic atlas
UB-VC8-09	Fixed-exponent continuation Return	blow-up alternative
UB-VC8-10	Defect-free continuation package	analytic Return chain
UB-VC8-11	Typed analytic bridge route	exact composition
UB-VC8-12	Finite replay soundness	analytic verifier theorem
UB-VC8-13	Analytic success-case closure	compactness-to-continuation theorem
UB-VC9-01	Ball-stable valuation and nonvanishing	exact finite-precision certificate
UB-VC9-02	Determinant-ball certificate	rank and valuation Return
UB-VC9-03	Unique tropical-minimum determinant	cancellation-free nonvanishing
UB-VC9-04	Finite p-adic logarithm truncation	certified logarithm residue
UB-VC9-05	Certified zero-free disc	analytic nonvanishing
UB-VC9-06	Finite Strassmann certificate	zero-count Return
UB-VC9-07	Certified Iwasawa μ–λ data	finite coefficient Return
UB-VC9-08	Certified regulator-minor Return	Leopoldt criterion
UB-VC9-09	Complete splitting and Hensel certificate	explicit local embeddings
UB-VC9-10	Explicit 61-adic regulator certificate	exact valuation-two witness
UB-VC9-11	Explicit Leopoldt closure	cubic and S_3 Return
UB-VC9-12	VC9 replay soundness	nonvanishing verifier theorem
UB-VC9-13	Certified-nonvanishing closure	fully executed arithmetic theorem
UB-VC10-01	Finite cyclotomic scan	obstruction-free range
UB-VC10-02	Finite logarithmic-norm certificate	effective analytic threshold
UB-VC10-03	Resultant valuation after the threshold	exact norm growth
UB-VC10-04	Exact square-presentation growth	Tor-free finite-layer length
UB-VC10-05	Exact Fitting-to-layer Return	controlled arithmetic specialization
UB-VC10-06	Finite-data Iwasawa growth closure	VC7L–VC9 composition
UB-VC10-07	Explicit exact layer sequence	complete numerical replay
UB-VC10-08	VC10 replay soundness	verifier theorem
UB-VC10-09	Effective Iwasawa-growth closure	complete declared-class theorem
UB-VC11-01	Transverse Bockstein leading-term theorem	finite leading-index Return

UB locator	Statement	Strength
UB-VC11-02	Finite-character twist Return	characterwise singular Return
UB-VC11-03	Exact singular-layer sequence	free/finite decomposition
UB-VC11-04	Exact cyclotomic rank and torsion formula	all-layer reduced Return
UB-VC11-05	Exact direct-sum singular profile	additive elementary-divisor Return
UB-VC11-06	Explicit $\mathbb{Z}_3[[T]]$ singular sequence	complete numerical replay
UB-VC11-07	VC11 replay soundness	singular-layer verifier theorem
UB-VC11-08	Cyclotomic-singular success-case closure	complete declared-class theorem

142 Explicit non-claims

Appendix UB makes none of the following claims:

1. every external mathematical object admits a finite projective-generator model;
2. every realization functor is conservative, fully faithful, monoidal, or exact;
3. field-valued homology or persistence determines integral, torsion, completed, or multiplicative structure;
4. finite slices classify arbitrary multiparameter persistence modules;
5. a finite tower prefix determines an arbitrary inverse limit or derived completion;
6. local objects descend without an effective cocycle theorem;
7. a Čech complex computes derived global sections without the displayed acyclicity or another convergence theorem;
8. a dualizable tensor factor is automatically conservative;
9. equality of characteristic ideals, determinants, ranks, barcodes, or Ext dimensions implies exact object isomorphism;
10. the finite-state Tachikawa theorem proves the general Tachikawa conjecture without a stable-complete finite state package;
11. the Universal Finite Bridge Closure Theorem proves a named open problem before all domain adapters and hypotheses are discharged;
12. scalar rank probes recover arbitrary morphisms, Yoneda classes, or non-box-decomposable representations;
13. enriched projective probes for finite posets imply finite probes for arbitrary infinite diagram categories;
14. the VC3 truncated-cycle theorem solves Tachikawa's second conjecture for arbitrary self-injective algebras;

15. the displayed uniform Nakayama period is asserted to be the smallest possible self-Ext detection degree;
16. every self-injective algebra is periodic, or every twisted-periodic algebra has finite outer twist order;
17. the VC4 displayed periodicity degree is always minimal for projectivity detection;
18. the periodicity of weighted surface, generalized-quaternion, or Dynkin preprojective algebras is newly proved here;
19. the VC5 fixed-point certificate proves primitive p-adic regulator nonvanishing without a subfield arithmetic input;
20. the S_3 subgroup-probe minimality is minimal among all conceivable arithmetic certificates;
21. the Frobenius or cubic-subfield defect relations are asserted to be literature-first results;
22. the VC6 strict descent theorem identifies naive descent data internal to arbitrary derived categories with effective higher-coherent descent;
23. pairwise local isomorphism types, without actual overlap maps and the triple cocycle, determine a global sheaf or complex;
24. a non-principal-good cover computes hypercohomology without an independent acyclicity or convergence certificate;
25. a finite quotient prefix, without a certified finite-state, surjective, or domain-specific tail theorem, determines an inverse limit;
26. finite-state equality of dimensions or ranks replaces equality of the typed bonding matrices;
27. VC7 proves primitive Iwasawa control, adic completeness, or Leopoldt regulator nonvanishing;
28. VC7L constructs primitive arithmetic p-adic L-functions, proves explicit reciprocity, main conjectures, general nonvanishing, or general Leopoldt;
29. VC8 proves unconditional three-dimensional Navier–Stokes regularity, manufactures a Serrin norm from energy or samples, or proves a new endpoint or epsilon-regularity theorem;
30. VC9 proves general Leopoldt, an effective general Baker–Brumer theorem, or nonvanishing from unresolved finite precision or an uncertified analytic tail;
31. VC10 proves a multivariable or noncommutative exact growth theorem, removes pseudo-null errors, supplies an arithmetic control theorem, or assigns finite length across an unresolved cyclotomic zero;
32. VC11 claims that one determinant determines every singular presentation, that pseudo-isomorphism preserves exact singular layers, or that positive-rank layers have an ordinary finite length;
33. GPT generation, successful compilation, or internal consistency is itself mathematical proof.

143 Final closure theorem

Theorem 143.1 (Appendix UB canonical closure)**.** *Appendix UB forms a single internally complete universal bridge library with:*

1. *one bridge-component type and one strength architecture;*
2. *positive finite-generator, coefficient, completion, Noetherian, finite-state, descent, monoidal, duality, coherence, and rigidity theorems;*
3. *explicit counterexamples and promotion ceilings;*
4. *exact interfaces to AK v20 and each principal HT transfer layer;*
5. *a composable UB–HT proof-DAG theorem;*
6. *the Universal Finite Bridge Closure Theorem;*
7. *nontrivial application theorems with clearly isolated domain obligations;*
8. *one fully executed finite-grid rank-probe success case with exact reconstruction, sharp MinCert, replay verification, and arbitrary-dimensional extension;*
9. *one fully executed finite-poset enriched-probe success case with exact morphism, exact-sequence, Yoneda, derived Return, and sharp atomic projective-generator minimality;*
10. *one fully executed Tachikawa success case for every self-injective truncated-cycle Nakayama algebra, with explicit syzygy permutation, one-degree self-Ext detection, finite verification, and representation-finite self-injective expansion;*
11. *one fully executed periodic-algebra Tachikawa success case, with natural bimodule-to-module syzygy transport, a twisted-orbit refinement, representation-infinite weighted-surface and quaternion-type degree-four closures, a Dynkin preprojective degree-six closure, and finite replay;*
12. *one fully executed finite Galois–Leopoldt success case, with exact fixed-field descent, unimodular S_3 kernel reconstruction, a minimal two-subfield zero certificate, Frobenius defect propagation, odd-dihedral one-subfield Return, and finite replay;*
13. *one fully executed finite Alexandrov descent success case, with explicit object and morphism gluing, sharp triple coherence, exact and all-degree Yoneda Return, bounded-derived descent, principal-good Čech hypercohomology, and finite replay;*
14. *one fully executed finite-state inverse-limit success case, with exact stable-image reconstruction, $\varprojlim^1$ vanishing, derived Milnor Return, finite apex and cone criteria, a surjective completion lane, and deterministic replay;*
15. *one fully executed p-adic L-Return success case, with compatible finite group-ring reconstruction, exact measure and Amice equivalence, sharp finite Fourier recovery, explicit one-level precision bounds, weight-branch Mellin Return, controlled pseudomeasures, Kubota–Leopoldt exact recovery, and typed arithmetic, algebraic, and Leopoldt interfaces;*
16. *one fully executed analytic compactness and continuation success case, with finite Serrin atlases, local Aubin–Lions compactness, nonlinear-defect annihi-*

lation, approximation-to-regularity, weak–strong uniqueness, blow-up alternative, continuation, and finite replay;

17. *one fully executed certified-nonvanishing success case, with complete-DVR balls, determinant and tropical-minor witnesses, finite logarithm truncation, Strassmann and zero-free-disc certificates, Iwasawa-order Return, a concrete* 61*-adic regulator certificate, and Leopoldt closure for a totally real cubic field and its* S_3*-Galois closure;*
18. *one fully executed effective Iwasawa-growth success case, with a finite cyclotomic scan, VC9-certified logarithmic norm and threshold, Tor-free square-presentation specialization, exact* $\mu p^n + \lambda e n + C_P$ *layer lengths, a complete* $\mathbb{Z}_3[[T]]$ *replay example, and a v21 Finite Iwasawa Return Kernel candidate;*
19. *one fully executed cyclotomic-singular success case, with a determinant insufficiency counterexample, transverse Bockstein leading-term Return, arbitrary finite cyclotomic support for exact cyclic elementary divisors, exact rank and reduced torsion formulas, and a complete rank-three* $\mathbb{Z}_3[[T]]$ *replay;*
20. *one verifier certificate and one theorem locator registry;*
21. *one Appendix-local derived-theorem pack containing certified implementation substitution, finite Pareto strength and blocker-frontier calculus, product Return, finite-state descent/derived-limit composition, field-extension reflection, and a unified regular–singular one-variable layer theorem.*

No additional appendix is required to define, compose, audit, or Return the UB components at the strengths proved here. New domain kits extend UB by supplying source realizations and rigidity theorems through the existing component contract; they do not require a new universal bridge grammar.

Proof. The component type, defect stack, and status precedence are fixed in Parts I and II. Parts II–VI prove the positive bridge components and their failure boundaries. Part VII defines and proves their composition with AK and HT, culminating in the finite closure theorem. The validated case parts supersede the former pre-validation application sketches and isolate the remaining domain-specific obligations. The first validated success-case part executes the bridge on box-decomposable finite-grid persistence modules, proves exact reconstruction and sharp additive MinCert, and supplies a finite replay verifier. The second validated success-case part executes the enriched projective probe on every finite-poset representation, proves full morphism, exact, Yoneda, and derived Return, proves atomic projective-generator minimality, and composes the first two validation tiers without strength inflation. The third validated success-case part classifies the truncated-cycle Nakayama modules, computes syzygy exactly, proves a uniform stable return degree, converts stable identity into a nonzero self-Ext witness, closes Tachikawa's second conjecture on the whole family, and supplies a finite verifier and representation-finite expansion. The fourth validated success-case part transports regular-bimodule periodicity to every module, proves the exact one-degree projectivity criterion and its twisted-orbit refinement, closes representation-infinite weighted-surface and quaternion-type families in degree four and Dynkin preprojective families in degree six, and records the Challenge-to-Stable gate without novelty inflation. The fifth validated success-case part realizes the Leopoldt kernel as a finite semisimple Galois module, identifies subfield kernels with fixed spaces, reconstructs the complete S_3 kernel through a unimodular probe matrix, proves a minimal zero certificate, derives Frobenius defect propagation, and isolates primitive regulator nonvanishing as the sole external arithmetic leaf. The sixth validated success-case part realizes finite Alexandrov sheaves as finite poset representations, gives an explicit effective gluing operator, proves sharp triple coherence, transports exact and all-degree Yoneda data through bounded derived categories, computes hypercohomology on principal-good covers by a finite Čech totalization, and supplies deterministic replay. The seventh validated success-case part reduces every typed finite-state tower to an eventual-period monodromy, identifies the inverse limit with its stable image, kills the Roos defect, computes derived inverse-limit cohomology through Milnor collapse, supplies finite apex and cone tests, and isolates external completeness or arithmetic control as separate domain obligations. The inserted VC7L part identifies compatible finite quotient distributions with Iwasawa elements, measures, and Amice series, proves exact character and weight specialization, reconstructs finite layers by sharp Fourier probes, derives a one-layer finite-precision bound from binomial valuations, controls pseudomeasure poles, recovers the Kubota–Leopoldt function at exact source strength, and types the missing explicit-reciprocity, characteristic-ideal, and Leopoldt leaves without importing them. The eighth validated success-case part builds finite Serrin atlases, passes approximation sequences through local compactness, annihilates nonlinear product defects, Returns regularity and weak–strong uniqueness from the frozen $L_t^6 L_x^6$ source criterion, derives the corresponding blow-up alternative and continuation theorem, and isolates the missing primitive norm estimate without strength inflation. The ninth validated success-case part proves complete-DVR ball, determinant, logarithm, Strassmann, and regulator certificates, executes an exact 61-adic regulator computation for a totally real cubic field, and composes it with VC5 to Return Leopoldt for the S_3-Galois closure. The tenth validated success-case part combines square presentations, finite cyclotomic scans, certified logarithmic norms, determinant resultants, and finite-layer control to prove an exact Iwasawa growth formula beyond an explicit threshold, and executes a complete $\mathbb{Z}_3[[T]]$ example. The eleventh validated success-case part proves the transverse Bockstein leading-term theorem, decomposes exact cyclic singular layers into free cyclotomic and finite resultant sectors, derives the closed rank–torsion formula for arbitrary finite cyclotomic support, and executes a complete rank-three $\mathbb{Z}_3[[T]]$ example. The governance part provides the canonical certificate, registry, nonclaims, and closure statement. The v21 derived-theorem Part additionally proves the certified implementation paracategory and substitution rule, finite strength and blocker frontiers, product Return, strict descent and field-extension transport for finite-state derived limits, and the exact mixed regular–singular one-variable layer profile. Thus every universal operation asserted by UB has a local theorem, a composition rule, a Return ceiling, and at least one fully closed external validation inside this single source. □

Remark 143.2 (Expansion policy). The next mathematical progress should occur through further validated domain cases, not by duplicating the universal component library. VC3–VC11 now show ten required patterns: replace a conditional finite-state assumption by an explicit source classification and transition calculation, replace an impossible module classification by a stronger algebra-level bimodule certificate that transports naturally to every module, reconstruct an arithmetic defect object from a finite fixed-field probe matrix while exposing the remaining primitive nonvanishing leaf, glue finite local objects and all derived structure through an explicitly certified cocycle, or replace an uncertified infinite tail by a finite typed state machine and stable-image monodromy, or convert compatible finite arithmetic distributions into exact Iwasawa, measure, power-series, and specialization Returns while exposing boundedness, reciprocity, and nonvanishing as separate source leaves, or pass a nonlinear approximation through compactness, defect annihilation, finite norm atlases, uniqueness, and continuation while exposing the primitive analytic estimate as the only unresolved leaf, or separate a non-Archimedean value from its precision ceiling and propagate the certified nonvanishing through regulator and Galois Return, or combine an Iwasawa determinant with a finite cyclotomic scan and logarithmic norm to obtain exact Tor-free layer growth beyond a certified threshold, or replace an invalid finite-length claim at a cyclotomic zero by a typed free sector together with Bockstein and reduced-resultant data. A domain adapter is accepted only when it instantiates the UB component tuple, supplies an external Return theorem, passes the finite closure contract, and leaves no core hypothesis hidden inside the requested result.

144 v21 ownership, validation, and release boundary

Verifier contract 144.1 (Release-facing UB ownership contract)**.** Final release assembly must reconcile Main Chapter 9 canonical theorem identities, Appendix DG proof locators, Appendix HT research-spine locators, UB implementation and validation locators, predecessor artifacts, componentwise migration status, source/target types, propagated assumptions, supported strengths, failure owners, target-indexed results, source bindings, replay artifacts, and final governance entries. A local UB closure declaration does not substitute for that cross-document pass.

Theorem 144.2 (v21 UB ownership closure)**.** *Appendix UB is namespace-compatible with v21 when every Main-owned generic statement is cited through its Main identity, every UB implementation identifies that owner, every Appendix-local theorem is explicitly local, every source-specific validation remains in its declared layer, and every retained predecessor statement has a predecessor locator and migration record. Under these conditions the component library and twelve-case portfolio introduce no conflicting theorem ownership.*

Classification 144.3 (Final construction-stage status)**.**

```
appendix-UB-v21-synchronized-and-component-extraction-audited
+ generic-Main-Core-ownership-separated
+ executable-component-contracts-and-regressions-strengthened
+ dual-Core-DG-HT-handoffs-typed
+ tail-descent-coherence-rigidity-and-target-results-separated
+ twelve-source-bound-validation-cases-preserved
+ appendix-local-derived-theorem-pack-proved-and-bounded
+ finite-state-descent-limit-and-mixed-Iwasawa-returns-added
+ predecessor-mathematics-and-locators-retained
+ benchmark-quarantine-and-validation-no-backflow-enforced
+ final-Claim-Register-and-release-closure-pending
```

Selected mathematical source regime

The constituent results used above belong to the standard regimes of homological algebra, descent, derived categories, Morita theory, and commutative algebra. Canonical reference families include:

- S. Mac Lane, *Categories for the Working Mathematician*, for categorical composition, naturality, and coherence;
- M. Atiyah and I. Macdonald, *Introduction to Commutative Algebra*, for Nakayama, support, and flatness;
- C. Weibel, *An Introduction to Homological Algebra*, for cones, derived functors, Kunneth, universal coefficients, and spectral sequences;
- B. Keller, derived Morita and dg-category results, for generator-based derived realization;
- A. Grothendieck and the fpqc descent tradition, for faithfully flat descent;
- the Stacks Project sections on descent, perfect complexes, derived completion, and base change;
- A. Neeman, *Triangulated Categories*, for thick generation and exact-functor propagation;
- M. Kashiwara and P. Schapira, sheaf and derived-category references, for Čech and hypercohomological descent;
- standard representation-theoretic treatments of Nakayama algebras, stable module categories, and dimension shifting, for the classical constituents used in VC3;
- A. Dugas and the periodic-resolution literature, for bimodule periodicity and self-injective algebras;
- K. Erdmann and A. Skowroński, *Weighted surface algebras*, for period-four representation-infinite symmetric families;
- periodicity results for Dynkin preprojective algebras, for the uniform degree-six VC4 specialization;
- Ferri–Johnston and the Brauer-relation literature, for fixed-field Leopoldt kernels, defect relations, Frobenius propagation, and the S_3 specialization used by VC5;
- standard finite-Alexandrov-space and sheaf-descent references, together with Čech hypercohomology, for the classical constituents used by VC6.
- standard Roos, Mittag–Leffler, Milnor inverse-limit, Fitting-decomposition, and adic-completion references for the classical constituents used by VC7.
- Mahler–Amice measure theory, completed group algebras, finite Fourier inversion, Kubota–Leopoldt interpolation, Coleman/Perrin–Riou regulator theory, and characteristic-ideal references for the classical constituents and external interfaces used by VC7L.
- J. Serrin and G. Prodi for the fixed-exponent regularity and weak–strong uniqueness source lane, together with Aubin–Lions–Simon compactness and standard Leray–Hopf continuation theory, for the classical constituents used by VC8.

- complete-DVR ball arithmetic, Hensel lifting, p-adic logarithms, Strassmann and Weierstrass theory, Baker–Brumer context, and computational p-adic regulator methods for the classical constituents and Challenge panorama used by VC9.
- one-variable Weierstrass preparation, determinant norms and resultants, local logarithm/exponential theory, square presentations, and classical Iwasawa growth for the constituents used by VC10; multivariable Cuoco–Monsky, Akashi-series, pseudo-null, and noncommutative Weierstrass methods are Challenge-only context.
- prime-power cyclotomic resultants, DVR Smith form, Bockstein leading terms, determinant-line Euler data, and truncated Euler characteristics for the constituents used by VC11; higher-Bockstein and noncommutative leading-term theories are panorama-only context.

These sources support the classical constituent theorems. The UB component contracts and integrated closure theorems are governed by the statements and proofs in this appendix.

End of Appendix UB – Canonical v21 Global-Reaudited Maximal Derived-Theorem Component and Validation Edition